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Data and Biospecimen Collection Nonresponse Bias Analysis for Wave 5

Inter-university Consortium for Political and Social Research
P.O. Box 1248
Ann Arbor, Michigan 48106
www.icpsr.umich.edu

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PATH

Population Assessment of Tobacco and Health

A collaboration between the NIH and FDA

PATH Study
Wave 5 Data and Biospecimen Collection Nonresponse Bias Analysis Report

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Brian O'laughlin

Contracting Officer, PATH Study

National Institute on Drug Abuse

National Institutes of Health

Office of Acquisitions

3WFN MSC 6012

301 N Stonestreet Ave

Bethesda, Maryland 20892

Submitted by:

Westat

An Employee-Owned Research Corporation®

1600 Research Boulevard

Rockville, Maryland 20850-3129

(301) 251-1500

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1. Introduction

Westat prepared this report as a reference document for researchers using the Population Assessment of Tobacco and Health (PATH) Study interview and biomarker data. The PATH Study is a nationally representative, longitudinal cohort study of adults and youth in the United States, collecting data on those ages 12 years and older in Waves 1-4 and Wave 5, and on those ages 12 to 17 in the Wave 4.5 special data collection. This report focuses on the Wave 5 response rates and potential nonresponse biases for select demographic and outcome measures. Similar reports are also available for Wave 1 through Wave 4.5, and can be found at the following websites for the National Addiction & HIV Data Archive Program (NAHDAP).

Waves 1-4: https://www.icpsr.umich.edu/icpsrweb/NAHDAP/studies/36231/datadocumentation

Wave 4.5: https://www.icpsr.umich.edu/icpsrweb/NAHDAP/studies/37519/datadocumentation

In this document, the term "participants" is used to indicate people who agreed to be a part of the PATH Study whether or not they completed an interview in a particular wave. The term "respondents" is used to indicate the subset of participants who actually completed an interview or, in the case of shadow youth (i.e., participants younger than 12 years old), verified their information with the study in a particular wave; where needed for clarity, this term may be qualified to indicate the type of respondent (e.g., "shadow youth respondent," "adult interview respondent," etc.).

At Wave 5, the PATH Study data support analyses of two cohorts. The Wave 1 Cohort includes all Wave 1 respondents. These study participants were selected to represent adults ages 18 and older and youth ages 9 to 17 in the U.S. civilian, noninstitutionalized population (CNP) at the time of Wave 1. The Wave 1 Cohort is to be used for longitudinal analysis at Wave 5. All Wave 1 respondents were eligible for Wave 5 data collection unless they were incarcerated, deceased, or resided outside of the U.S. at the time of Wave 5.

The Wave 4 Cohort consists of two components. The first component is the Wave 4 adult and youth respondents from the Wave 1 Cohort who were in the U.S. CNP at the time of Wave 4. The second component is a replenishment sample designed to supplement the Wave 1 sample. The Wave 4 Cohort serves the purpose of both longitudinal and cross-sectional analyses at Wave 5.

This report summarizes the Wave 5 nonresponse bias analyses conducted separately for the two cohorts. The Wave 5 analyses for the Wave 1 Cohort use an "all-waves" perspective, whereas the analyses for previous waves used a "single-wave" approach (see Section 3.2.1 for an explanation of this change and why it better serves the research community). The analyses for the Wave 4 Cohort use a single-wave perspective and are similar at Wave 5 to those conducted for the Wave 1 Cohort at Wave 2. The potential for nonresponse bias in Wave 5 estimates based on the Wave 1 Cohort is evaluated conditioning on Wave 1 response, while potential nonresponse bias in estimates based on the Wave 4 Cohort is assessed conditioning on Wave 4 response. With this in mind, a brief summary of the findings from the Wave 1 nonresponse bias analyses and from the Wave 4 nonresponse bias analyses for the Wave 4 Cohort is provided below.

Wave 1 Cohort Nonresponse Bias Analyses at Wave 1

- For the Wave 1 household screener and adult interview, most demographic and socio-economic characteristics of the respondents in Wave 1 aligned with estimates from the 2013 American Community Survey 1-year Public Use Microdata Sample (2013 ACS) when using the basic design weight, also known as the inverse-of-probability-of-selection (IPS) weight for the PATH Study. Exceptions were found for single-person households, sex, education, and ethnicity when comparing Wave 1 estimates using the IPS weight to the 2013 ACS estimates. Estimates of cigarette-smoking rates among adults in Wave 1 were within the range of estimates found in other national health studies. When the estimates were adjusted for nonresponse using the Wave 1 cross-sectional weight, they more closely approximated the 2013 ACS estimates for demographic and socio-economic characteristics and adult cigarette-smoking rates remained essentially the same.

- For the Wave 1 youth interview, most demographic characteristics of respondents were consistent with the estimates from the 2013 ACS, with the exception of ethnicity, when using the IPS weight. When the estimates were adjusted for nonresponse among youth, they more closely approximated the 2013 ACS estimates, but the ever-tried cigarette-smoking rates for all youth in Wave 1 remained 2 to 12 percentage points lower than those found in other national studies.

- For the Wave 1 urine and blood collections, most of the demographic and socio-economic characteristics of specimen providers generally aligned with estimates of these characteristics from the 2013 ACS, when using the IPS weight. In addition, when the estimates were adjusted for interview nonresponse, they were found to approximate the 2013 ACS estimates more closely.

Wave 4 Cohort Nonresponse Bias Analyses at Wave 4

- When using the Wave 4 cross-sectional weight, the weighted demographic and socio-economic estimates for the Wave 4 adult interview for the Wave 4 Cohort were nearly identical to corresponding estimates based on the 2016 ACS 1-year PUMS (2016 ACS). However, adults with health insurance were slightly underrepresented by the Wave 4 Cohort weighted estimate. Weighted estimates of cigarette-smoking rates among adults in the Wave 4 Cohort at Wave 4 were within the range of estimates found in other national health studies.

- When using the Wave 4 cross-sectional weight, demographic estimates based on all Wave 4 Cohort youth at Wave 4 were nearly identical to corresponding estimates based on the 2016 ACS. The PATH Study ever-tried cigarette-smoking rates for all Wave 4 Cohort youth in Wave 4 remained 1 to 6 percentage points lower than those found in other national studies, when comparing weighted estimates.

This report is organized as follows: Chapter 2 provides an overview of the PATH Study sample design. Chapter 3 describes the methodology used for the analyses in this report. Chapters 4 and 5 present the results of nonresponse bias analyses for the Wave 1 Cohort interview data and Wave 4 Cohort interview data, respectively. Chapter 6 presents unweighted biospecimen collection response rates, ignoring cohort membership. Chapter 7 summarizes the findings and discusses their implications.

2. Overview of Sample Design and Data Collection

This chapter provides an overview of the sample design for the PATH Study. The PATH Study is a nationally representative, longitudinal cohort study of tobacco-use behaviors and related health outcomes among adults and youth in the United States. Interviews were conducted with respondents ages 12 years and older in Waves 1-4 and Wave 5, and with those ages 12 to 17 in the Wave 4.5 special data collection. The study's design allows for the longitudinal assessment of within-person and between-person change in the use of tobacco products, including initiation, cessation, relapse, and transitions between products, as well as in related health effects associated with use patterns.

At Wave 1, a four-stage, stratified probability sample design was used to select adults ages 18 and older and youth ages 12 to 17 from the U.S. CNP; an additional "shadow sample" of youth ages 9 to 11 was selected to be interviewed after they turn 12 years of age in later waves. All Wave 1 sample respondents together form the Wave 1 Cohort.

At Wave 4, the original set of Wave 1 sample respondents was replenished with a probability sample of adults, youth, and shadow youth ages 10 to 11[1] selected from the U.S. CNP at the time of Wave 4. The replenishment effort had two objectives: supplementation of the Wave 1 sample to address attrition and the changing composition of the study's adult respondents as Wave 1 youth (who were not oversampled with respect to tobacco use) reach age 18, and selection of a shadow sample for fielding in future waves. The replenishment effort also gave persons who had been overseas, in the military, or in an institutional setting at the time of Wave 1 (but were no longer at Wave 4) a chance of selection for the study. Because this "replenishment sample" was designed to supplement the Wave 1 sample, it is not intended to be used for estimation purposes on its own; rather, it was intended to be combined for estimation and analysis purposes with Wave 4 adult and youth respondents from the Wave 1 sample who were in the CNP at the time of Wave 4. This combined set of Wave 4 respondents forms the Wave 4 Cohort.

[1] The difference in the shadow youth age range between the Wave 1 sample (ages 9 to 11) and the Wave 4 replenishment sample (ages 10 to 11) reflects the planned timings of (a) future data collections among the PATH Study sample and (b) efforts to replenish the sample periodically. After Wave 4, data collection with all study participants is planned to occur biennially.

Figure 1 illustrates the components of the Wave 1 and Wave 4 Cohorts. Table 2-1 shows the sample sizes for the two cohorts.

Figure 1. Illustration of the relationship between the Wave 1 Cohort and the Wave 4 Cohort

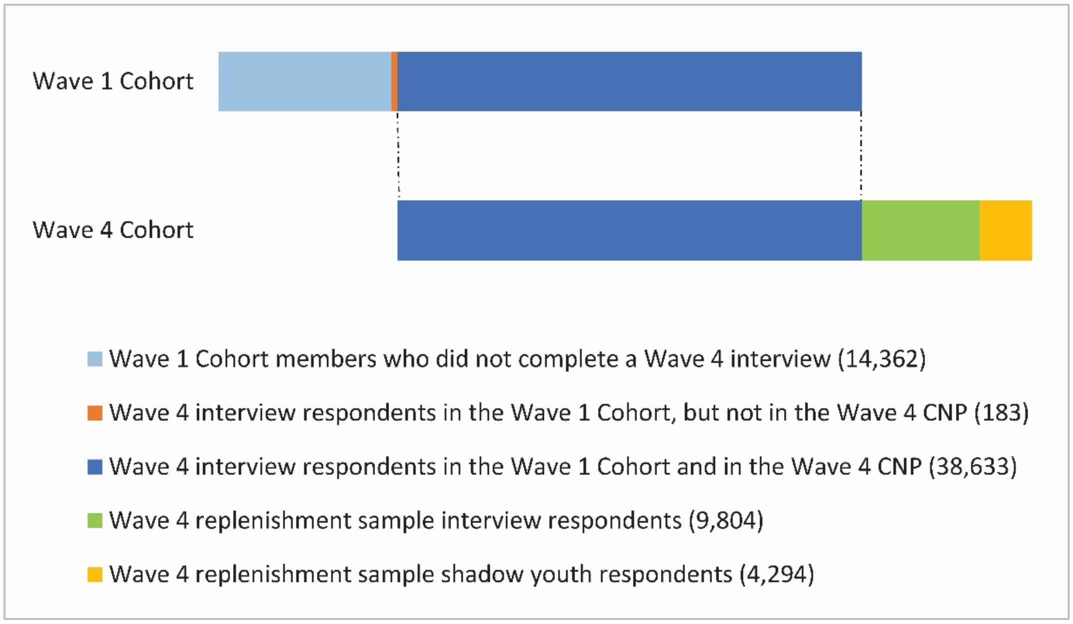

Table 2-1. Wave 1 Cohort and Wave 4 Cohort sample sizes

Sample	Wave 1 Cohort	Wave 4 Cohort
Wave 1 Cohort members who did not complete a Wave 4 interview	14,362	
Wave 4 interview respondents in the Wave 1 Cohort, but not in the Wave 4 CNP	183	
Wave 4 interview respondents in the Wave 1 Cohort and in the Wave 4 CNP	38,633	38,633
Wave 4 replenishment sample interview respondents		9,804
Wave 4 replenishment sample shadow youth respondents		4,294
Total	53,178	52,731

Key features of the Wave 1 and Wave 4 sample designs are summarized below. Further details can be found in Chapter 2 of the PATH Study Restricted Use Files (RUFs) User Guide (https://www.icpsr.umich.edu/icpsrweb/NAHDAP/studies/36231/datadocumentation).

2.1 Wave 1

The target population of the PATH Study at Wave 1 is the U.S. CNP 9 years of age and older at the time of Wave 1 (2013–2014). Active duty military personnel and those residing in an institutional setting at the time of Wave 1 were excluded. College students living away from home during the school year were identified as members of their permanent residence (e.g., parents' home). For Wave 1, a four-stage, stratified area probability sample design was used with a two-phase design for sampling adults at the final stage. The sampling rates for adults varied by age, race, and tobacco-use status. At the first stage, a stratified sample of geographical primary sampling units (PSUs) was selected, in which a PSU was a county or group of counties. For the second stage, within each selected PSU, smaller geographical segments (consisting of one or more census blocks) were formed and then a systematic sample of these segments was drawn. At the third stage, a systematic sample of addresses within sampled segments was drawn from listings of addresses; the main source of these addresses was the U.S. Postal Service (USPS) Computerized Delivery Sequence Files (CDSFs). The CDSFs provide very high coverage of the residential addresses in the U.S. The fourth stage was the random selection of persons within sampled households.

For within-household selection, a roster of all the members in the sampled household was constructed using the household screener. An adult household member, the household screener respondent, was asked to provide the age for each household member, as well as the race, active military service status, and tobacco usage for each adult household member.[2] This information was used for sampling three main groups of interest:

- Adults ages 18 and older (up to two adults per household);
- Children ages 12 to 17 (i.e., "youth," generally up to two per household);[3] and
- Children ages 9 to 11 (i.e., "shadow youth," generally up to two per household) to be interviewed in later waves after reaching 12 years of age.

Two-phase sampling was used for adult selection due to potential misreporting by the household screener respondent of the tobacco-use status of adult household members. Phase 1 sampling depended on the age, race, and tobacco-use information obtained from the household informant

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[2] The household screener collected information on adult household members' use of several different types of tobacco products. For example, it collected information on current use of products with relatively high prevalence or well-established use, such as cigarettes, cigars, and pipes; and on ever use of products with relatively low prevalence or newly emerging use, such as electronic cigarettes or e-cigarettes.

[3] Given a special analytic interest in multiple-birth youth (e.g., twins), the shadow youth and youth sampling procedures were modified when households containing multiple-birth youths were encountered so that the multiple-birth youths would have relatively higher probabilities of selection. This resulted in some households with more than two children selected for the youth and/or shadow youth samples.

during the household screener. Phase 2 sampling was based on self-reported age, race, and tobacco-use status, obtained by interviewing the individuals sampled at Phase 1. Sampling rates for the two phases were designed to oversample young adults (ages 18 to 24) and adult tobacco users of all ages, thereby increasing precision for estimates relating to these subgroups.

PATH Study Wave 1 interview data and biospecimen collections began September 12, 2013 and ended December 14, 2014. For Wave 1, 32,320 adult interviews and 13,651 youth interviews were completed. All adult interview respondents were asked to provide urine and blood specimens; 21,801 provided a urine specimen and 14,520 provided a blood specimen. A subsample of the adults who provided urine form the Wave 1 Biospecimen Core. The Wave 1 Data and Biospecimen Collection Nonresponse Bias Analysis Report is available on the NAHDAP website.

2.2 Wave 2 and Wave 3

There was no additional sampling for Wave 2 or Wave 3 of the PATH Study. Wave 2 was the first follow-up wave of Wave 1 respondents. Wave 1 respondents were eligible for Wave 2 if they were still residents of the U.S. and not incarcerated.

During the Wave 2 data collection period, attempts were made to contact the Wave 1 adult and youth respondents as well as members of the shadow youth sample established at Wave 1. Shadow youth who turned age 12 by Wave 2 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 2. Similarly, persons in the youth sample at Wave 1 who reached age 18 by Wave 2 were asked to complete the adult interview and to provide urine and blood specimens.

The PATH Study Wave 2 interview data and biospecimen collections began October 23, 2014 and ended October 30, 2015. For Wave 2, 28,362 adult interviews and 12,172 youth interviews were completed. The study subsampled 14,465 adults for urine collection at Wave 2 from adults who completed an adult interview at Wave 1; among these adults, 12,561 completed the Wave 2 interview and 12,109 provided urine specimens at Wave 2. The study also collected 1,587 urine specimens and 908 blood specimens from the 1,915 persons who had completed the PATH Study adult interview for the first time (referred to as "first-time adult interview respondents") at Wave 2. The Wave 2

Data and Biospecimen Collection Nonresponse Bias Analysis Report is available on the NAHDAP website.

Wave 3 was the second follow-up wave of the PATH Study for the Wave 1 Cohort. Wave 1 respondents were eligible for Wave 3 if they were still residents of the U.S. and not incarcerated. Wave 2 nonrespondents were fielded for Wave 3 unless they were deceased, had moved out of the U.S. permanently, had specifically requested withdrawal from the study, were firm or hostile refusers at Wave 2, had a language barrier,[4] or had a physical or mental disability or chronic illness that prevented participation in the study. Shadow youth from previous waves who turned age 12 by Wave 3 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 3. Similarly, youth from previous waves who reached age 18 by Wave 3 were asked to complete the adult interview and to provide urine and blood specimens.

The PATH Study Wave 3 interview data and biospecimen collections began October 19, 2015 and ended October 23, 2016. For Wave 3, 28,148 adult interviews and 11,814 youth interviews were completed. The study subsampled 13,700 adult respondents for urine collection at Wave 3 from adults who had completed an adult interview at a prior wave; among these adults, 13,338 provided urine specimens at Wave 3. The study also collected 1,641 urine specimens and 835 blood specimens from the 1,907 first-time adult interview respondents in Wave 3. The Wave 3 Data and Biospecimen Collection Nonresponse Bias Analysis Report is available on the NAHDAP website.

2.3 Wave 4

For members of the Wave 1 Cohort, Wave 4 of the PATH Study was the third follow-up wave. Wave 1 respondents were eligible for Wave 4 if they were still residents of the U.S. and not incarcerated. Nonrespondents from previous waves were fielded for Wave 4 unless they were deceased, had moved out of the U.S. permanently, had specifically requested withdrawal from the study, were firm or hostile refusers at a previous wave, had a language barrier, or had a physical or

[4] The PATH Study interviews can be conducted in English or Spanish. At Wave 2, the cases assigned a final nonresponse status due to language problems included two groups: (a) Wave 1 respondents who mentioned difficulty completing the Wave 1 interview in English or Spanish as their main reason for not completing an interview at Wave 2; and (b) Wave 1 shadow youth who did not feel comfortable completing an interview in English or Spanish when asked to do so by the study for the first time.

mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at both Wave 2 and Wave 3 were not fielded for data collection.[5] (Note that refusal is a specific type of nonresponse, so some nonrespondents to Wave 2 and Wave 3 who did not refuse at both those waves were fielded for Wave 4.) Shadow youth from previous waves who turned age 12 by Wave 4 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 4. Similarly, youth from previous waves who reached age 18 by Wave 4 were asked to complete the adult interview and to provide urine and blood specimens. Unlike in previous waves, youth interview respondents were asked to provide a urine specimen.

At Wave 4, an additional sample was selected from the U.S. CNP to account for the aging of Wave 1 respondents and sample attrition among the Wave 1 Cohort. As part of the Wave 4 replenishment effort, adults, youth, and shadow youth were sampled within the existing PATH Study sample segments from among the addresses not selected for Wave 1.

The Wave 4 within-household sampling procedures mirrored those used at Wave 1. However, the sampling rates were designed to bring the Wave 4 adult and youth sample sizes up to Wave 1 levels by sampling domain. This required consideration for the expected combined effect of the aging of Wave 1 respondents and the loss of sample size due to attrition on the PATH Study sample by the time of Wave 4. In particular, it was necessary to oversample adult tobacco users given that Wave 1 youth respondents who became adults since Wave 1 (and those who will continue to do so) were not oversampled with respect to tobacco-use status. All adult interview respondents from the replenishment sample were asked to provide urine and blood specimens, and all youth interview respondents from this sample were asked to provide a urine specimen.

The PATH Study Wave 4 interview data and biospecimen collections began December 1, 2016 and ended January 3, 2018. Among the Wave 1 Cohort members, 27,757 adult interviews and 11,059 youth interviews were completed at Wave 4. Among Wave 4 Cohort members (i.e., after combining the Wave 1 Cohort and Wave 4 replenishment samples and restricting to those in the U.S. CNP at the time of Wave 4), 33,644 adult interviews and 14,793 youth interviews were completed.

[5] For youth, refusal could have come from the youth or their parent/guardian.

The study subsampled 14,811 adult respondents for urine collection at Wave 4 from adults who had completed an adult interview at a prior wave; among these adults, 14,471 provided urine specimens. The study also collected 1,660 urine specimens and 890 blood specimens from the 1,900 members of the Wave 1 Cohort who were first-time adult interview respondents at Wave 4. Of the youth respondents from the Wave 1 Cohort, 9,892 provided a urine specimen. Among the Wave 4 Cohort interview respondents, the study collected 20,928 urine and 3,602 blood specimens from adults and 13,095 urine specimens from youth. The Wave 4 Data and Biospecimen Collection Nonresponse Bias Analysis Report is available on the NAHDAP website.

2.4 Wave 4.5

Wave 4.5 was a special data collection of the PATH Study; its focus was on continuing the annual collection of data among youth ages 12 to 17 (and their parents). There was no additional sampling for Wave 4.5 and no biospecimens were collected.

Wave 4.5 of the PATH Study was the fourth follow-up wave for age-eligible members of the Wave 1 Cohort and the first follow-up wave for age-eligible participants selected as part of the Wave 4 replenishment sample. All study participants were eligible for Wave 4.5 if they were ages 12 to 17, still residents of the U.S., and not incarcerated. Nonrespondents from previous waves were asked to participate in Wave 4.5 unless they were deceased, had moved out of the U.S. permanently, had specifically requested withdrawal from the study, were firm or hostile refusers at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at any two consecutive waves were not fielded for data collection.[6] Shadow youth from previous waves who turned age 12 by Wave 4.5 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 4.5.

The PATH Study Wave 4.5 special data collection began December 1, 2017 and ended December 1, 2018. The PATH Study completed 13,131 youth interviews at Wave 4.5; 8,761 of these youth were members of the Wave 1 Cohort and 12,918 were members of the Wave 4 Cohort. The Wave 4.5 Data Collection Nonresponse Bias Analysis Report is available on the NAHDAP website.

[6] For youth, refusal could have come from the youth or their parent/guardian.

2.5 Wave 5

Wave 5 was the fourth or fifth follow-up wave for those in the Wave 1 Cohort and the first or second follow-up wave for those in the Wave 4 replenishment sample (depending on whether or not the study participant was part of the Wave 4.5 special data collection). There was no additional sampling for Wave 5, and all study members were eligible if they were still residents of the U.S. and not incarcerated. Nonrespondents from previous waves were asked to participate in Wave 5 unless they were deceased, had moved out of the U.S. permanently, had specifically requested withdrawal from the study, were firm or hostile refusers at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at any two consecutive waves were not fielded for data collection.[7] Shadow youth from previous waves who turned age 12 by Wave 5 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 5 and to provide a urine specimen. In addition, urine specimens were requested from Wave 5 youth who completed their first youth interview at the Wave 4.5 special data collection (when no biospecimens were requested) or who consented to urine specimen collection at Wave 4.[8] Similarly, participants who completed their first adult interview at Wave 5 were asked to provide urine and blood specimens. Urine specimens were also requested from all adults belonging to the Wave 1 Biomarker Core.

The PATH Study Wave 5 interview data and biospecimen collections began December 1, 2018 and ended November 30, 2019. Among the Wave 1 Cohort members, 28,970 adult interviews and 6,760 youth interviews were completed at Wave 5. Among Wave 4 Cohort members, 32,687 adult interviews and 11,976 youth interviews were completed.

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[7] For youth, refusal could have come from the youth or their parent/guardian. A refusal at Wave 4.5 was counted in the calculation of "two consecutive waves" of refusal for participants who were part of the special data collection. Otherwise, Waves 3 and 4 were the last two consecutive waves that were considered in the calculation for Wave 5.

[8] Note that approximately 60 Wave 5 youth respondents who consented to urine specimen collection at Wave 4 did not ultimately provide a usable specimen at Wave 4. In addition to the main groups of Wave 5 youth respondents described above, urine specimens were also requested from approximately 60 youth respondents who did not complete the Wave 4 youth interview or did not consent to urine specimen collection at Wave 4, where the reason for nonresponse or lack of consent was something other than a refusal. These Wave 5 youth respondents are included in the biospecimen analyses in this report.

There were 8,167 Wave 5 adult respondents who were members of the Wave 1 Biomarker Core. Among these adults, 7,988 provided urine specimens at Wave 5. The study also collected 4,114 urine specimens and 2,040 blood specimens from the 4,433 first-time adult interview respondents at Wave 5. Urine specimens were collected from 7,682 of the 7,905 Wave 5 youth respondents who consented to urine specimen collection at Wave 4. The study also collected 2,849 urine specimens from 3,251 Wave 5 youth respondents who completed their first youth interview at Wave 4.5 or Wave 5.[9]

A summary of the results of the data and biospecimen collection efforts for Waves 1 to 5, by cohort and age group, is provided in Table 2-2.

Table 2-2. PATH Study data and biospecimen collection summary by wave, cohort, and age group

Wave	Data collection start date	Data collection end date	Cohort	Age group	Interviews conducted	Urine specimens collected[a]	Blood specimens collected[b]
1	September 12, 2013	December 14, 2014	Wave 1 Cohort	Adults	32,320	21,801	14,520
				Youth	13,651	N/A	N/A
2	October 23, 2014	October 30, 2015	Wave 1 Cohort	Adults	28,362	13,696	908
				Youth	12,172	N/A	N/A
3	October 19, 2015	October 23, 2016	Wave 1 Cohort	Adults	28,148	14,979	835
				Youth	11,814	N/A	N/A
4	December 1, 2016	January 3, 2018	Wave 1 Cohort	Adults	27,757	16,131	890
				Youth	11,059	9,892	N/A
			Wave 4 Cohort	Adults	33,644	20,928	3,602
				Youth	14,793	13,095	N/A
4.5	December 1, 2017	December 1, 2018	Wave 1 Cohort	Youth	8,761	N/A	N/A
			Wave 4 Cohort	Youth	12,918	N/A	N/A
5	December 1, 2018	November 30, 2019	Wave 1 Cohort	Adults	28,970	11,270	1,640
				Youth	6,760	5,948	N/A
			Wave 4 Cohort	Adults	32,687	11,429	1,929
				Youth	11,976	10,490	N/A

[a] The PATH Study requested urine specimens from first-time adult interview respondents, subsamples of other adult respondents, all youth at Wave 4, and a subsample of youth respondents at Wave 5.

[b] The PATH Study only requested blood specimens from first-time adult interview respondents.

[9] The biospecimen analyses presented in Chapter 6 (and the Wave 5 counts provided in Table 2-2) also reflect the results of urine specimen collection among the approximately 60 Wave 5 youth respondents who did not fall into one of the main collection groups. See the previous footnote to this report.

3. Methodology

This chapter first discusses the factors affecting nonresponse bias, and then describes the methods used for evaluating potential nonresponse bias in this report.

3.1 Factors Affecting Nonresponse Bias

Bias is the difference between a survey estimate and the actual population value. Although nonresponse bias can be a major concern in multi-stage household surveys, nonresponse does not necessarily lead to nonresponse bias in survey estimates. Assuming nonresponse to be a fixed property of an individual in the population, the nonresponse bias of an estimate can be expressed mathematically to show the relationship between the bias and two factors—the amount of nonresponse and the difference between respondents and nonrespondents (Groves, 2006):

$$Bias(\bar{y}_r) = (1 - r)(\bar{Y}_r - \bar{Y}_m)$$

where $\bar{y}_r$ is the estimated population mean based on the sample respondents only, r is the response rate in the target population, $\bar{Y}_r$ is the mean for respondents in the target population, and $\bar{Y}_m$ is the mean for nonrespondents in the target population.

That is, the magnitude of nonresponse bias depends on the correlation between response propensity and the measure of interest. Within the same survey, different estimates can be subject to different levels of nonresponse bias. Some measures, unrelated to the propensity to respond, can be immune from the biasing effect of nonresponse; others, in the same survey, can be subject to large biases (Groves, 2006). For example, if cigarette smokers were less inclined to respond to the PATH Study than nonsmokers, then the estimated prevalence of cigarette smoking would be subject to nonresponse bias, prior to any weighting adjustments for nonresponse. However, if cigarette smoking were unrelated to, say, number of years lived in the U.S., then the estimated average years of residency may be unaffected by the different response propensities of smokers and nonsmokers. If propensity to respond is completely random, then nonresponse reduces sample sizes for analysis but does not lead to bias, even if the overall response rate is lower than desired.

In practice, survey practitioners often attempt to decrease potential nonresponse bias by not only increasing the overall response rate but also improving cooperation from "difficult to reach" subgroups. In addition, effective statistical adjustments can also help reduce potential nonresponse biases in some survey estimates.

3.2 Analyses for Evaluating Wave 5 Potential Nonresponse Bias

It is not always possible to measure the actual bias due to nonresponse; however, various approaches can be used to help identify potential sources of nonresponse bias. At Wave 5, the Wave 1 Cohort data serve the purpose of longitudinal analysis, while the Wave 4 Cohort data support both cross-sectional and longitudinal analyses. Separate sets of analyses were conducted for the two cohorts to assess potential nonresponse bias in estimates using interview data from Wave 5 of the PATH Study.

3.2.1 Weights and Corresponding Analysis Perspective Used by Cohort

As described in Chapter 5 of the PATH Study RUFs User Guide, the study provides three different types of weights for researchers analyzing Wave 5 data: a single-wave weight, an all-waves weight, and a special-collection all-waves weight. Separately for the Wave 1 Cohort and the Wave 4 Cohort, a single-wave weight was created for all cohort members who responded at Wave 5. For the Wave 1 Cohort only, an all-waves weight was created for Wave 5 respondents who also responded in all previous waves, ignoring the Wave 4.5 special data collection. (For the Wave 4 Cohort at Wave 5, the single-wave weight is an all-waves weight when the Wave 4.5 special data collection is ignored.) A special collection all-waves weight was created for each cohort, and applies to Wave 5 respondents who also responded at all previous waves relevant to the cohort, including the Wave 4.5 special data collection. However, many PATH Study participants were not eligible for the Wave 4.5 special data collection due to the age restrictions. Therefore, unless explicitly noted, participation in the Wave 4.5 special data collection was ignored for the purpose of the analyses in this report.

In terms of longitudinal analyses using Wave 1 Cohort data, the single-wave weight has less utility than the all-waves weight. The single-wave weight allows researchers to relate a behavior or outcome at a later wave (Wave *w*) to characteristics of the same person at the first wave of a cohort. An example of an analysis that would use the single-wave weight for Wave 5 data from the Wave 1

Cohort would be estimating the proportion of Wave 5 e-cigarette users who used e-cigarettes at Wave 1. This is a longitudinal analysis using the persons who have data at both Wave 1 and Wave 5. However, if it is desired to relate Wave *w* outcomes to characteristics from an earlier wave that is not the first wave of a cohort then the all-waves weight is needed. An example of an all-waves analysis would be a logistic regression estimating a person's e-cigarette use at Wave 5 from the person's usage status and demographic characteristics at Waves 1 through 4.

As a result, and as waves of data collection accrue, the preponderance of analyses for the Wave 1 Cohort are likely to use all-waves respondents and their corresponding weights. To address the majority of analyses the research community will conduct using Wave 5 data from this cohort, the evaluation of potential nonresponse bias in estimates based on the Wave 5 interview data for the Wave 1 Cohort will be from an all-waves perspective.

In previous Nonresponse Bias Analysis Reports, the Wave 1 Cohort was assessed from a single-wave perspective. Although changing to an all-waves perspective will benefit the research community, a likely consequence is lower interview response rates for the Wave 1 Cohort as compared to a single-wave approach. As discussed further in Section 4.1.1, for a Wave 1 Cohort member to be classified as an all-waves respondent at Wave 5, he or she must not only complete a Wave 5 interview but also have been a respondent in each of Waves 2 to 4. This reduces the numerator of the response rate compared to a single-wave approach (which counts all Wave 5 respondents from the Wave 1 Cohort in the numerator). Although some Wave 1 Cohort participants who did not respond at Wave 5 will also be classified differently using the all-waves perspective due to their response statuses at previous waves (e.g., some eligible nonrespondents at Wave 5 will be classified as ineligible or of unknown eligibility in an all-waves approach), this is unlikely to offset the reduction in the response rate numerator.

The Wave 4 Cohort will be evaluated from a single-wave perspective. As noted above, when the Wave 4.5 special data collection is ignored, the single-wave and all-waves approaches are equivalent for this cohort at Wave 5. In addition, it is strongly recommended that researchers use the Wave 4 Cohort, along with the Wave 4 Cohort single-wave weight, for any cross-sectional analyses using Wave 5 data (like those presented in Sections 5.1.3, 5.1.4, 5.2.3, and 5.2.4).

3.2.2 Comparison of Response Rates Across Subgroups

Although a response rate does not yield direct estimates of potential nonresponse biases on key measures, examining response rates by subgroups (for example, males versus females) may reveal sources of potential nonresponse bias. Wave 5 interview response rates were calculated for the Wave 1 Cohort and the Wave 4 Cohort, separately, as described below. The response rate calculations were based on the formula provided by the Office of Management and Budget in its "Standards and Guidelines for Statistical Surveys" (2006).

For the Wave 1 Cohort, the Wave 5 interview all-waves response rates are conditioning on Wave 1 response, so the subgroups are based on Wave 1 information. In contrast, the Wave 5 interview single-wave response rates for the Wave 4 Cohort are conditioning on Wave 4 response, so the subgroups are based on Wave 4 information, apart from the recruitment wave subgroup (which distinguishes cohort members who were recruited at Wave 1 from those recruited at Wave 4). This should be kept in mind when reviewing the response rates in Sections 4.1.1. 4.2.1, 5.1.1, and 5.2.1.

As in Waves 1 through 4, participants ages 18 and older were asked to complete an adult interview and participants ages 12 to 17 were asked to complete a youth interview. A Wave 5 nonrespondent does not have a Wave 5 interview date, so his/her Wave 5 age and interview type (i.e., adult interview versus youth interview) were determined using the latest available date of birth or age information. Each study participant was assigned an "anniversary month" for Wave 5, based primarily on the interview date(s) of the study member(s) in his or her household at prior waves. The age classification date for a Wave 5 nonrespondent was 1 month after the last day of his/her anniversary month or the final date of the Wave 5 data collection, whichever was earlier. A nonrespondent was included in the Wave 5 youth analyses if his/her age was determined to be between 12 and 17 on the age classification date, or included in the Wave 5 adult analyses if his/her age was determined to be 18 or older on the age classification date.

Both weighted and unweighted response rates are reported for the adult and youth interviews. The unweighted response rate measures the success of field operations in obtaining responses from the sampled persons. The weighted response rate estimates the proportion of the population represented by the sampled persons that would have responded if they all had been asked to participate in the study, and thus provides a measure of the potential impact of nonresponse on the study estimates. The basic design weight associated with the Wave 1 sample selection, also known as the Wave 1 IPS weight, was used for calculating the Wave 1 Cohort weighted all-waves response rates. A Wave 4 IPS weight does not exist for the Wave 4 Cohort. As discussed in Chapter 2, the Wave 4 Cohort consists of the Wave 4 adult and youth respondents from the Wave 1 sample (with the exception of those not in the U.S. CNP at Wave 4) and the respondents from the replenishment sample. The Wave 1 sample and the Wave 4 replenishment sample were selected at different times from two different sampling frames, so the probabilities of selection (and IPS weights) were computed for the two samples separately. However, the information needed to create the IPS weight for the Wave 4 Cohort is not available. This is because an adjustment to the IPS weights computed for the separate samples is needed for persons with multiple chances of selection for the Wave 4 Cohort. The information needed to make this adjustment (e.g., Wave 1 age, race, and tobacco-use status from which the probability of an adult being selected in Wave 1 could be determined) is known only for some replenishment sample respondents and is unknown for persons from the replenishment sample who did not respond at Wave 4. Therefore, the Wave 4 cross-sectional weight was used for calculating the Wave 4 Cohort weighted single-wave response rates.

3.2.3 Comparison to "Frame" Data

In a longitudinal study, the information collected in the baseline wave can be viewed as "frame" information for future waves, and thus used for evaluating potential nonresponse bias (Bose and West, 2002; Javitz and Wagner, 2005; Brownstein et al., 2009). For the PATH Study Wave 1 and Wave 4 Cohorts, separately, estimates based on Wave 5 interview respondents were compared to the "frame" information from the cohort's baseline wave (i.e., Wave 1 or Wave 4) for Wave 5 adults and youth. For the Wave 1 Cohort, this comparison helps identify characteristics that might be associated with bias due to nonresponse at any time since Wave 1, after compensating for possible undercoverage of the target population at Wave 1 (due to sampling frame deficiencies) and Wave 1 nonresponse. For the Wave 4 Cohort, the comparison helps identify characteristics that might be associated with nonresponse bias due to attrition between Wave 4 and Wave 5, after compensating for possible undercoverage and Wave 4 nonresponse. Participants known to have been ineligible at any wave are excluded from the Wave 1 Cohort all-waves response rates and nonresponse bias analyses in this report. This is analogous to the single-wave approach, e.g., participants known to be ineligible at Wave 5 are excluded from the Wave 4 Cohort single-wave analyses. In this sense, the focus of both analyses is on assessing bias due to nonresponse when the participant was known to be eligible or eligibility was unknown.[10]

Wave 1 Cohort

For the Wave 1 Cohort, the "frame" information included Wave 1 demographic and socio-economic characteristics, as well as Wave 1 tobacco-use status. The estimates were based on all the Wave 1 respondents who were eligible to be Wave 5 all-waves respondents and were computed using the Wave 1 cross-sectional weight (which was designed to reduce potential nonresponse bias at Wave 1).

Two sets of Wave 1 Cohort estimates were obtained from Wave 5 all-waves respondents. The first set of estimates was based on the Wave 1 cross-sectional weight. Comparing this set of estimates to the "frame" data shows the representativeness of the Wave 5 all-waves respondents before statistical adjustment for any nonresponse occurring since Wave 1. The second set of estimates was based on the Wave 5 all-waves weight that applies to Wave 5 respondents in the Wave 1 Cohort who also responded in Waves 2, 3, and 4. This weight was designed to reduce potential bias due to nonresponse at any time since Wave 1, so the second set of estimates shows the extent to which the statistical weighting adjustment might have helped reduce potential nonresponse bias at Wave 5. For reference, the terms "before Wave 5 weighting adjustment" and "after Wave 5 weighting adjustment" are used to refer to the two sets of estimates and comparisons in Sections 4.1.2 and 4.2.2. To facilitate interpretation of these two sets of estimates and comparisons, summary descriptions of the PATH Study statistical weighting adjustments for the Wave 1 Cohort are provided in the next two paragraphs.

Weighting adjustments are often used to account for differential response propensities across population subgroups. Among numerous sources, the Handbook on Household Surveys by the United Nations (2005, Chapter 6) and Särndal and Lundström (2005) discuss the methods and theory of using weight adjustments for nonresponse. For Wave 1, these adjustments were conducted at the household level and at the person level. The Wave 1 household-level weighting adjustments calibrated the estimates to household-level population estimates for census region and household

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[10] To avoid repetition, this interpretation of nonresponse is not accompanied by an explanatory footnote every time it is mentioned.

composition and size from the 2013 ACS. Such weighting adjustments also correct for disparities among other characteristics that might be associated with the variables involved in the weighting adjustments. After accounting for household-level nonresponse, households with at least one person sampled for the PATH Study were identified, and each sampled person within a household was assigned the corresponding household weight with an adjustment to account for his/her within-household probability of selection. The resulting weight was then adjusted to account for nonresponse to the Wave 1 adult or youth interview or shadow youth recruitment. After this adjustment for nonresponse, the weights were calibrated using a raking process to person-level population estimates from the 2013 ACS. Outlier values of the sample weights were trimmed and the weights were re-raked after any such trimming. More details about the PATH Study Wave 1 weight construction can be found in Section A.1 of the PATH Study RUFs User Guide.

For the Wave 1 Cohort, the Wave 4 all-waves nonresponse-adjusted weights served as the initial weights for developing the all-waves weights for respondents to the Wave 5 interview who also responded in all previous waves, ignoring the Wave 4.5 special data collection. These initial weights were adjusted to account for nonresponse to the Wave 5 interview and the resulting weights were raked to control totals. Some of the control totals came from the 2013 ACS; others involving tobacco use were sample-based rather than population-based and reflected estimated Wave 1 population characteristics. Raking to sample-based control totals, often employed in longitudinal studies (see, for example, Brick, Lê, and West (2003)), can limit drifting in some important baseline characteristics that might arise through the applications of nonresponse adjustments over time. Lundström and Särndal (1999) provide theoretical discussions about sample-based calibration together with empirical evidence that such calibration can help reduce both variance and nonresponse bias. General discussions about the calibration method can be found in Särndal and Lundström (2005) and Särndal (2007). More details about the PATH Study Wave 5 all-waves weight construction for the Wave 1 Cohort appear in Section A.6.1 of the PATH Study RUFs User Guide.

Wave 4 Cohort

For the Wave 4 Cohort, the "frame" information included Wave 4 demographic characteristics, as well as Wave 4 tobacco-use status. The estimates were based on all the Wave 4 Cohort members who were eligible for the Wave 5 interview and were computed using the Wave 4 cross-sectional weight (which was designed to reduce potential nonresponse bias at Wave 4).

Two sets of Wave 4 Cohort estimates were obtained from Wave 5 interview respondents: "before Wave 5 weighting adjustment" and "after Wave 5 weighting adjustment." The first set of estimates was based on the Wave 4 cross-sectional weight. Comparing this set of estimates to the "frame" data shows the representativeness of the Wave 5 responding sample before any statistical adjustment for attrition between Waves 4 and 5.[11] The second set of estimates was based on the Wave 5 single-wave weight for the Wave 4 Cohort, and shows the extent to which the statistical weighting adjustment might have helped reduce potential nonresponse bias at Wave 5.

To create the Wave 4 cross-sectional weight for the Wave 4 Cohort, a sequence of weighting steps involving the Wave 1 Cohort single-wave weight and the Wave 4 replenishment sample weight was implemented, including nonresponse adjustments, preliminary raking, compositing, and final raking and trimming. The control totals were created using the 2016 ACS. For shadow youth and youth, raking was done using cross-classifications of census region, single-year of age, race/ethnicity, and sex. (Missing values in the variables used for raking were imputed.)

The Wave 4 cross-sectional weights served as the initial weights for developing the Wave 4 Cohort single-wave weights for Wave 5 interview respondents. These weights were adjusted to account for nonresponse to the Wave 5 interview and the resulting weights were raked to control totals. Some of the control totals came from the 2016 ACS; others involving tobacco use were sample-based rather than population-based and reflected estimated Wave 4 population characteristics. More details about the construction of the Wave 4 cross-sectional weight and the Wave 5 single-wave weight for the Wave 4 Cohort appear in Sections A.4.3 and A.6.5, respectively, of the PATH Study RUFs User Guide.

[11] As noted in Section 3.2.1, the Wave 4.5 special data collection was ignored for this report.

3.2.4 Wave 4 Cohort: Comparison to External Data Sources

For the first three waves of the PATH Study, the Wave 1 Cohort can be used to produce cross-sectional estimates (which are approximate in nature for Waves 2 and 3). For Waves 4, 4.5, and 5 the Wave 4 Cohort serves the purpose of cross-sectional analysis (which is approximate in nature for Waves 4.5 and 5). A widely used approach for assessing potential nonresponse bias in cross-sectional estimates is benchmarking to external sources. A strength of this method is that estimates from external sources are independent of the PATH Study. On the other hand, key outcomes from the PATH Study are not necessarily available from external sources; the measurements may not be exactly the same across studies, and the external sources may be subject to coverage and nonresponse errors as well (Groves, 2006).

Estimates for adults and youth were obtained using the Wave 5 interview data and Wave 5 single-wave weight for the Wave 4 Cohort. Several PATH Study measures of cigarette smoking and electronic nicotine delivery system (ENDS)[12] use were compared to those from other national studies of tobacco and health (for the most similar time frames for which data were available). The PATH Study collects data on a range of tobacco-use behaviors; many of these variables are not available in other studies. However, responses to the PATH Study questions on cigarette smoking and ENDS use can be compared with estimates from other national studies that ask about these behaviors.

The external data sources used for analyses of cigarette smoking were the Tobacco Use Supplement to the Current Population Survey 2018-2019 (TUS-CPS 2018-2019), the National Health Interview Survey 2018 (NHIS 2018), the National Health and Nutrition Examination Survey 2017-2018 (NHANES 2017-2018), the National Survey on Drug Use and Health 2018 (NSDUH 2018), the National Youth Tobacco Survey 2019 (NYTS 2019), the National Youth Risk Behavior Survey 2017 (NYRBS 2017), and Monitoring the Future 2017 (MTF 2017). The external data sources used for comparison of ENDS use were TUS-CPS 2018-2019, NYTS 2019, NYRBS 2017, and MTF 2017. Estimates from these surveys were obtained using public use files, with the exception of MTF 2017. The variance stratum and cluster variables, which are needed to account appropriately for MTF's

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[12] Although the term "electronic nicotine products" is used in the PATH Study instrument to represent all electronic nicotine devices such as e-cigarettes, e-cigars, and e-hookahs, this report uses the more technical term "ENDS" because it is more commonly used in the tobacco-use literature.

complex sample design when estimating the variance of estimates and corresponding confidence intervals, were available only on restricted use files.

When computing estimates of youth cigarette-smoking or youth ENDS use, the data from the other national studies (NHANES 2017-2018, NSDUH 2018, NYTS 2019, NYRBS 2017, and MTF 2017) were restricted to ages 12 to 17 to match the PATH Study definition of youth. Appendices A and B describe the questions used to define cigarette use and ENDS use on these surveys as well as the PATH Study, and outlines differences in their target populations.

For both the PATH Study and the external data sources for cigarette smoking and ENDS use, item nonresponse was handled by excluding respondents with missing values for an item from the counts and estimates regarding that item. The proportions of item-missing values were generally very low (below 1 percent for adult estimates at the overall level and mostly below 3.1 percent for youth estimates at the overall level) in both the PATH Study and the studies that were used for comparison purposes. The exceptions for youth were a 5.5 percent missing data rate for ever tried cigarette smoking and 10.6 percent missing for past 30-day cigarette use from NHANES 2017-2018 data, as well as 20.9 percent missing for ever tried cigarette smoking and 12.8 percent missing for past 30-day ENDS use among high school students from NYRBS 2017.[13]

3.2.5 Analyses About Biospecimen Collections

Biospecimens provide a basis for the assessment of between-person differences and within-person changes over time in markers of tobacco exposure. They also allow for the detection of health conditions and disease processes potentially associated with the use of tobacco products. At Wave 5, all first-time adult interview respondents were asked to provide urine and blood specimens. In addition, members of the Wave 1 Biomarker Core were asked to provide a urine specimen. All first-time youth interview respondents were asked to provide a urine specimen, along with youth respondents who completed their first youth interview at Wave 4.5 (when no biospecimens were requested) or who consented to urine specimen collection at Wave 4. Field interviewers collected the urine specimens; on separate visits, phlebotomists collected the blood specimens.

[13] However, note that the NSDUH 2018 public use file included only "usable" cases that met specified minimum item response requirements.

Unweighted biospecimen collection response rates were calculated ignoring membership of the Wave 1 Cohort and/or Wave 4 Cohort. Ideally, weighted biospecimen collection response rates would be calculated separately for the subsamples of PATH Study participants belonging to special groups, such as the Wave 1 Biomarker Core. However, the Wave 5 biomarker weights for these groups were not available when the analyses for this report were conducted. Unweighted response rates were calculated, both overall and by subgroups, to assess the operational success of the urine and blood specimen collections among the Wave 5 respondents selected to provide a biospecimen. All Wave 5 biospecimen collection response rates presented in this report are conditioning on Wave 5 adult interview or youth interview completion.

3.3 Estimation Method and Software Package

Fay's balanced repeated replication (BRR) method with a factor of 0.3 was used for variance estimation to account for the impact of the stratification and clustering involved in the PATH Study's sample design and the weighting adjustments made to the IPS weights. SAS/STAT software version 15.1 was used to calculate all the point estimates. Confidence intervals were estimated using the modified Wilson approach (Wilson, 1927; SAS Institute, 2018). The preferred approach for testing whether two point estimates differ is to examine the confidence interval for the difference between the two point estimates; if the 95 percent confidence interval does not include zero, it can be concluded that the difference between the two estimates is statistically significant at the 0.05 significance level. This is discussed in Heeringa, West, and Berglund (2010, Section 5.6.1). Another approach is to examine whether the confidence intervals for the two point estimates overlap; if the confidence intervals for two proportions do not overlap, then the difference between the two proportions is considered statistically significant. However, Schenker and Gentleman (2001) show that using the second approach results in a conservative test. For the analyses presented in this report, the first approach was used for all comparisons that involve only PATH Study estimates and the second, more conservative approach was used to compare estimates between the PATH Study and external sources (in Sections 5.1.3, 5.1.4, 5.2.3, and 5.2.4). No adjustments were made for multiple comparisons because all the statistical tests were based on pre-planned (i.e., not post-hoc) comparisons and are presented in this report.

4. Results of Wave 1 Cohort Nonresponse Bias Analyses at Wave 5

This chapter presents results of nonresponse bias analyses for the Wave 1 Cohort interview data at Wave 5. The interview nonresponse bias analyses for the Wave 1 Cohort at Wave 5 are all conditioning on Wave 1 response. Sections 4.1.1 and 4.2.1 describe how the interview all-waves response rates vary across select subgroups. Sections 4.1.2 and 4.2.2 compare demographic and socio-economic characteristics as well as tobacco-use status between those eligible to be Wave 5 all-waves respondents and Wave 5 all-waves respondents. The results are presented separately for adults and youth.

4.1 Adult Interview

4.1.1 All-Waves Response Rates Conditioning on Wave 1 Response

Among those in the Wave 1 Cohort, unweighted and weighted all-waves response rates were calculated for the Wave 5 adult interview using the following formulas:

$$RR_{A\_AW} = C_{A\_AW} / \left(C_{A\_AW} + N_{A\_AW} + e_{A\_AW} \times U_{A\_AW}\right)$$

$$e_{A\_AW} = (C_{A\_AW} + N_{A\_AW}) / \left(C_{A\_AW} + N_{A\_AW} + I_{A\_AW}\right)$$

where

RR_{A_AW}	=	Wave 5 adult interview all-waves response rate;
C_{A_AW}	=	number of Wave 5 adult interview complete cases that were also a respondent in each of Waves 2 through 4;
N_{A_AW}	=	number of Wave 5 adults known to be eligible in each wave, but a nonrespondent at least once for Waves 2 through 5;
U_{A_AW}	=	number of Wave 5 adults with unknown eligibility status at least once for Waves 2 through 5 and never ineligible;
e_{A_AW}	=	estimated proportion of eligible cases among the Wave 5 adults with unknown eligibility status (from an all-waves perspective); and
I_{A_AW}	=	number of Wave 5 adults who were ineligible at least once for Waves 2 through 5.

Unweighted counts and weighted counts based on the Wave 1 IPS weight were obtained for the all-waves response status categories C_{A_AW}, N_{A_AW}, U_{A_AW}, and I_{A_AW} for the unweighted all-waves response rates and weighted all-waves response rates, respectively.

Table 4-1 provides all-waves response rates for the Wave 5 adult interview. In addition to the overall row, response rates are shown by Wave 1 sex, age, race/ethnicity, census region, and tobacco-use status. Persons with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic.

The unweighted and weighted all-waves response rates for the Wave 5 adult interview were 63.5 percent and 65.1 percent, respectively. For reasons discussed in Section 3.2.1, the unweighted and weighted single-wave response rates for the Wave 5 adult interview were higher than the all-waves response rates, at 69.0 percent and 69.4 percent, respectively.

For many subgroup categories (e.g., males) the weighted all-waves response rate was at least one percentage point higher than the unweighted all-waves response rate, with Non-Hispanic white alone adults having the greatest difference between weighted (64.7 percent) and unweighted (62.0 percent) all-waves response rates. Table 4-1 shows some differences in the all-waves response rates across the various subgroups (e.g., male versus female). The weighted all-waves response rate was lower among males (62.5 percent) than among females (67.3 percent); this pattern is consistent with most household surveys (Groves and Couper, 1998; Stoop, 2005). It is generally considered harder to reach younger adults, and although the Wave 5 weighted all-waves response rate for the Wave 1 "under 18" age group (65.0 percent) was comparable to those for older age groups (64.2 percent for "25-44," 68.8 percent for "45-64," and 64.2 percent for "65 and above"), the rate for the "18-24" age group was noticeably lower (59.2 percent). Non-Hispanic Black alone adults had a moderately higher weighted all-waves response rate (68.5 percent) than the other Wave 1 race/ethnicity groups (65.7 percent for Hispanic, 64.7 percent for Non-Hispanic White alone, and 63.0 percent for Non-Hispanic other race or multiple races). Across the census regions, those living in the Midwest at Wave 1 had the highest weighted all-waves response rate (66.9 percent) while those in the Northeast had the lowest (62.3 percent). Finally, the weighted all-waves response rates were 5.0 percentage points lower for Wave 1 adult current established tobacco users than for Wave 1 adults who were not current established tobacco users, and 6.0 percentage points lower for Wave 1 youth who had ever used tobacco than for Wave 1 youth who were never users.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts. In particular, Cunradi et al. (2005) and Young et al. (2006) have found that smokers were less likely to be retained in subsequent waves of surveys than were nonsmokers. The differences in response rates among the demographic and tobacco-use subgroups in Table 4-1 do not cause serious concern about potential nonresponse bias in the Wave 5 adult interview estimates for all-waves respondents from the Wave 1 Cohort. However, it is pertinent to examine the degree to which the Wave 5 weighting adjustments addressed differences in response propensity. The results of these analyses are discussed in the next section.

Table 4-1. Wave 5 adult interview all-waves response rates conditioning on Wave 1 response

Wave 1 characteristic[a]	C_{A_AW}: Completed Wave 5 adult interview and respondent in each of Waves 2 to 4 (n)	I_{A_AW}: Ineligible at least once for Waves 2 to 5 (n)	N_{A_AW}: Nonrespondent at least once for Waves 2 to 5 and eligible each wave (n)	U_{A_AW}: Unknown eligibility at least once for Waves 2 to 5 and never ineligible (n)	RR_{A_AW}: Unweighted all-waves response rate[b] (%)	RR_{A_AW}: Weighted all-waves response rate (%)
Overall	26,184	1,829	4,290	11,431	63.5	65.1
Sex						
Male	12,618	1,114	2,261	6,153	61.2	62.5
Female	13,359	711	2,022	5,260	65.8	67.3
Age group						
Under 18	7,259	195	1,343	2,617	65.0	65.0
18-24	5,042	310	845	2,913	58.3	59.2
25-44	6,744	367	981	3,175	62.7	64.2
45-64	5,528	494	788	2,007	67.6	68.8
65 and above	1,607	463	333	707	64.0	64.2
Race/ethnicity[c]						
Hispanic	5,442	338	746	2,257	65.3	65.7
Non-Hispanic White alone	14,502	1,001	2,673	6,591	62.0	64.7
Non-Hispanic Black alone	3,840	272	466	1,412	68.2	68.5
Non-Hispanic other race or multiple races	1,998	163	333	954	62.0	63.0
Census region						
Northeast	3,867	254	659	1,992	60.3	62.3
Midwest	6,350	438	1,032	2,342	66.2	66.9
South	9,749	726	1,601	4,471	62.7	64.5
West	6,218	411	998	2,626	64.1	65.9
Current established tobacco use[d] (Wave 1 adults only)						
Current established tobacco user	8,010	916	1,350	4,093	61.2	61.5
Not current established tobacco user	10,409	633	1,510	4,458	64.4	66.5
Ever tobacco use[e] (Wave 1 youth only)						
Ever user	1,604	79	340	726	60.7	60.8
Never user	5,435	110	958	1,762	66.9	66.8

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] The unweighted adult interview all-waves response rate equals $C_{A\_AW}/(C_{A\_AW}+N_{A\_AW}+U_{A\_AW}*((C_{A\_AW}+N_{A\_AW})/(C_{A\_AW}+N_{A\_AW}+I_{A\_AW})))$, where $(C_{A\_AW}+N_{A\_AW})/(C_{A\_AW}+N_{A\_AW}+I_{A\_AW})$ is the estimated proportion of eligible cases among those with unknown eligibility at least once for Waves 2 to 5 who were never ineligible. The weighted adult interview all-waves response rate was calculated using a similar approach except that the Wave 1 IPS weight was used to obtain weighted counts.

[c] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

[d] A 'current established user' of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 1 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, and e-cigarettes.

[e] An 'ever user' of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 1 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, e-cigarettes, bidis, and kreteks.

4.1.2 Comparison Between Wave 5 All-Waves Respondents and Those Eligible to be Wave 5 All-Waves Respondents (Among the Wave 1 Cohort)

This section compares weighted estimates between the Wave 5 adult interview all-waves respondents in the Wave 1 Cohort and the Wave 1 respondents who were eligible to be Wave 5 adult interview all-waves respondents. The weighted estimates cover demographic and socio-economic characteristics (shown in Table 4-2) as well as tobacco-use status (shown in Tables 4-3 and 4-4).

Section 3.2.3 describes the methods used for obtaining the results in Tables 4-2, 4-3, and 4-4. Each table shows two sets of comparisons between the Wave 5 adult interview all-waves respondents and those eligible to be Wave 5 adult interview all-waves respondents. The first set is for the "before Wave 5 weighting adjustment" comparison, and the second set is for the "after Wave 5 weighting adjustment" comparison.

Table 4-2 compares Wave 1 demographic and socio-economic characteristics between the Wave 5 adult interview all-waves respondents in the Wave 1 Cohort and those Wave 1 respondents eligible to be Wave 5 adult interview all-waves respondents. In Table 4-2, the percentages sum to 100 percent[14] over the categories associated with each characteristic. For example, for both the Wave 5 all-waves respondents and those eligible to be Wave 5 all-waves respondents, the estimated percentage of males and the estimated percentage of females add to 100 percent. Although the confidence intervals for some estimated differences in the first set of comparisons in Table 4-2 did not include zero,[15] most of the differences between the Wave 5 all-waves respondents and those eligible to be Wave 5 all-waves respondents were not large enough to be substantively meaningful. The most noticeable underrepresentation among the Wave 5 adult all-waves respondents was the male population, which tends to have a lower response propensity than the female population in most household surveys (Groves and Couper, 1998; Stoop, 2005). Adults ages 45 to 64 at Wave 1 were overrepresented among the Wave 5 all-waves respondents, prior to Wave 5 weighting adjustments.

[14] The sum may be not exactly 100 percent due to rounding.

[15] Testing whether or not a 100(1-α) percent confidence interval for a difference between two population proportions includes zero is equivalent to a two-sided test of the null hypothesis that the difference is zero at the α significance level (see, for example, Hanushek and Jackson, 1977).

Table 4-2 has two columns labeled "Unweighted count." These columns represent the numerator for calculating an unweighted percentage for the category associated with the characteristic. Tables 4-3 and 4-4 each have two columns labeled "Sample size." These columns represent the denominator for calculating an unweighted estimate of Wave 1 current established tobacco use (Table 4-3) or Wave 1 ever tobacco use (Table 4-4) for the category associated with the characteristic. These definitions generalize to all tables in this report that use a column heading of "Unweighted count" or "Sample size."

Table 4-2. Comparison of Wave 1 demographic and socio-economic characteristics between Wave 1 respondents who were eligible to be Wave 5 adult all-waves respondents and Wave 1 respondents who were Wave 5 adult all-waves respondents

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 5 adult all-waves respondents: Unweighted count	Wave 1 respondents who were eligible to be Wave 5 adult all-waves respondents: A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Wave 1 respondents who were Wave 5 adult all-waves respondents: Unweighted count	Before Wave 5 weighting adjustment: B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Before Wave 5 weighting adjustment: Difference in weighted percentages [B – A] [95% confidence interval]	After Wave 5 weighting adjustment: C: Weighted percentage, using Wave 5 all-waves weight [95% confidence interval]	After Wave 5 weighting adjustment: Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	21,032	48.1% [47.6%, 48.5%]	12,618	46.1% [45.5%, 46.7%]	-2.0% [-2.5%, -1.5%]	48.1% [47.5%, 48.7%]	0.0% [-0.1%, 0.1%]
Female	20,821	51.9% [51.5%, 52.4%]	13,539	53.9% [53.3%, 54.5%]	2.0% [1.5%, 2.5%]	51.9% [51.3%, 52.5%]	0.0% [-0.1%, 0.1%]
Age group							
Under 18	11,219	8.4% [8.1%, 8.6%]	7,259	8.5% [8.2%, 8.8%]	0.1% [-0.0%, 0.3%]	8.3% [8.0%, 8.6%]	-0.1% [-0.1%, -0.0%]
18-24	8,800	12.1% [11.8%, 12.5%]	5,042	11.1% [10.8%, 11.5%]	-1.0% [-1.3%, -0.7%]	12.2% [11.8%, 12.6%]	0.0% [-0.0%, 0.1%]
25-44	10,900	32.2% [31.7%, 32.6%]	6,744	31.8% [31.2%, 32.4%]	-0.4% [-0.9%, 0.1%]	32.3% [31.7%, 32.9%]	0.1% [-0.0%, 0.2%]
45-64	8,323	32.0% [31.6%, 32.5%]	5,528	34.0% [33.4%, 34.7%]	2.0% [1.4%, 2.6%]	32.1% [31.6%, 32.7%]	0.1% [0.0%, 0.2%]
65 and above	2,647	15.3% [15.0%, 15.6%]	1,607	14.5% [14.0%, 15.1%]	-0.7% [-1.2%, -0.3%]	15.1% [14.7%, 15.5%]	-0.2% [-0.3%, -0.0%]
Race/ethnicity[a]							
Hispanic	8,445	15.9% [15.5%, 16.2%]	5,442	16.0% [15.4%, 16.5%]	0.1% [-0.5%, 0.6%]	15.9% [15.5%, 16.4%]	0.0% [-0.1%, 0.1%]
Non-Hispanic White alone	23,766	65.1% [64.6%, 65.5%]	14,502	64.7% [64.2%, 65.3%]	-0.3% [-0.9%, 0.2%]	65.0% [64.4%, 65.5%]	-0.1% [-0.2%, 0.1%]
Non-Hispanic Black alone	5,718	11.4% [11.1%, 11.7%]	3,840	12.0% [11.6%, 12.4%]	0.6% [0.2%, 0.9%]	11.4% [11.0%, 11.8%]	0.0% [-0.1%, 0.1%]
Non-Hispanic other race or multiple races	3,285	7.6% [7.4%, 7.9%]	1,998	7.3% [7.0%, 7.7%]	-0.3% [-0.6%, -0.0%]	7.7% [7.4%, 8.0%]	0.1% [-0.1%, 0.2%]

Table 4-2. Comparison of Wave 1 demographic and socio-economic characteristics between those who were eligible to be Wave 5 adult all-waves respondents and those who were Wave 5 adult all-waves respondents (continued)

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 5 adult all-waves respondents		Wave 1 respondents who were Wave 5 adult all-waves respondents				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 5 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	6,518	18.1% [17.7%, 18.4%]	3,867	17.4% [16.9%, 17.8%]	-0.7% [-1.1%, -0.3%]	18.1% [17.6%, 18.5%]	0.0% [-0.1%, 0.1%]
Midwest	9,724	21.4% [21.1%, 21.8%]	6,350	22.2% [21.6%, 22.8%]	0.7% [0.1%, 1.3%]	21.4% [20.9%, 21.9%]	0.0% [-0.1%, 0.1%]
South	15,821	37.1% [36.6%, 37.5%]	9,749	36.7% [35.9%, 37.5%]	-0.4% [-1.1%, 0.4%]	37.1% [36.5%, 37.7%]	0.0% [-0.1%, 0.1%]
West	9,842	23.4% [23.0%, 23.8%]	6,218	23.8% [23.1%, 24.4%]	0.4% [-0.2%, 0.9%]	23.4% [22.9%, 23.9%]	0.0% [-0.1%, 0.1%]
Education							
Less than high school or GED	5,892	16.0% [15.6%, 16.5%]	3,636	15.5% [14.9%, 16.0%]	-0.6% [-1.1%, -0.1%]	16.0% [15.5%, 16.5%]	0.0% [-0.2%, 0.1%]
High school	7,182	24.2% [23.7%, 24.7%]	4,244	23.0% [22.3%, 23.6%]	-1.2% [-1.8%, -0.6%]	24.2% [23.6%, 24.8%]	0.0% [-0.2%, 0.1%]
Some college, no degree	10,863	31.4% [30.9%, 31.9%]	6,672	31.3% [30.7%, 32.0%]	-0.1% [-0.6%, 0.5%]	31.3% [30.7%, 32.0%]	-0.1% [-0.2%, 0.1%]
Bachelor degree and above	6,555	28.4% [27.9%, 28.9%]	4,290	30.3% [29.6%, 30.9%]	1.8% [1.3%, 2.4%]	28.5% [27.9%, 29.2%]	0.1% [-0.0%, 0.2%]

Table 4-2. Comparison of Wave 1 demographic and socio-economic characteristics between those who were eligible to be Wave 5 adult all-waves respondents and those who were Wave 5 adult all-waves respondents (continued)

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 5 adult all-waves respondents: Unweighted count	Wave 1 respondents who were eligible to be Wave 5 adult all-waves respondents: A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Wave 1 respondents who were Wave 5 adult all-waves respondents: Unweighted count	Before Wave 5 weighting adjustment: B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Before Wave 5 weighting adjustment: Difference in weighted percentages [B – A] [95% confidence interval]	After Wave 5 weighting adjustment: C: Weighted percentage, using Wave 5 all-waves weight [95% confidence interval]	After Wave 5 weighting adjustment: Difference in weighted percentages [C – A] [95% confidence interval]
Health insurance (Wave 1 adults only)							
Yes	24,476	85.4% [84.8%, 85.9%]	15,311	86.4% [85.7%, 87.1%]	1.0% [0.6%, 1.4%]	85.5% [84.7%, 86.1%]	0.1% [-0.3%, 0.5%]
No	5,830	14.6% [14.1%, 15.2%]	3,440	13.6% [12.9%, 14.3%]	-1.0% [-1.4%, -0.6%]	14.5% [13.9%, 15.3%]	-0.1% [-0.5%, 0.3%]

[a] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

Table 4-3. Comparison of Wave 1 “current established tobacco-use” estimates between Wave 1 adults eligible to be Wave 5 all-waves respondents and Wave 1 adults who were Wave 5 all-waves respondents*

	Wave 1 adults who were eligible to be Wave 5 all-waves respondents		Wave 1 adults who were Wave 5 all-waves respondents				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
Wave 1 characteristic[a]	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 5 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	29,930	23.5% [22.9%, 24.1%]	18,419	21.9% [21.1%, 22.6%]	-1.7% [-2.1%, -1.3%]	23.4% [22.8%, 24.1%]	-0.1% [-0.2%, -0.0%]
Sex							
Male	14,916	29.2% [28.4%, 30.1%]	8,722	26.8% [25.7%, 28.0%]	-2.4% [-3.1%, -1.8%]	29.0% [28.1%, 30.0%]	-0.2% [-0.4%, -0.0%]
Female	14,890	18.3% [17.6%, 19.0%]	9,682	17.7% [16.9%, 18.5%]	-0.6% [-1.0%, -0.2%]	18.3% [17.5%, 19.1%]	0.0% [-0.1%, 0.1%]
Age group							
18-24	8,662	28.8% [27.4%, 30.2%]	4,966	26.3% [24.6%, 28.1%]	-2.5% [-3.4%, -1.6%]	28.8% [27.4%, 30.3%]	0.0% [-0.2%, 0.2%]
25-44	10,684	28.7% [27.7%, 29.6%]	6,619	26.5% [25.3%, 27.8%]	-2.1% [-2.8%, -1.4%]	28.4% [27.3%, 29.6%]	-0.2% [-0.7%, 0.3%]
45-64	8,012	22.5% [21.6%, 23.4%]	5,326	21.2% [20.1%, 22.3%]	-1.3% [-1.9%, -0.7%]	22.4% [21.3%, 23.5%]	-0.1% [-0.6%, 0.4%]
65 and above	2,461	9.9% [8.8%, 11.2%]	1,504	9.2% [7.8%, 10.7%]	-0.8% [-1.6%, 0.0%]	9.7% [8.3%, 11.3%]	-0.2% [-1.0%, 0.6%]
Race/ethnicity[b]							
Hispanic	5,082	17.2% [16.2%, 18.3%]	3,198	16.6% [15.4%, 18.0%]	-0.6% [-1.3%, 0.1%]	17.2% [16.0%, 18.6%]	0.0% [-0.2%, 0.2%]
Non-Hispanic White alone	17,952	25.0% [24.1%, 25.9%]	10,921	22.8% [21.8%, 23.9%]	-2.2% [-2.7%, -1.8%]	24.9% [24.0%, 25.8%]	-0.1% [-0.3%, 0.0%]
Non-Hispanic Black alone	4,132	26.2% [24.9%, 27.6%]	2,739	26.7% [24.8%, 28.7%]	0.5% [-0.6%, 1.6%]	26.4% [24.8%, 28.1%]	0.2% [-0.1%, 0.5%]
Non-Hispanic other race or multiple races	2,225	19.4% [17.7%, 21.1%]	1,312	16.3% [14.4%, 18.4%]	-3.1% [-4.5%, -1.7%]	18.6% [16.6%, 20.8%]	-0.8% [-1.3%, -0.3%]
Census region							
Northeast	4,680	21.9% [20.5%, 23.4%]	2,750	20.2% [18.2%, 22.2%]	-1.7% [-2.8%, -0.7%]	21.8% [20.3%, 23.4%]	-0.1% [-0.3%, 0.2%]
Midwest	7,119	26.3% [24.6%, 28.1%]	4,614	25.1% [23.2%, 27.1%]	-1.2% [-2.0%, -0.5%]	26.2% [24.4%, 28.0%]	-0.2% [-0.4%, 0.1%]
South	11,247	25.4% [24.0%, 26.8%]	6,844	23.6% [22.0%, 25.3%]	-1.8% [-2.6%, -1.0%]	25.2% [23.8%, 26.7%]	-0.1% [-0.3%, 0.1%]
West	6,784	19.4% [17.8%, 21.0%]	4,211	17.5% [15.9%, 19.2%]	-1.9% [-2.5%, -1.2%]	19.3% [17.7%, 20.9%]	-0.1% [-0.3%, 0.1%]

* A ‘current established user’ of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 1 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, and e-cigarettes.

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

Table 4-4. Comparison of Wave 1 "ever tobacco-use" estimates between Wave 1 youth eligible to be Wave 5 all-waves respondents and Wave 1 youth who were Wave 5 all-waves respondents (among Wave 5 adults)*

Wave 1 characteristic[a]	Wave 1 youth who were eligible to be Wave 5 adult all-waves respondents		Wave 1 youth who were Wave 5 adult all-waves respondents				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 5 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	10,825	24.6% [23.6%, 25.7%]	7,039	22.7% [21.5%, 24.0%]	-1.9% [-2.6%, -1.2%]	24.4% [23.3%, 25.6%]	-0.2% [-0.6%, 0.2%]
Sex							
Male	5,466	26.0% [24.8%, 27.3%]	3,545	24.1% [22.6%, 25.7%]	-1.9% [-3.0%, -0.9%]	25.6% [24.2%, 27.1%]	-0.4% [-1.0%, 0.2%]
Female	5,343	23.2% [22.0%, 24.6%]	3,486	21.4% [19.7%, 23.1%]	-1.9% [-2.8%, -0.9%]	23.3% [21.7%, 24.9%]	0.0% [-0.7%, 0.7%]
Race/ethnicity[b]							
Hispanic	3,053	24.5% [22.6%, 26.5%]	2,066	22.7% [20.6%, 25.0%]	-1.8% [-3.0%, -0.6%]	24.8% [22.8%, 26.9%]	0.3% [-0.7%, 1.3%]
Non-Hispanic White alone	5,242	26.3% [24.8%, 28.0%]	3,260	23.9% [22.1%, 25.8%]	-2.5% [-3.5%, -1.4%]	25.5% [23.9%, 27.3%]	-0.8% [-1.4%, -0.1%]
Non-Hispanic Black alone	1,411	21.0% [18.9%, 23.3%]	982	21.5% [18.9%, 24.4%]	0.5% [-1.1%, 2.0%]	22.8% [20.1%, 25.8%]	1.8% [0.3%, 3.2%]
Non-Hispanic other race or multiple races	961	21.2% [18.8%, 23.9%]	632	18.8% [15.9%, 22.0%]	-2.4% [-4.7%, -0.2%]	20.1% [17.2%, 23.4%]	-1.1% [-3.4%, 1.1%]
Census region							
Northeast	1,641	23.9% [21.3%, 26.6%]	1,007	21.8% [19.1%, 24.9%]	-2.0% [-3.8%, -0.3%]	23.3% [20.3%, 26.6%]	-0.5% [-2.1%, 1.1%]
Midwest	2,370	25.1% [23.0%, 27.2%]	1,598	24.3% [21.8%, 26.9%]	-0.8% [-2.2%, 0.6%]	25.8% [23.4%, 28.3%]	0.7% [-0.5%, 1.9%]
South	4,084	25.2% [23.6%, 26.9%]	2,618	22.5% [20.5%, 24.7%]	-2.7% [-3.8%, -1.5%]	24.5% [22.4%, 26.6%]	-0.8% [-1.8%, 0.3%]
West	2,730	23.9% [21.4%, 26.5%]	1,816	22.1% [19.6%, 24.9%]	-1.8% [-3.3%, -0.2%]	24.0% [21.3%, 26.8%]	0.1% [-1.4%, 1.6%]

* An 'ever user' of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 1 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, e-cigarettes, bidis, and kreteks.

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

For the “before Wave 5 weighting adjustments” in Table 4-2, the most noticeable underrepresentation among the Wave 5 adult all-waves respondents was the male population, which tends to have lower response propensity than the female population in most household surveys (Groves and Couper, 1998; Stoop, 2005). The most evident overrepresentations were of the Wave 1 “45-64” age group (with corresponding small underrepresentation in the “18-24,” “25-44,” and “65 and above” age groups) and the Wave 1 “Bachelor degree and above” education group (with consequent slight underrepresentation in the “Less than high school or GED” and “High school” education groups). Non-Hispanic Black alone adults, adults from the Midwest census region, and adults with health insurance at Wave 1 were slightly overrepresented among the Wave 5 all-waves respondents, while non-Hispanic adults of other or multiple races and adults from the Northeast at Wave 1 were slightly underrepresented.

For the “after Wave 5 weighting adjustment” differences, the estimated confidence intervals (in the last column of Table 4-2) either included zero or were in the proximity of zero for each of the Wave 1 characteristics examined: sex, age, race/ethnicity, census region, education, and health insurance status.

Tables 4-3 and 4-4 show the comparisons for the Wave 1 tobacco-use measures between those eligible to be Wave 5 adult interview all-waves respondents and Wave 5 adult interview all-waves respondents, among the Wave 1 Cohort. The Wave 1 respondents who were eligible to be Wave 5 adult interview all-waves respondents were divided into two groups based on their Wave 1 interview type (i.e., adult versus youth) and analyzed separately. Table 4-3 covers “current established tobacco use” for the Wave 1 adult respondents and Table 4-4 covers “ever tobacco use” for the Wave 1 youth respondents. Besides the overall estimates, these tables also show the estimated Wave 1 “current established tobacco use” and “ever tobacco-use” rates for some demographic subgroups.

For the “before Wave 5 weighting adjustment” estimates, most of the confidence intervals did not include zero, with the differences generally indicating an underrepresentation of Wave 1 tobacco users among the Wave 5 adult interview all-waves respondents. For the “after Wave 5 weighting adjustment” estimates, most of the confidence intervals either included or were in the proximity of zero. However, using the Wave 5 all-waves weight, the Wave 5 adult interview all-waves respondents tended to underrepresent Wave 1 current established tobacco users among non-Hispanic adults of other or multiple races (see Table 4-3), as well as Wave 1 ever users of tobacco among non-Hispanic

White alone adults who were youth at Wave 1 (see Table 4-4). There was also a slight overrepresentation of Wave 1 ever users of tobacco among non-Hispanic Black alone adults who were youth at Wave 1, after Wave 5 weighting adjustments (see Table 4-4).

Overall, these results reflect positively on the use of both demographic variables and tobacco-use measures from Wave 1 for calibrating the Wave 5 all-waves weight for the Wave 1 Cohort. However, weighting adjustments are implemented within weighting cells and the groupings of participants used for these cells cannot account for each possible interaction of characteristics nor align with every possible analytic subgroup. For example, the Wave 5 all-waves weight for the Wave 1 Cohort is calibrated using age at Wave 1, yet many Wave 5 analyses (including those in this report) will be based on Wave 5 age. Further discussion of this issue in relation to the aging of study participants appears in Section 4.2.2.

Assuming that the Wave 1 demographic, socio-economic, and tobacco-use characteristics in Tables 4-2, 4-3, and 4-4 are correlated with key tobacco- and health-related outcome measures in Wave 5 of the PATH Study, these results suggest little bias in the adult interview estimates for the Wave 1 Cohort due to nonresponse occurring at any time since Wave 1. However, they do hint at the challenges faced by panel studies as waves of nonresponse accrue.

4.2 Youth Interview

4.2.1 All-Waves Response Rates Conditioning on Wave 1 Response

Among those in the Wave 1 Cohort, unweighted and weighted all-waves response rates were calculated for the Wave 5 youth interview using the following formulas:

$$RR_{Y\_AW} = C_{Y\_AW} / \left(C_{Y\_AW} + N_{Y\_AW} + e_{Y\_AW} \times U_{Y\_AW} \right)$$

$$e_{Y\_AW} = (C_{Y\_AW} + N_{Y\_AW}) / \left(C_{Y\_AW} + N_{Y\_AW} + I_{Y\_AW} \right)$$

where

RR_{Y_AW} = Wave 5 youth interview all-waves response rate;

C_{Y_AW} = number of Wave 5 youth interview complete cases that were also a respondent in each of Waves 2 through 4;

N_{Y_AW} = number of Wave 5 youth known to be eligible in each wave, but a nonrespondent at least once for Waves 2 through 5;

U_{Y_AW} = number of Wave 5 youth with unknown eligibility status at least once for Waves 2 through 5 and never ineligible;

e_{Y_AW} = estimated proportion of eligible cases among the Wave 5 youth with unknown eligibility status (from an all-waves perspective); and

I_{Y_AW} = number of Wave 5 youth who were ineligible at least once for Waves 2 through 5.

Unweighted counts and weighted counts based on the Wave 1 IPS weight were obtained for the response status categories C_{Y_AW}, N_{Y_AW}, U_{Y_AW}, and I_{Y_AW} for the unweighted all-waves response rates and weighted all-waves response rates, respectively.

Table 4-5 provides all-waves response rates for the Wave 5 youth interview. In addition to the overall row, response rates are shown by Wave 1 sex, age, race/ethnicity, census region, and tobacco-use status. Persons with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic.

Table 4-5. Wave 5 youth interview all-waves response rates conditioning on Wave 1 response

Wave 1 Characteristic[a]	C_{Y_AW}: Completed Wave 5 youth interview and respondent in each of Waves 2 to 4 (n)	I_{Y_AW}: Ineligible at least once for Waves 2 to 5 (n)	N_{Y_AW}: Nonrespondent at least once for Waves 2 to 5 and eligible each wave (n)	U_{Y_AW}: Unknown eligibility at least once for Waves 2 to 5 and never ineligible (n)	RR_{Y_AW}: Unweighted all-waves response rate[b] (%)	RR_{Y_AW}: Weighted all-waves response rate (%)
Overall	6,386	70	778	2,210	68.3	68.4
Sex						
Male	3,267	45	415	1,151	67.8	67.9
Female	3,107	24	363	1,052	68.8	68.9
Age group						
Under 12	4,750	53	619	1,781	66.6	66.7
12-13	1,634	17	158	426	73.8	73.5
Race/ethnicity[c]						
Hispanic	1,821	28	234	612	68.5	68.7
Non-Hispanic White alone	2,993	17	374	1,113	66.9	67.0
Non-Hispanic Black alone	918	10	98	268	71.6	71.4
Non-Hispanic other race or multiple races	592	13	68	198	69.3	69.5
Census region						
Northeast	884	12	109	364	65.4	65.4
Midwest	1,421	17	176	481	68.6	68.6
South	2,396	15	311	810	68.2	68.2
West	1,685	26	182	555	69.8	70.0
Ever tobacco use[d] (Wave 1 youth only)						
Ever user	82	3	15	19	71.0	71.9
Never user	1,459	13	133	376	74.3	74.0

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values or unusual circumstances. For example, the sum of the counts across the age categories is not equal to the count in the overall row because a small number of Wave 5 youth were ages 14 to 17 at Wave 1.

[b] The unweighted youth interview all-waves response rate equals $C_{Y\_AW}/(C_{Y\_AW}+N_{Y\_AW}+U_{Y\_AW}*((C_{Y\_AW}+N_{Y\_AW})/(C_{Y\_AW}+N_{Y\_AW}+I_{Y\_AW})))$, where $(C_{Y\_AW}+N_{Y\_AW})/(C_{Y\_AW}+N_{Y\_AW}+I_{Y\_AW})$ is the estimated proportion of eligible cases among those with unknown eligibility at least once for Waves 2 to 5 who were never ineligible. The weighted youth interview all-waves response rate was calculated using a similar approach except that the Wave 1 IPS weight was used to obtain weighted counts.

[c] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

[d] An 'ever user' of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 1 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, e-cigarettes, bidis, and kreteks.

The unweighted and weighted all-waves response rates for the Wave 5 youth interview were 68.3 percent and 68.4 percent, respectively; the unweighted and weighted all-waves response rates for each subgroup were also very similar to each other. For reasons discussed in Section 3.2.1, the unweighted and weighted single-wave response rates for the Wave 5 youth interview were higher than the all-waves response rates, at 72.1 percent and 72.3 percent, respectively.

The Wave 5 weighted all-waves response rates varied slightly by sex, with females being one percentage point higher than males, but to a greater extent by Wave 1 age and race/ethnicity. For Wave 1 age, the weighted all-waves response rate for the "under 12" age group (66.7 percent) was noticeably lower than for the "12-13" age group (73.5 percent).[16] This is likely because the persons in the "under 12" age group were shadow youth at Wave 1, recruited to be interviewed in future waves after turning 12. As mentioned in Section 2.2, the Wave 1 shadow youth became a part of the Wave 1 Cohort as a result of their parent/guardian agreeing at Wave 1 to their child's participation in the study. Thus, unlike other cohort members, this required no willingness from the shadow youth themselves to complete an interview at Wave 1. (In Waves 2 through 4.5, the weighted single-wave response rate was also lower among youth who were shadow youth at Wave 1.) Weighted all-waves response rates also differed somewhat across Wave 1 race/ethnicity groups, with the highest among non-Hispanic Black alone youth (71.4 percent) and lowest among non-Hispanic White alone youth (67.0 percent). Similar to the response rates observed by Wave 1 census region for adults, those in the Northeast had the lowest weighted all-waves response rate (65.4 percent). However, for the youth interview, those in the West region had the highest weighted all-waves response rate (70.0 percent). For tobacco use, the weighted all-waves response rate among Wave 1 youth was 2.1 percentage points lower for the Wave 1 "ever user" group than for the Wave 1 "never user" group.

As mentioned in Section 4.1.1, variation in response rates by subgroups is to be expected in large-scale data collection efforts. None of the differences among the demographic and tobacco-use subgroups in Table 4-5 causes serious concern about potential nonresponse bias in the Wave 5 youth interview estimates for all-waves respondents from the Wave 1 Cohort. However, it is

[16] In theory, all participants who were classified in the "14-17" age group at Wave 1 should have reached adulthood by Wave 5. There were five Wave 1 Cohort participants in the "14-17" age group at Wave 1 who were still youth at Wave 5 as the result of age corrections since Wave 1. Due to the small sample size, there is no row for the "14-17" age group row in Tables 4-5, 4-6, and 4-7; however, these five cases were included in the other subgroup and overall calculations.

pertinent to examine the degree to which the Wave 5 weighting adjustments addressed differences in response propensity. The results of these analyses are discussed in the next section.

4.2.2 Comparison Between Those Eligible to be Wave 5 All-Waves Respondents and Wave 5 All-Waves Respondents (Among the Wave 1 Cohort)

This section compares weighted estimates between the Wave 1 respondents who were eligible to be Wave 5 youth interview all-waves respondents and the Wave 5 youth interview all-waves respondents in the Wave 1 Cohort. The weighted estimates cover demographic characteristics (shown in Table 4-6) and tobacco-use status (shown in Table 4-7). The structure of these tables is similar to the tables in Section 4.1.2 for the adult interview.

Table 4-6. Comparison of Wave 1 demographic characteristics between Wave 1 respondents who were eligible to be Wave 5 youth all-waves respondents and Wave 1 respondents who were Wave 5 youth all-waves respondents

	Wave 1 respondents who were eligible to be Wave 5 youth all-waves respondents		Wave 1 respondents who were Wave 5 youth all-waves respondents				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
Wave 1 characteristic	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 5 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	4,833	51.2% [50.2%, 52.2%]	3,267	50.7% [49.5%, 52.0%]	-0.5% [-1.2%, 0.3%]	51.1% [49.9%, 52.3%]	-0.1% [-0.2%, 0.1%]
Female	4,522	48.8% [47.8%, 49.8%]	3,107	49.3% [48.0%, 50.5%]	0.5% [-0.3%, 1.2%]	48.9% [47.7%, 50.1%]	0.1% [-0.1%, 0.2%]
Age group							
Under 12	7,150	75.2% [74.3%, 76.1%]	4,750	73.4% [72.3%, 74.4%]	-1.8% [-2.5%, -1.2%]	74.2% [73.1%, 75.2%]	-1.1% [-1.3%, -0.8%]
12-13	2,218	24.7% [23.9%, 25.6%]	1,634	26.6% [25.5%, 27.7%]	1.8% [1.2%, 2.5%]	25.8% [24.7%, 26.9%]	1.1% [0.8%, 1.3%]
Race/ethnicity[a]							
Hispanic	2,667	23.6% [22.7%, 24.5%]	1,821	23.5% [22.5%, 24.5%]	-0.1% [-0.9%, 0.7%]	23.6% [22.6%, 24.7%]	0.0% [-0.2%, 0.2%]
Non-Hispanic White alone	4,480	52.8% [51.8%, 53.8%]	2,993	52.0% [50.8%, 53.2%]	-0.8% [-1.7%, 0.0 %]	52.7% [51.5%, 53.9%]	-0.1% [-0.3%, 0.1%]
Non-Hispanic Black alone	1,284	13.6% [12.9%, 14.3%]	918	14.2% [13.3%, 15.1%]	0.6% [-0.0%, 1.2%]	13.6% [12.8%, 14.5%]	0.0% [-0.1%, 0.1%]
Non-Hispanic other race or multiple races	858	10.0% [9.4%, 10.6%]	592	10.3% [9.6%, 11.1%]	0.4% [-0.1%, 0.8%]	10.1% [9.4%, 10.9%]	0.1% [-0.0%, 0.2%]

Table 4-6. Comparison of Wave 1 demographic characteristics between Wave 1 respondents who were eligible to be Wave 5 youth all-waves respondents and Wave 1 respondents who were Wave 5 youth all-waves respondents (continued)

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 5 youth all-waves respondents		Wave 1 respondents who were Wave 5 youth all-waves respondents				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 5 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	1,357	16.4% [15.7%, 17.1%]	884	15.8% [15.0%, 16.8%]	-0.5% [-1.3%, 0.2%]	16.5% [15.6%, 17.4%]	0.1% [-0.1%, 0.3%]
Midwest	2,078	21.5% [20.7%, 22.4%]	1,421	21.7% [20.7%, 22.8%]	0.2% [-0.5%, 0.9%]	21.5% [20.5%, 22.5%]	0.0% [-0.2%, 0.1%]
South	3,517	38.1% [37.1%, 39.1%]	2,396	37.8% [36.6%, 38.9%]	-0.3% [-1.4%, 0.7%]	38.0% [36.9%, 39.2%]	0.0% [-0.3%, 0.2%]
West	2,422	24.0% [23.1%, 24.9%]	1,685	24.7% [23.6%, 25.8%]	0.7% [-0.0%, 1.4%]	24.0% [23.0%, 25.1%]	0.0% [-0.1%, 0.2%]

[a] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

Table 4-7. Comparison of Wave 1 "ever tobacco-use" estimates between Wave 1 youth who were eligible to be Wave 5 youth all-waves respondents and Wave 1 youth who were Wave 5 youth all-waves respondents*

	Wave 1 youth who were eligible to be Wave 5 youth all-waves respondents		Wave 1 youth who were Wave 5 youth all-waves respondents				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
Wave 1 characteristic[a]	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 5 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	2,084	5.3% [4.4%, 6.4%]	1,541	5.1% [4.1%, 6.3%]	-0.2% [-0.8%, 0.3%]	5.4% [4.3%, 6.7%]	0.1% [-0.5%, 0.6%]
Sex							
Male	1,042	6.4% [5.0%, 8.1%]	757	6.7% [5.1%, 8.8%]	0.3% [-0.6%, 1.3%]	7.1% [5.4%, 9.3%]	0.7% [-0.2%, 1.7%]
Female	1,027	4.3% [3.2%, 5.7%]	773	3.4% [2.4%, 5.0%]	-0.8% [-1.6%, -0.1%]	3.6% [2.5%, 5.3%]	-0.6% [-1.4%, 0.2%]
Age group[b]							
12-13	2,079	5.4% [4.5%, 6.4%]	1,539	5.1% [4.1%, 6.3%]	-0.2% [-0.8%, 0.3%]	5.4% [4.3%, 6.7%]	0.1% [-0.5%, 0.6%]
Race/ethnicity[c]							
Hispanic	596	4.2% [2.8%, 6.3%]	440	4.7% [3.1%, 7.1%]	0.5% [-0.2%, 1.3%]	5.0% [3.2%, 7.8%]	0.8% [-0.1%, 1.8%]
Non-Hispanic White alone	937	6.0% [4.7%, 7.8%]	658	5.5% [4.0%, 7.5%]	-0.6% [-1.5%, 0.3%]	5.8% [4.3%, 7.9%]	-0.2% [-1.1%, 0.7%]
Non-Hispanic Black alone	288	3.9% † [1.8%, 8.5%]	231	4.2% † [1.7%, 10.1%]	0.3% [-1.0%, 1.6%]	4.2% † [1.6%, 10.3%]	0.2% [-1.2%, 1.7%]
Non-Hispanic other race or multiple races	197	6.2% [3.6%, 10.5%]	160	6.4% [3.5%, 11.3%]	0.2% [-1.3%, 1.7%]	7.0% [4.0%, 12.0%]	0.8% [-0.9%, 2.5%]

Table 4-7. Comparison of Wave 1 “ever tobacco-use” estimates between Wave 1 youth who were eligible to be Wave 5 youth all-waves respondents and Wave 1 youth who were Wave 5 youth all-waves respondents* (continued)

	Wave 1 youth who were eligible to be Wave 5 youth all-waves respondents		Wave 1 youth who were Wave 5 youth all-waves respondents				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
Wave 1 characteristic[a]	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 5 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	302	4.9% [2.9%, 8.0%]	207	4.9% † [2.5%, 9.2%]	0.0% [-1.7%, 1.7%]	4.9% † [2.5%, 9.4%]	0.1% [-1.7%, 1.9%]
Midwest	443	4.9% [3.3%, 7.3%]	325	5.0% [3.1%, 7.9%]	0.1% [-1.1%, 1.3%]	5.4% [3.4%, 8.4%]	0.5% [-0.7%, 1.7%]
South	795	6.5% [5.0%, 8.4%]	599	6.0% [4.3%, 8.3%]	-0.5% [-1.4%, 0.5%]	6.4% [4.6%, 8.9%]	-0.1% [-1.0%, 0.9%]
West	544	4.2% [2.9%, 6.3%]	410	3.9% [2.4%, 6.3%]	-0.3% [-0.9%, 0.3%]	4.1% [2.6%, 6.5%]	-0.1% [-0.8%, 0.6%]

*An ‘ever user’ of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 1 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, e-cigarettes, bidis, and kreteks.

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Table 4-7 includes a subset of the youth in Table 4-6 because there are no Wave 1 interview data for the Wave 5 youth who were shadow youth at Wave 1.

[c] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

† Estimate should be interpreted with caution because it has low precision. It is based on a denominator sample size of less than 50, or the coefficient of variation of the estimate or its complement is larger than 30 percent.

Table 4-6 compares Wave 1 demographic characteristics between the Wave 5 youth interview all-waves respondents in the Wave 1 Cohort and the Wave 1 respondents eligible to be Wave 5 youth interview all-waves respondents. In Table 4-6, as in Table 4-2, the percentages sum to 100 percent[17] over the categories associated with each characteristic. For the "before Wave 5 weighting adjustment" distributions, the only slight underrepresentation among the Wave 5 youth all-waves respondents was the Wave 1 "under 12" age group (with corresponding slight overrepresentation among the "12-13" age group).

For the "after Wave 5 weighting adjustment" differences, the estimated confidence intervals (in the last column of Table 4-6) also included zero for sex, race/ethnicity, and census region. For Wave 1 age, the underrepresentation of the "under 12" age group (and consequent overrepresentation of the "12-13" age group) improved after Wave 5 weighting adjustments but persisted to some degree. Note that only the Wave 1 12-year-olds among the Wave 1 "12-13" age group would be expected to be youth at Wave 5, but some Wave 1 13-year-olds were still a youth. Moreover, some Wave 1 12-year-olds were adults at Wave 5 and so were included in the adult tables in Section 4.1. Separate analyses, not shown here, confirmed that the Wave 1 age group distribution was preserved when the Wave 5 all-waves weight was applied to all Wave 5 all-waves respondents (i.e., youth and adults) together. That is, the results in Table 4-6 need to be interpreted in light of the fact that not all study participants were exactly 5 years older at the time of Wave 5, compared to their Wave 1 age.

Table 4-7 shows the comparison of Wave 1 "ever tobacco-use" rates between the Wave 5 youth interview all-waves respondents and the Wave 1 youth participants eligible to be Wave 5 youth interview all-waves respondents. (Table 4-7 includes a subset of the youth in Table 4-6 because there are no Wave 1 interview data for the Wave 5 youth who were shadow youth at Wave 1.) Prior to Wave 5 weighting adjustments, Table 4-7 shows a slight underrepresentation of Wave 1 ever users of tobacco among females who were youth at Wave 1. However, for both the "before Wave 5 weighting adjustment" and the "after Wave 5 weighting adjustment" estimates, differences between the all-waves respondents and those eligible to be all-waves respondents were no more than 0.8 percent. The largest differences after Wave 5 weighting adjustments were for the Wave 1 "ever

[17] The sum may be not exactly 100 percent due to rounding.

tobacco-use" measure among Hispanic youth and non-Hispanic youth of other or multiple races; however, all of the confidence intervals included zero.

Assuming that the Wave 1 demographic and tobacco-use characteristics in Tables 4-6 and 4-7 are correlated with key tobacco- and health-related outcome measures in Wave 5 of the PATH Study, these results indicate little if any bias in the youth interview estimates for the Wave 1 Cohort due to nonresponse at any time since Wave 1.

5. Results of Wave 4 Cohort Nonresponse Bias Analyses at Wave 5

This chapter presents results of nonresponse bias analyses for the Wave 4 Cohort interview data at Wave 5. As discussed in Section 3.2.1, the Wave 4 Cohort is comprised of two groups of study members recruited approximately 3 years apart. Depending on their eligibility for the Wave 4.5 special data collection, Wave 5 was the fourth or fifth follow-up attempt for those sampled at Wave 1, whereas members of the Wave 4 replenishment sample were asked to participate in the PATH Study for the second or third time. The interview nonresponse bias analyses for the Wave 4 Cohort at Wave 5 are all conditioning on Wave 4 response. Sections 5.1.1 and 5.2.1 describe how the interview single-wave response rates vary across select subgroups, including recruitment wave. Sections 5.1.2 and 5.2.2 compare demographic and socio-economic characteristics as well as tobacco-use status between those eligible for the Wave 5 interview and the Wave 5 respondents. Four sections compare cross-sectional estimates of cigarette smoking (Sections 5.1.3 and 5.2.3) and ENDS use (Sections 5.1.4 and 5.2.4) between the PATH Study Wave 4 Cohort at Wave 5 and external sources. All results are presented separately for adults and youth.

5.1 Adult Interview

5.1.1 Single Wave Response Rates Conditioning on Wave 4 Response

This section summarizes the Wave 4 Cohort single-wave response rates for the Wave 5 adult interview, conditioning on Wave 4 response. Unweighted and weighted single-wave response rates were calculated for the Wave 5 adult interview among those in the Wave 4 Cohort using the following formulas:

$$RR_{Y\_AW} = C_{Y\_AW} / \left(C_{Y\_AW} + N_{Y\_AW} + e_{Y\_AW} \times U_{Y\_AW} \right)$$

$$e_{Y\_AW} = \left(C_{Y\_AW} + N_{Y\_AW} \right) / \left(C_{Y\_AW} + N_{Y\_AW} + I_{Y\_AW} \right)$$

where

RR_{A_SW} = Wave 5 adult interview single-wave response rate;

C_{A_SW} = number of Wave 5 adult interview complete cases;

N_{A_SW} = number of Wave 5 adult interview nonrespondents known to be eligible;

U_{A_SW} = number of Wave 5 adult interview nonrespondents with unknown eligibility status;

e_{A_SW} = estimated proportion of eligible cases among the Wave 5 adult interview nonrespondents with unknown eligibility status; and

I_{A_SW} = number of Wave 5 adult interview ineligible cases.

Unweighted counts and weighted counts based on the Wave 4 cross-sectional weight were obtained for the response status categories C_{A_SW}, N_{A_SW}, U_{A_SW}, and I_{A_SW} for the unweighted single-wave response rates and weighted single-wave response rates, respectively. Ideally, weighted single-wave response rates for the Wave 4 Cohort would use an IPS weight, as has been the practice for Wave 1 Cohort weighted response rates. However, the information needed to create the IPS weight for the Wave 4 Cohort is not available (see Section 3.2.1). The Wave 4 cross-sectional weight was used for the weighted response rates presented in Table 5-1 because the sum of these weights provides the closest approximation to the Wave 4 Cohort target population at Wave 4, i.e., the U.S. CNP ages 10 and older as of Wave 4.

Table 5-1 provides single-wave response rates for the Wave 5 adult interview. In addition to the overall row, response rates are shown by Wave 4.5 response status and recruitment wave as well as the following Wave 4 characteristics: sex, age, race/ethnicity, census region, and tobacco-use status. Persons with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic.

The unweighted and weighted single-wave response rates for the Wave 5 adult interview were 86.4 percent and 88.0 percent, respectively. The unweighted and weighted response rates for most subgroups were similar to each other. The largest difference was for non-Hispanic white alone adults and adults located in the West census region at Wave 4, where each of these subgroups had a weighted single-wave response rate that was 2.2 percentage points higher than the unweighted single-wave response rate. The weighted response rates varied modestly by Wave 4 sex, race/ethnicity, census region, current established tobacco use (among adults who were adults at Wave 4), and ever tobacco use (among adults who were youth at Wave 4). The response rates varied to a greater extent by Wave 4 age, and substantially by Wave 4.5 response status and recruitment wave. Adults ages 45 to 64 at Wave 4 had the highest weighted single-wave response rate (90.3 percent) among the age groups, while those ages 18 to 24 at Wave 4 had the lowest (82.5 percent). (This is similar to the pattern observed across Wave 1 age groups for the Wave 1 Cohort weighted all-waves response rates in Table 4-1.) The weighted single-wave response rate for the Wave 5 adult interview was 91.3 percent for respondents to the Wave 4.5 special data collection and 41.1 percent for Wave 4.5 nonrespondents. In terms of recruitment wave, the weighted single-wave response rate for Wave 4 Cohort adults recruited in Wave 1 was 12.9 percentage points higher than for those recruited as part of the Wave 4 replenishment sample. A possible explanation for the large differences in subgroup response rates for both of these characteristics is that Wave 4 Cohort members who have responded to multiple previous waves, particularly those approached for the Wave 4.5 special data collection, may be more cooperative than other cohort members. The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years and all completed an interview at Wave 4. The Wave 4 replenishment sample members of the Wave 4 Cohort do not have the same history with the study and Wave 5 was the first follow-up wave for some of these participants.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts. None of the differences among the demographic and tobacco-use subgroups in Table 5-1 causes serious concern about potential nonresponse bias in the Wave 5 adult interview estimates for the Wave 4 Cohort. However, it is pertinent to examine the effect of the Wave 5 weighting adjustments for the Wave 4 Cohort, given the differences in response propensity associated with recruitment wave and prior participation in the study. The results of these analyses are discussed in the next section.

Table 5-1. Wave 5 adult interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort)

Characteristic[a]	C_{A_sw}: Completed Wave 5 adult interview (n)	I_{A_sw}: Ineligible at Wave 5 (n)	N_{A_sw}: Nonresponse known to be eligible at Wave 5 (n)	U_{A_sw}: Nonresponse with unknown eligibility status at Wave 5 (n)	RR_{A_sw}: Unweighted single-wave response rate[b] (%)	RR_{A_sw}: Weighted single-wave response rate (%)
Overall	32,687	664	3,274	1,925	86.4	88.0
Wave 4.5 response status						
Respondent	1,982	9	127	58	91.5	91.3
Nonrespondent	149	4	133	65	43.1	41.1
Not a part of Wave 4.5	30,556	651	3,014	1,802	86.5	88.1
Wave 4 sex						
Male	15,866	383	1,752	1,047	85.1	86.9
Female	16,780	279	2,022	876	87.6	89.1
Wave 4 age group						
Under 18	4,151	39	475	241	85.3	85.0
18-24	9,215	114	1,103	782	83.1	82.5
25-44	9,731	150	817	637	87.1	88.1
45-64	7,012	171	596	206	89.8	90.3
65 and above	2,576	190	283	58	88.4	87.8
Wave 4 race/ethnicity[c]						
Hispanic	6,907	134	606	490	86.4	87.8
Non-Hispanic White alone	17,900	331	2,004	907	86.1	88.3
Non-Hispanic Black alone	4,825	111	345	311	88.1	88.4
Non-Hispanic other race or multiple races	2,543	64	266	181	85.2	86.5
Wave 4 census region						
Northeast	4,757	87	508	271	86.0	88.1
Midwest	7,787	158	790	399	86.8	88.2
South	12,326	274	1,181	807	86.2	87.6
West	7,817	145	795	447	86.4	88.6
Wave 4 current established tobacco use[d] (Wave 4 adults only)						
Current established tobacco user	10,495	315	1,003	771	85.7	86.4
Not current established tobacco user	17,949	304	1,784	905	87.0	88.7
Wave 4 ever tobacco use[e] (Wave 4 youth only)						
Ever user	1,456	21	145	106	85.4	84.9
Never user	2,416	16	293	104	85.9	85.5

Table 5-1. Wave 5 adult interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort) (continued)

Characteristic[a]	C_{A_sw}: Completed Wave 5 adult interview (n)	I_{A_sw}: Ineligible at Wave 5 (n)	N_{A_sw}: Nonresponse known to be eligible at Wave 5 (n)	U_{A_sw}: Nonresponse with unknown eligibility status at Wave 5 (n)	RR_{A_sw}: Unweighted single-wave response rate[b] (%)	RR_{A_sw}: Weighted single-wave response rate (%)
Recruitment wave						
Wave 1	27,348	529	2,153	1,290	88.9	89.9
Wave 4	5,339	135	1,121	635	75.4	77.0

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] The unweighted adult interview single-wave response rate equals $C_{A\_sw}/(C_{A\_sw}+N_{A\_sw}+U_{A\_sw}*((C_{A\_sw}+N_{A\_sw})/(C_{A\_sw}+N_{A\_sw}+I_{A\_sw})))$, where $(C_{A\_sw}+N_{A\_sw})/(C_{A\_sw}+N_{A\_AW}+I_{A\_sw})$ is the estimated proportion of eligible cases among those with unknown eligibility. The weighted adult interview single-wave response rate was calculated using a similar approach except that the Wave 4 cross-sectional weight was used to obtain weighted counts.

[c] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

[d] A 'current established user' of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 4 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

[e] An 'ever user' of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 4 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), bidis, and kreteks.

5.1.2 Comparison Between Wave 5 Respondents and Those Eligible for Wave 5 Interview (Among the Wave 4 Cohort)

This section compares weighted estimates between all the Wave 5 adult interview respondents in the Wave 4 Cohort and the members of the Wave 4 Cohort who were eligible for the Wave 5 adult interview. The weighted estimates cover demographic and socio-economic characteristics (shown in Table 5-2) as well as tobacco-use status (shown in Tables 5-3 and 5-4). The structure of these tables is similar to the tables in Section 4.1.2.

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents who were eligible for Wave 5 adult interview and Wave 5 adult interview respondents (Wave 4 Cohort)

Wave 4 characteristic	Wave 4 respondents who were eligible for Wave 5 adult interview: Unweighted count	Wave 4 respondents who were eligible for Wave 5 adult interview: A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Wave 4 respondents who completed Wave 5 adult interview: Unweighted count	Before Wave 5 weighting adjustment: B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Before Wave 5 weighting adjustment: Difference in weighted percentages [B – A] [95% confidence interval]	After Wave 5 weighting adjustment: C: Weighted percentage, using Wave 5 single-wave weight [95% confidence interval]	After Wave 5 weighting adjustment: Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	18,665	48.2% [47.7%, 48.7%]	15,866	47.5% [47.0%, 48.1%]	-0.7% [-0.9%, -0.4%]	48.1% [47.6%, 48.7%]	0.0% [-0.0%, 0.0%]
Female	19,176	51.8% [51.3%, 52.3%]	16,780	52.5% [51.9%, 53.0%]	0.7% [0.4%, 0.9%]	51.9% [51.3%, 52.4%]	0.0% [-0.0%, 0.0%]
Age group							
Under 18	4,867	3.3% [3.2%, 3.5%]	4,151	3.2% [3.0%, 3.4%]	-0.1% [-0.2%, -0.1%]	3.3% [3.1%, 3.5%]	0.0% [-0.0%, -0.0%]
18-24	11,100	12.1% [11.7%, 12.4%]	9,215	11.3% [11.0%, 11.7%]	-0.8% [-0.9%, -0.6%]	12.1% [11.7%, 12.4%]	0.0% [-0.0%, 0.0%]
25-44	11,185	32.2% [32.7%, 33.7%]	9,731	33.2% [32.7%, 33.7%]	0.0% [-0.2%, 0.3%]	32.2% [32.7%, 33.7%]	0.0% [0.0%, 0.0%]
45-64	7,814	33.1% [32.6%, 33.6%]	7,012	34.0% [33.5%, 34.5%]	0.9% [0.6%, 1.2%]	33.1% [32.6%, 33.6%]	0.0% [0.0%, 0.0%]
65 and above	2,917	18.3% [17.9%, 18.7%]	2,576	18.3% [17.8%, 18.7%]	0.0% [-0.3%, 0.2%]	18.3% [17.9%, 18.7%]	0.0% [-0.0%, 0.0%]
Race/ethnicity[a]							
Hispanic	8,003	16.2% [15.8%, 16.6%]	6,907	16.1% [15.7%, 16.5%]	-0.1% [-0.3%, 0.2%]	16.2% [15.8%, 16.6%]	-0.0% [-0.0%, 0.0%]
Non-Hispanic White alone	20,811	64.3% [63.8%, 64.8%]	17,900	64.5% [63.9%, 65.0%]	0.2% [-0.1%, 0.4%]	64.3% [63.8%, 64.9%]	0.0% [-0.0%, 0.1%]
Non-Hispanic Black alone	5,481	11.4% [11.1%, 11.7%]	4,825	11.4% [11.1%, 11.8%]	0.0% [-0.1%, 0.2%]	11.4% [11.0%, 11.7%]	0.0% [-0.1%, 0.0%]
Non-Hispanic other race or multiple races	2,990	8.2% [7.9%, 8.4%]	2,543	8.0% [7.7%, 8.3%]	-0.1% [-0.4%, 0.1%]	8.1% [7.8%, 8.4%]	0.0% [-0.1%, 0.0%]

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents who were eligible for Wave 5 adult interview and Wave 5 adult interview respondents (Wave 4 Cohort) (continued)

Wave 4 characteristic	Wave 4 respondents who were eligible for Wave 5 adult interview		Wave 4 respondents who completed Wave 5 adult interview				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 5 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	5,536	17.8% [17.4%, 18.2%]	4,757	17.8% [17.4%, 18.3%]	0.0% [-0.3%, 0.3%]	17.8% [17.4%, 18.2%]	0.0% [-0.0%, -0.0%]
Midwest	8,976	21.0% [20.6%, 21.4%]	7,787	21.1% [20.6%, 21.5%]	0.0% [-0.2%, 0.3%]	21.0% [20.6%, 21.5%]	0.0% [-0.0%, 0.0%]
South	14,314	37.5% [37.0%, 38.0%]	12,326	37.3% [36.8%, 37.8%]	-0.2% [-0.6%, 0.1%]	37.5% [37.0%, 38.0%]	0.0% [-0.0%, 0.0%]
West	9,059	23.7% [23.2%, 24.1%]	7,817	23.8% [23.4%, 24.3%]	0.2% [-0.1%, 0.4%]	23.7% [23.2%, 24.1%]	0.0% [0.0%, 0.0%]
Education							
Less than high school or GED	6,134	15.6% [15.2%, 16.0%]	5,279	15.5% [15.0%, 15.9%]	-0.1% [-0.3%, 0.1%]	15.6% [15.2%, 16.0%]	0.0% [-0.0%, 0.1%]
High school	7,967	23.7% [23.3%, 24.2%]	6,763	23.2% [22.7%, 23.7%]	-0.5% [-0.8%, -0.3%]	23.7% [23.2%, 24.2%]	0.0% [-0.1%, 0.0%]
Some college, no degree	11,728	31.2% [30.7%, 31.7%]	10,154	31.2% [30.7%, 31.7%]	0.0% [-0.3%, 0.2%]	31.2% [30.7%, 31.8%]	0.0% [-0.0%, 0.1%]
Bachelor degree and above	7,045	29.5% [29.0%, 30.0%]	6,228	30.2% [29.7%, 30.7%]	0.7% [0.4%, 0.9%]	29.5% [29.0%, 30.0%]	0.0% [-0.1%, 0.0%]

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents who were eligible for Wave 5 adult interview and Wave 5 adult interview respondents (Wave 4 Cohort) (continued)

	Wave 4 respondents who were eligible for Wave 5 adult interview		Wave 4 respondents who completed Wave 5 adult interview					
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment		
Wave 4 characteristic	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 5 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Health insurance (Wave 4 adults only)							
Yes	27,187	88.1% [87.5%, 88.6%]	23,660	88.5% [87.9%, 89.1%]	0.4% [0.2%, 0.7%]	88.3% [87.6%, 88.9%]	0.2% [-0.0%, 0.4%]
No	5,478	11.9% [11.4%, 12.5%]	4,585	11.5% [10.9%, 12.1%]	-0.4% [-0.7%, -0.2%]	11.7% [11.1%, 12.4%]	-0.2% [-0.4%, 0.0%]

[a] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

Table 5-3. Comparison of Wave 4 "current established tobacco-use" estimates between Wave 4 adults who were eligible for Wave 5 adult interview and Wave 4 adults who completed Wave 5 adult interview (Wave 4 Cohort)*

	Wave 4 adults who were eligible for Wave 5 adult interview		Wave 4 adults who completed Wave 5 adult interview				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
Wave 4 characteristic[a]	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 5 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	32,907	22.6% [22.1%, 23.2%]	28,444	22.2% [21.6%, 22.8%]	-0.5% [-0.7%, -0.2%]	22.7% [22.2%, 23.3%]	0.1% [-0.0%, 0.2%]
Sex							
Male	16,089	28.0% [27.1%, 28.9%]	13,689	27.5% [26.5%, 28.5%]	-0.5% [-0.8%, -0.1%]	28.1% [27.1%, 29.0%]	0.1% [-0.1%, 0.3%]
Female	16,785	17.7% [17.1%, 18.4%]	14,724	17.4% [16.8%, 18.1%]	-0.3% [-0.6%, -0.1%]	17.8% [17.2%, 18.5%]	0.1% [-0.1%, 0.2%]
Age group							
18-24	11,059	22.5% [21.3%, 23.7%]	9,183	21.7% [20.5%, 23.0%]	-0.8% [-1.2%, -0.3%]	22.5% [21.3%, 23.7%]	0.0% [-0.4%, 0.3%]
25-44	11,154	28.4% [27.4%, 29.4%]	9,704	27.8% [26.8%, 29.0%]	-0.6% [-0.9%, -0.2%]	28.6% [27.5%, 29.7%]	0.2% [-0.1%, 0.5%]
45-64	7,795	23.6% [22.6%, 24.6%]	6,995	23.1% [22.1%, 24.2%]	-0.4% [-0.8%, -0.1%]	23.7% [22.7%, 24.8%]	0.1% [-0.2%, 0.4%]
65 and above	2,897	10.5% [9.5%, 11.7%]	2,560	10.4% [9.3%, 11.6%]	-0.2% [-0.6%, 0.3%]	10.5% [9.4%, 11.7%]	0.0% [-0.4%, 0.3%]
Race/ethnicity[b]							
Hispanic	6,489	17.1% [16.0%, 18.3%]	5,611	16.8% [15.7%, 18.0%]	-0.3% [-0.7%, 0.1%]	17.1% [16.0%, 18.3%]	-0.1% [-0.4%, 0.3%]
Non-Hispanic White alone	18,549	24.4% [23.6%, 25.2%]	16,003	23.8% [22.9%, 24.7%]	-0.6% [-0.9%, -0.3%]	24.4% [23.6%, 25.3%]	-0.0% [-0.1%, 0.2%]
Non-Hispanic Black alone	4,819	25.5% [24.1%, 26.8%]	4,241	25.4% [23.9%, 26.9%]	-0.1% [-0.7%, 0.5%]	25.7% [24.3%, 27.2%]	0.3% [-0.1%, 0.6%]
Non-Hispanic other race or multiple races	2,547	15.8% [14.3%, 17.4%]	2,164	15.5% [13.9%, 17.2%]	-0.3% [-1.1%, 0.5%]	16.2% [14.7%, 18.0%]	0.5% [-0.3%, 1.2%]
Census region							
Northeast	4,863	20.7% [19.1%, 22.5%]	4,199	20.2% [18.5%, 22.0%]	-0.5% [-1.1%, 0.0%]	20.9% [19.2%, 22.6%]	-0.1% [-0.3%, 0.5%]
Midwest	7,870	25.9% [24.2%, 27.6%]	6,829	25.2% [23.6%, 26.8%]	-0.7% [-1.3%, -0.1%]	25.7% [24.1%, 27.4%]	-0.2% [-0.4%, 0.1%]
South	12,444	24.9% [23.6%, 26.2%]	10,731	24.5% [23.1%, 26.0%]	-0.3% [-0.7%, 0.0%]	25.1% [23.8%, 26.4%]	0.2% [-0.1%, 0.5%]
West	7,730	17.7% [16.4%, 19.1%]	6,685	17.3% [15.9%, 18.9%]	-0.4% [-0.7%, 0.0%]	17.8% [16.5%, 19.3%]	0.1% [-0.1%, 0.4%]

* A 'current established user' of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 4 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American

Table 5-4. Comparison of Wave 4 “ever tobacco-use” estimates between Wave 4 youth who were eligible for Wave 5 adult interview and Wave 4 youth who completed Wave 5 adult interview (Wave 4 Cohort)*

	Wave 4 youth who were eligible for Wave 5 adult interview		Wave 4 youth who completed Wave 5 adult interview				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
Wave 4 characteristic[a]	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 5 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	4,520	37.6% [35.7%, 39.6%]	3,872	37.4% [35.4%, 39.5%]	-0.2% [-0.9%, 0.5%]	37.6% [35.6%, 39.7%]	0.0% [-0.5%, 0.4%]
Sex							
Male	2,327	39.8% [37.3%, 42.4%]	1,973	39.6% [37.0%, 42.3%]	-0.2% [-1.2%, 0.7%]	39.7% [37.1%, 42.4%]	-0.1% [-0.7%, 0.6%]
Female	2,184	35.3% [32.8%, 37.8%]	1,890	35.1% [32.5%, 37.8%]	-0.1% [-1.1%, 0.8%]	35.3% [32.7%, 37.9%]	0.0% [-0.5%, 0.6%]
Race/ethnicity[b]							
Hispanic	1,341	38.5% [35.7%, 41.3%]	1,162	38.4% [35.3%, 41.6%]	-0.1% [-1.3%, 1.2%]	39.0% [35.7%, 42.4%]	0.5% [-0.6%, 1.6%]
Non-Hispanic White alone	2,102	40.0% [37.1%, 42.9%]	1,775	39.8% [36.7%, 42.9%]	-0.2% [-1.3%, 0.8%]	39.8% [36.9%, 42.8%]	-0.2% [-0.8%, 0.4%]
Non-Hispanic Black alone	587	32.6% [28.1%, 37.4%]	518	33.7% [28.9%, 38.8%]	1.1% [-0.3%, 2.6%]	33.5% [28.8%, 38.6%]	0.9% [-0.4%, 2.3%]
Non-Hispanic other race or multiple races	408	30.0% [24.9%, 35.6%]	352	28.7% [24.1%, 33.7%]	-1.3% [-4.0%, 1.4%]	28.8% [24.3%, 33.7%]	-1.2% [-3.7%, 1.2%]
Census region							
Northeast	601	40.8% [36.5%, 45.2%]	500	40.6% [35.8%, 45.6%]	-0.2% [-2.2%, 1.8%]	40.5% [35.7%, 45.4%]	-0.3% [-2.2%, 1.6%]
Midwest	1,030	40.7% [36.9%, 44.6%]	899	39.5% [35.5%, 43.8%]	-1.1% [-2.5%, 0.2%]	39.7% [35.7%, 43.7%]	-1.0% [-2.3%, 0.2%]
South	1,691	35.7% [33.1%, 38.5%]	1,448	36.2% [33.3%, 39.1%]	0.4% [-0.6%, 1.4%]	36.4% [33.5%, 39.3%]	0.6% [-0.3%, 1.6%]
West	1,198	35.6% [30.8%, 40.7%]	1,025	35.3% [30.7%, 40.2%]	-0.3% [-1.7%, 1.1%]	35.6% [30.8%, 40.7%]	0.0% [-1.2%, 1.2%]

* An ‘ever user’ of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 4 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), bidis, and kreteks.

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

For the "before Wave 5 weighting adjustments" in Table 5-2, the largest overrepresentation among Wave 5 adult interview respondents was of the Wave 4 "45-64" age group (with corresponding underrepresentations of the "under 18" and "18-24" age groups). Females, the Wave 4 "bachelor degree and above" education group, and those with health insurance at Wave 4 were also slightly overrepresented among the Wave 5 adult interview respondents, while the Wave 4 "high school" education group was slightly underrepresented.

For the "after Wave 5 weighting adjustment" differences, the estimated confidence intervals (in the last column of Table 5-2) either included zero or were in the proximity of zero for each of the Wave 4 characteristics examined: sex, age, race/ethnicity, census region, education, and health insurance status.

Tables 5-3 and 5-4 show the comparisons for the Wave 4 tobacco-use measures between those eligible for the Wave 5 adult interview and all Wave 5 adult interview respondents, among the Wave 4 Cohort. (As in Section 4.1.2, separate analyses were conducted based on the participant's interview type at Wave 4.) For the "before Wave 5 weighting adjustment" estimates, the differences between the Wave 5 adult respondents and those eligible for the Wave 5 adult interview were no more than 1.3 percentage points. About half of the confidence intervals for those who were adults at Wave 4 were just below zero, indicating a slight underrepresentation of Wave 4 current established tobacco users among the Wave 5 respondents (see Table 5-3). However, all of the confidence intervals included zero for those who were youth at Wave 4 (see Table 5-4). For the "after Wave 5 weighting adjustment" estimates, the differences between the adult respondents and those eligible for the adult interview were mostly negligible and all of the confidence intervals included zero.

Overall, these results reflect the use of both demographic variables and tobacco-use measures from Wave 4 for calibrating the Wave 5 single-wave weight for the Wave 4 Cohort. Assuming that the Wave 4 demographic, socio-economic, and tobacco-use characteristics in Tables 5-2, 5-3, and 5-4 are correlated with key tobacco- and health-related outcome measures in Wave 5 of the PATH Study, these results indicate little if any nonresponse bias in the adult interview estimates for the Wave 4 Cohort due to attrition from Wave 4 to Wave 5.

5.1.3 Comparison of Adult Cigarette-Use Estimates to Other National Studies

This section compares adult cigarette-smoking estimates based on the Wave 5 adult interview respondents from the Wave 4 Cohort of the PATH Study to similar estimates based on data from TUS-CPS 2018-2019, NHIS 2018, NHANES 2017-2018, and NSDUH 2018.

Table 5-5 presents estimates of the prevalence of current cigarette smoking[18] based on the Wave 4 Cohort, for the adult population as a whole and for subgroups. These estimates are accompanied by 95 percent confidence intervals. The point estimates for the PATH Study were calculated using the Wave 5 single-wave weight for the Wave 4 Cohort. The corresponding replicate weights were used to calculate variances and confidence intervals. Point estimates and 95 percent confidence intervals are reported for the other national studies as well.

Table 5-5 shows that the PATH Study estimates of adult current cigarette-smoking rates were similar to estimates from NHANES 2017-2018 and from NSDUH 2018 using the modified definition of a current cigarette smoker (see footnote to Table 5-5 or Appendix A for details); these three studies had overlapping confidence intervals for almost all the estimates in the table. Estimates from Wave 5 of the PATH Study tended to be higher than those from TUS-CPS 2018-2019 and NHIS 2018, and lower than those from NSDUH 2018 using the original definition of a current cigarette smoker. Table 5-5 shows no evidence of nonresponse bias in the PATH Study with respect to current cigarette-smoking behavior among adults, in the sense that the PATH Study's estimates were all within the range of estimates from comparable surveys.

[18] For the PATH Study, following common practice for tobacco surveys, a current cigarette smoker is someone who (1) has smoked at least 100 cigarettes in his or her lifetime and (2) currently smokes cigarettes every day or some days. The questions used to define current cigarette smoking for each survey are provided in Appendix A.

Table 5-5. Comparison of adult cigarette-smoking estimates from PATH Study Wave 5 adult interview (Wave 4 Cohort) and other national studies

Cigarette-smoking measure and subgroup	PATH Study Wave 5 (2018-2019) sample size	Percentage from PATH Study Wave 5 (2018-2019) [95% confidence interval]	Percentage from TUS-CPS 2018-2019 [95% confidence interval]	Percentage from NHIS 2018 [95% confidence interval]	Percentage from NHANES 2017-2018 [95% confidence interval]	Percentage from NSDUH 2018, original definition[a] [95% confidence interval]	Percentage from NSDUH 2018, modified definition[a] [95% confidence interval]
Current smoker, overall	32,663	16.5% [15.9%, 17.0%]	11.4% [11.2%, 11.6%]	13.7% [13.1%, 14.4%]	17.0% [14.5%, 19.8%]	18.8% [18.2%, 19.4%]	17.3% [16.7%, 17.8%]
Current smoker, male	15,852	18.2% [17.4%, 19.0%]	12.9% [12.6%, 13.3%]	15.6% [14.8%, 16.5%]	19.1% [16.4%, 22.1%]	21.0% [20.1%, 21.9%]	19.4% [18.6%, 20.3%]
Current smoker, female	16,769	14.8% [14.2%, 15.5%]	10.0% [9.7%, 10.3%]	12.0% [11.3%, 12.7%]	15.1% [12.4%, 18.2%]	16.7% [16.1%, 17.4%]	15.2% [14.6%, 15.9%]
Current smoker, age 18-24	11,349	9.9% [9.2%, 10.6%]	7.4% [6.7%, 8.2%]	7.8% [6.4%, 9.5%]	12.2% [8.5%, 17.3%]	N/A	N/A
Current smoker, age 25-44	11,060	21.7% [20.8%, 22.7%]	12.5% [12.2%, 12.9%]	16.5% [15.4%, 17.6%]	23.5% [19.3%, 28.3%]	N/A	N/A
Current smoker, age 45-64	7,126	19.1% [18.2%, 20.1%]	14.3% [13.9%, 14.7%]	16.3% [15.3%, 17.3%]	17.1% [14.3%, 20.4%]	N/A	N/A
Current smoker, age 65 and above	3,128	7.7% [6.8%, 8.7%]	7.2% [6.9%, 7.6%]	8.4% [7.7%, 9.2%]	8.6% [6.8%, 10.9%]	N/A	N/A
Current smoker, Hispanic	6,899	13.2% [12.2%, 14.3%]	7.5% [7.0%, 8.0%]	9.8% [8.5%, 11.3%]	12.5% [9.4%, 16.5%]	14.3% [13.2%, 15.5%]	11.9% [11.0%, 13.0%]
Current smoker, non-Hispanic White alone	17,891	17.2% [16.5%, 18.0%]	12.6% [12.3%, 12.9%]	15.0% [14.3%, 15.7%]	17.6% [14.7%, 21.1%]	20.2% [19.4%, 21.0%]	19.1% [18.4%, 19.9%]
Current smoker, non-Hispanic Black alone	4,819	20.1% [18.9%, 21.3%]	12.6% [11.9%, 13.3%]	14.6% [12.9%, 16.4%]	20.6% [16.7%, 25.0%]	19.9% [18.4%, 21.4%]	17.4% [15.9%, 19.1%]
Current smoker, non-Hispanic other race or multiple races	2,541	11.7% [10.4%, 13.0%]	8.7% [8.1%, 9.4%]	11.2% [9.5%, 13.3%]	16.3% [12.0%, 21.7%]	15.2% [13.4%, 17.2%]	13.4% [11.6%, 15.3%]
Current every-day smoker, overall	32,672	12.4% [11.9%, 12.9%]	8.7% [8.5%, 8.9%]	10.3% [9.8%, 10.7%]	13.6% [11.3%, 16.2%]	N/A	N/A
Current some-days smoker, overall	32,667	4.0% [3.8%, 4.3%]	2.7% [2.6%, 2.8%]	3.5% [3.2%, 3.8%]	3.5% [2.8%, 4.2%]	N/A	N/A

[a] NSDUH's definition of a current cigarette smoker is someone who has smoked part or all of a cigarette in the past 30 days, which is more expansive than the definition used in the other surveys. However, NSDUH contains questions on lifetime smoking and current smoking. The modified definition uses these questions to construct a measure of "current smoking" that is comparable to that of the other surveys (Ryan et al., 2012). The construction of this variable is described in Appendix A. Detailed age information was not available in the public use file for NSDUH 2018.

The disparities in Table 5-5 can be explained by a number of potential reasons. In addition to the varying degrees of sampling and measurement errors, the surveys differ in question order, context, mode of administration, and year of data collection. The TUS-CPS estimates of cigarette-smoking prevalence are generally lower than estimates from the other surveys, which may be due to the proxy responses used in the TUS-CPS. The rotation group structure of the TUS-CPS may result in underestimates of cigarette-smoking prevalence, as smokers are more likely to drop out over the course of the panel survey (Song, 2013). The PATH Study and NSDUH both use audio computer-assisted self-interview (ACASI) administration for the tobacco-use questions so that the interviewer does not see responses to the questions. In contrast, TUS-CPS, NHIS, and NHANES have direct questioning by an interviewer: NHIS and NHANES are conducted in person, and TUS-CPS is conducted in person and by telephone. The contexts and purposes of these surveys also differ: CPS is a general survey on unemployment, NHIS and NHANES are general health surveys, and NSDUH is a cross-sectional survey on substance use (including tobacco use) and health, including mental health. Unlike the cross-sectional prevalence surveys, the PATH Study uses a longitudinal cohort design to assess within-person changes and between-person differences in tobacco-use behaviors and health over time. Other differences among the questions used in the instruments of these different studies are outlined in Appendix A.

5.1.4 Comparison of Adult ENDS-Use Estimates to Other National Studies

Table 5-6 compares estimates for adult ever use and current use of ENDS based on the Wave 5 adult interview respondents from the Wave 4 Cohort of the PATH Study to similar estimates based on data from TUS-CPS 2018-2019 and NHIS 2018. As for cigarette smoking, the primary measure of ENDS use among adults is whether the adult currently uses ENDS.[19] (See Appendix B for details.)

Table 5-6 presents estimates of the prevalence of current ENDS use based on the Wave 4 Cohort, for the adult population as a whole and for subgroups. These estimates are accompanied by 95 percent confidence intervals. The point estimates for the PATH Study were calculated using the Wave 5 single-wave weight for the Wave 4 Cohort. The corresponding replicate weights were used

[19] For the PATH Study, a current ENDS user is someone who (1) at any time has used ENDS fairly regularly and (2) currently uses ENDS every day or some days. The questions used to define ENDS use for each survey are provided in Appendix B.

to calculate variances and confidence intervals. Point estimates and 95 percent confidence intervals are reported for the other national studies as well.

Table 5-6. Comparison of adult ENDS-use estimates from PATH Study Wave 5 adult interview (Wave 4 Cohort) and other national studies

ENDS-use measure and subgroup	PATH Study Wave 5 (2018-2019) sample size	Percentage from PATH Study Wave 5 (2018-2019) [95% confidence interval]	Percentage from TUS-CPS 2018-2019 [95% confidence interval]	Percentage from NHIS 2018 [95% confidence interval]
Current ENDS user, overall	32,672	4.8% [4.5%, 5.0%]	2.3% [2.2%, 2.4%]	3.2% [3.0%, 3.5%]
Current ENDS user, male	15,856	5.7% [5.4%, 6.1%]	2.8% [2.7%, 3.0%]	4.3% [3.8%, 4.8%]
Current ENDS user, female	16,774	3.9% [3.6%, 4.2%]	1.8% [1.7%, 1.9%]	2.3% [2.0%, 2.6%]
Current ENDS user, age 18-24	11,351	13.7% [12.9%, 14.4%]	5.3% [4.8%, 6.0%]	7.6% [6.2%, 9.3%]
Current ENDS user, age 25-44	11,064	6.3% [5.8%, 6.8%]	3.0% [2.8%, 3.2%]	4.3% [3.8%, 4.8%]
Current ENDS user, age 45-64	7,126	2.6% [2.2%, 3.0%]	1.6% [1.5%, 1.8%]	2.1% [1.8%, 2.5%]
Current ENDS user, age 65 and above	3,131	0.9% [0.6%, 1.3%]	0.6% [0.5%, 0.7%]	0.8% [0.6%, 1.1%]
Current ENDS user, Hispanic	6,904	3.2% [2.7%, 3.8%]	1.5% [1.3%, 1.8%]	2.5% [1.9%, 3.4%]
Current ENDS user, non-Hispanic White alone	17,890	5.6% [5.2%, 5.9%]	2.8% [2.6%, 2.9%]	3.7% [3.3%, 4.1%]
Current ENDS user, non-Hispanic Black alone	4,823	3.2% [2.7%, 3.8%]	1.1% [0.9%, 1.4%]	1.6% [1.2%, 2.3%]
Current ENDS user, non-Hispanic other race or multiple races	2,540	4.6% [3.8%, 5.5%]	1.9% [1.6%, 2.3%]	3.4% [2.6%, 4.5%]
Current every-day ENDS user, overall	32,675	2.5% [2.4%, 2.7%]	1.1% [1.0%, 1.2%]	1.4% [1.2%, 1.6%]
Current some-days ENDS user, overall	32,673	2.2% [2.1%, 2.4%]	1.2% [1.1%, 1.3%]	1.9% [1.7%, 2.1%]

The PATH Study current ENDS-use estimates were higher than those from TUS-CPS 2018-2019 and NHIS 2018, with only a few confidence intervals overlapping, and particularly so for the "18-24" age group. However, this subgroup finding is consistent with the relatively high estimates for past 30-day ENDS use among 12th graders from the PATH Study and MTF 2017 (see Table 5-11 in Section 5.2.4), who are on the cusp of becoming young adults. As discussed in Section 5.1.3, some differences in the estimates may be attributable to differences in context, question order and wording, mode of administration, time frames of the surveys, and measurement error. It is uncertain which factors are the key drivers of the differences in adult ENDS-use estimates observed between the PATH Study, TUS-CPS, and NHIS.

5.2 Youth Interview

5.2.1 Single Wave Response Rates Conditioning on Wave 4 Response

This section summarizes the Wave 4 Cohort single-wave response rates for the Wave 5 youth interview, conditioning on Wave 4 response. Unweighted and weighted single-wave response rates were calculated for the Wave 5 youth interview among those in the Wave 4 Cohort using the following formulas:

$$RR_{Y\_SW} = C_{Y\_SW} / \left(C_{Y\_SW} + N_{Y\_SW} + e_{Y\_SW} \times U_{Y\_SW}\right)$$

$$e_{Y\_SW} = (C_{Y\_SW} + N_{Y\_SW}) / \left(C_{Y\_SW} + N_{Y\_SW} + I_{Y\_SW}\right)$$

where

RR_{Y_SW} = Wave 5 youth interview single-wave response rate;

C_{Y_SW} = number of Wave 5 youth interview complete cases;

N_{Y_SW} = number of Wave 5 youth interview nonrespondents known to be eligible;

U_{Y_SW} = number of Wave 5 youth interview nonrespondents with unknown eligibility status;

e_{Y_SW} = estimated proportion of eligible cases among the Wave 5 youth interview nonrespondents with unknown eligibility status; and

I_{Y_SW} = number of Wave 5 youth interview ineligible cases.

Unweighted counts and weighted counts based on the Wave 4 cross-sectional weight were obtained for the response status categories C_{Y_SW}, N_{Y_SW}, U_{Y_SW}, and I_{Y_SW} for the unweighted single-wave response rates and weighted single-wave response rates, respectively. As discussed in Section 5.1.1, the Wave 4 cross-sectional weight was used for the weighted response rates presented in Table 5-7 because the sum of these weights provides the closest approximation to the Wave 4 Cohort target population at Wave 4, i.e., the U.S. CNP ages 10 and older as of Wave 4.

Table 5-7 provides single-wave response rates for the Wave 5 youth interview. In addition to the overall row, response rates are shown by Wave 4.5 response status and recruitment wave as well as the following Wave 4 characteristics: sex, age, race/ethnicity, census region, and tobacco-use status. Persons with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic.

The unweighted and weighted single-wave response rates for the Wave 5 youth interview were 84.8 percent and 83.5 percent, respectively; the unweighted and weighted response rates for each subgroup were also similar to each other. The weighted response rates varied modestly by Wave 4 sex, race/ethnicity, census region, and ever tobacco-use status. However, the single-wave response rates by Wave 4 age, Wave 4.5 response status, and recruitment wave showed more variation. For Wave 4 age, the weighted response rate for the "under 12" age group (72.6 percent) was noticeably lower than for the "12-13" and "14-17" age groups (88.9 percent and 89.2 percent, respectively). (A similar pattern has been observed for the youth interview response rates by age group among the Wave 1 Cohort.) For the Wave 4 Cohort, this is likely because the persons in the "under 12" age group were shadow youth at Wave 4 and about half of these participants were first asked to complete a PATH Study interview at Wave 5; the other half were first asked to complete an interview at the Wave 4.5 special data collection. The weighted response rate for those who were first asked to complete a youth interview at Wave 5 was 70.8 percent (see the "not a part of Wave 4.5" subgroup in Table 5-7). The Wave 4.5 youth interview weighted response rate was 78.9 percent for those who were first asked to complete a youth interview as part of the special data collection (see the "under 12" age group in Table 5-1 of the Wave 4.5 Data Collection Nonresponse Bias Analysis Report).

In terms of the Wave 4.5 response status subgroups, the Wave 5 weighted single-wave youth interview response rate was much higher for Wave 4.5 respondents (93.6 percent) than Wave 4.5

nonrespondents (25.4 percent) and those not a part of the Wave 4.5 special data collection (70.8 percent, as noted above). The last two rows of Table 5-7 partition the Wave 4 Cohort members into their two distinct recruitment waves. The Wave 5 youth interview weighted response rate was 90.8 percent for those recruited in Wave 1 and 76.5 percent for those recruited in Wave 4. A possible explanation for the large differences in subgroup response rates for both of these characteristics is that Wave 4 Cohort members who have responded to multiple previous waves, particularly those approached for the Wave 4.5 special data collection, may be more cooperative than other cohort members. The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years and all completed an interview at Wave 4, including the Wave 1 shadow youth. The Wave 4 replenishment sample members of the Wave 4 Cohort do not have the same history with the study and include Wave 4 shadow youth, who did not have to complete an interview at Wave 4 in order to become a study participant.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts. However, it is pertinent to examine the effect of the Wave 5 weighting adjustments for the Wave 4 Cohort, given the differences in response propensity associated with age and recruitment wave. The results of these analyses are discussed in the next section.

Table 5-7. Wave 5 youth interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort)

Characteristic[a]	C_{Y_sw}: Completed Wave 5 youth interview (n)	I_{Y_sw}: Ineligible at Wave 5 (n)	N_{Y_sw}: Nonresponse known to be eligible at Wave 5 (n)	U_{Y_sw}: Nonresponse with unknown eligibility status at Wave 5 (n)	RR_{Y_sw}: Unweighted single-wave response rate[b] (%)	RR_{Y_sw}: Weighted single-wave response rate (%)
Overall	11,976	48	1,508	637	84.8	83.5
Wave 4.5 response status						
Respondent	10,073	22	471	176	94.0	93.6
Nonrespondent	335	21	612	285	27.3	25.4
Not a part of Wave 4.5	1,568	5	425	176	72.3	70.8
Wave 4 sex						
Male	6,225	29	783	324	84.9	83.8
Female	5,717	18	721	311	84.7	83.3
Wave 4 age group						
Under 12	3,177	12	748	343	74.5	72.6
12-13	4,233	14	369	143	89.2	88.9
14-17	4,566	22	391	150	89.4	89.2
Wave 4 race/ethnicity[c]						
Hispanic	3,504	17	373	216	85.6	84.5
Non-Hispanic White alone	5,401	12	780	238	84.1	83.1
Non-Hispanic Black alone	1,484	7	157	101	85.2	82.6
Non-Hispanic other race or multiple races	1,153	7	158	65	83.8	83.0
Wave 4 census region						
Northeast	1,660	6	246	96	82.9	81.5
Midwest	2,653	7	387	120	84.0	83.2
South	4,518	21	520	263	85.2	83.8
West	3,145	14	355	158	86.0	84.7
Wave 4 ever tobacco use[e] (Wave 4 youth only)						
Ever user	1,144	8	102	46	88.6	88.7
Never user	7,270	21	624	232	89.5	89.1

Table 5-7. Wave 5 youth interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort) (continued)

Characteristic[a]	C_{Y_SW}: Completed Wave 5 youth interview (n)	I_{Y_SW}: Ineligible at Wave 5 (n)	N_{Y_SW}: Nonresponse known to be eligible at Wave 5 (n)	U_{Y_SW}: Nonresponse with unknown eligibility status at Wave 5 (n)	RR_{Y_SW}: Unweighted single-wave response rate[b] (%)	RR_{Y_SW}: Weighted single-wave response rate (%)
Recruitment wave						
Wave 1	6,638	22	478	175	91.1	90.8
Wave 4	5,338	26	1,030	462	78.2	76.5

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] The unweighted youth interview single-wave response rate equals $C_{Y\_SW}/(C_{Y\_SW}+N_{Y\_SW}+U_{Y\_SW}*((C_{Y\_SW}+N_{Y\_SW})/(C_{Y\_SW}+N_{Y\_SW}+I_{Y\_SW})))$, where $(C_{Y\_SW}+N_{Y\_SW})/(C_{Y\_SW}+N_{Y\_SW}+I_{Y\_SW})$ is the estimated proportion of eligible cases among those with unknown eligibility. The weighted youth interview single-wave response rate was calculated using a similar approach except that the Wave 4 cross-sectional weight was used to obtain weighted counts.

[c] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

[d] An 'ever user' of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 4 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), bidis, and kreteks.

5.2.2 Comparison Between Wave 5 Respondents and Those Eligible for Wave 5 Interview (Among the Wave 4 Cohort)

This section compares weighted estimates between the Wave 5 youth interview respondents in the Wave 4 Cohort and the members of the Wave 4 Cohort who were eligible for the Wave 5 youth interview. The weighted estimates cover demographic characteristics (shown in Table 5-8) and tobacco-use status (shown in Table 5-9). The structure of these tables is similar to the tables in Section 4.2.2.

Table 5-8. Comparison of Wave 4 demographic characteristics between Wave 4 respondents who were eligible for Wave 5 youth interview and Wave 5 youth interview respondents (Wave 4 Cohort)

Wave 4 characteristic	Wave 4 respondents who were eligible for Wave 5 youth interview		Wave 4 respondents who completed Wave 5 youth interview				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 5 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	7,332	51.0% [50.1%, 51.8%]	6,225	51.1% [50.2%, 52.0%]	0.1% [-0.2%, 0.5%]	51.0% [50.1%, 51.9%]	0.0% [-0.1%, 0.1%]
Female	6,749	49.0% [48.2%, 49.9%]	5,717	48.9% [48.0%, 49.8%]	-0.1% [-0.5%, 0.2%]	49.0% [48.1%, 49.9%]	0.0% [-0.1%, 0.1%]
Age group							
Under 12	4,268	33.4% [32.7%, 34.2%]	3,177	29.0% [28.2%, 29.9%]	-4.4% [-4.8%, -4.0%]	33.3% [32.5%, 34.2%]	-0.1% [-0.2%, -0.1%]
12-13	4,745	33.1% [32.4%, 33.9%]	4,233	35.3% [34.4%, 36.1%]	2.1% [1.8%, 2.5%]	33.0% [32.2%, 33.9%]	-0.1% [-0.1%, -0.1%]
14-17	5,107	33.4% [32.6%, 34.2%]	4,556	35.7% [34.8%, 36.5%]	2.3% [1.9%, 2.6%]	33.6% [32.8%, 34.5%]	0.2% [0.2%, 0.3%]
Race/ethnicity[a]							
Hispanic	4,093	24.6% [23.9%, 25.4%]	3,504	25.0% [24.2%, 25.8%]	0.3% [-0.0%, 0.7%]	24.7% [23.9%, 25.5%]	0.0% [-0.0%, 0.1%]
Non-Hispanic White alone	6,419	51.9% [51.1%, 52.8%]	5,401	51.8% [50.9%, 52.7%]	-0.2% [-0.6%, 0.3%]	52.0% [51.0%, 52.9%]	0.0% [-0.1%, 0.1%]
Non-Hispanic Black alone	1,742	13.1% [12.6%, 13.7%]	1,484	13.0% [12.4%, 13.6%]	-0.1% [-0.4%, 0.2%]	13.1% [12.5%, 13.7%]	-0.1% [-0.1%, 0.0%]
Non-Hispanic other race or multiple races	1,376	10.3% [9.8%, 10.8%]	1,153	10.2% [9.7%, 10.8%]	0.0% [-0.4%, 0.3%]	10.3% [9.7%, 10.9%]	0.0% [-0.1%, 0.1%]

Table 5-8. Comparison of Wave 4 demographic characteristics between Wave 4 respondents who were eligible for Wave 5 youth interview and Wave 5 youth interview respondents (Wave 4 Cohort) (continued)

	Wave 4 respondents who were eligible for Wave 5 youth interview		Wave 4 respondents who completed Wave 5 youth interview				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
Wave 4 characteristic	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 5 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	2,002	16.2% [15.6%, 16.9%]	1,660	15.9% [15.2%, 16.5%]	-0.4% [-0.8%, 0.0%]	16.2% [15.6%, 16.9%]	0.0% [-0.1%, 0.0%]
Midwest	3,160	21.2% [20.5%, 21.9%]	2,653	21.1% [20.4%, 21.9%]	-0.1% [-0.5%, 0.4%]	21.2% [20.5%, 21.9%]	0.0% [-0.0%, 0.0%]
South	5,301	38.5% [37.7%, 39.4%]	4,518	38.7% [37.8%, 39.6%]	0.1% [-0.3%, 0.6%]	38.5% [37.7%, 39.4%]	0.0% [-0.0%, 0.1%]
West	3,658	24.0% [23.3%, 24.7%]	3,145	24.4% [23.6%, 25.1%]	0.3% [-0.1%, 0.8%]	24.0% [23.3%, 24.8%]	0.0% [-0.0%, 0.1%]

[a] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

Table 5-9. Comparison of Wave 4 "ever tobacco-use" estimates between Wave 4 youth who were eligible for Wave 5 youth interview and Wave 4 youth who completed Wave 5 youth interview (Wave 4 Cohort)*

	Wave 4 youth who were eligible for Wave 5 youth interview		Wave 4 youth who completed Wave 5 youth interview				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
Wave 4 characteristic[a]	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 5 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	9,418	13.3% [12.6%, 14.0%]	8,414	13.2% [12.5%, 14.0%]	-0.1% [-0.3%, 0.2%]	13.4% [12.6%, 14.2%]	0.1% [-0.0%, 0.3%]
Sex							
Male	4,836	14.9% [13.9%, 16.0%]	4,322	14.9% [13.8%, 16.1%]	0.0% [-0.4%, 0.3%]	15.2% [14.0%, 16.3%]	0.2% [-0.1%, 0.6%]
Female	4,552	11.6% [10.6%, 12.7%]	4,065	11.5% [10.5%, 12.7%]	-0.1% [-0.5%, 0.3%]	11.6% [10.6%, 12.7%]	0.0% [-0.4%, 0.3%]
Age group[b]							
12-13	4,636	6.6% [5.8%, 7.5%]	4,136	6.7% [5.9%, 7.6%]	0.1% [-0.1%, 0.3%]	6.7% [5.9%, 7.6%]	0.1% [-0.1%, 0.3%]
14-17	4,782	20.2% [19.0%, 21.4%]	4,278	20.0% [18.7%, 21.3%]	-0.2% [-0.6%, 0.2%]	20.2% [19.0%, 21.5%]	0.1% [-0.3%, 0.4%]
Race/ethnicity[c]							
Hispanic	2,749	13.4% [12.1%, 14.8%]	2,480	13.4% [12.1%, 14.9%]	0.0% [-0.4%, 0.5%]	13.6% [12.2%, 15.1%]	0.2% [-0.3%, 0.7%]
Non-Hispanic White alone	4,145	14.3% [13.3%, 15.5%]	3,655	14.2% [13.1%, 15.4%]	-0.1% [-0.5%, 0.2%]	14.4% [13.2%, 15.5%]	0.0% [-0.3%, 0.3%]
Non-Hispanic Black alone	1,200	9.7% [7.8%, 11.8%]	1,092	10.0% [8.1%, 12.3%]	0.4% [-0.0%, 0.7%]	10.1% [8.2%, 12.4%]	0.5% [-0.0%, 0.9%]
Non-Hispanic other race or multiple races	897	12.6% [10.5%, 15.1%]	803	12.5% [10.3%, 15.2%]	-0.1% [-0.9%, 0.7%]	12.6% [10.4%, 15.2%]	0.0% [-0.7%, 0.7%]

Table 5-9. Comparison of Wave 4 “ever tobacco-use” estimates between Wave 4 youth who were eligible for Wave 5 youth interview and Wave 4 youth who completed Wave 5 youth interview (Wave 4 Cohort) (continued)*

Wave 4 characteristic[a]	Wave 4 youth who were eligible for Wave 5 youth interview		Wave 4 youth who completed Wave 5 youth interview				
				Before Wave 5 weighting adjustment		After Wave 5 weighting adjustment	
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 5 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	1,340	13.7% [12.0%, 15.7%]	1,181	14.3% [12.4%, 16.4%]	0.5% [-0.1%, 1.2%]	14.5% [12.6%, 16.6%]	0.7% [0.1%, 1.4%]
Midwest	2,051	13.1% [11.7%, 14.6%]	1,810	13.1% [11.7%, 14.8%]	0.0% [-0.4%, 0.4%]	13.3% [11.8%, 15.0%]	0.2% [-0.3 %, 0.6%]
South	3,575	13.5% [12.4%, 14.6%]	3,210	13.3% [12.1%, 14.5%]	-0.2% [-0.7%, 0.2%]	13.4% [12.2%, 14.6%]	-0.1% [-0.5%, 0.3%]
West	2,452	12.8% [11.1%, 14.9%]	2,213	12.6% [10.7%, 14.8%]	-0.2% [-0.7%, 0.2%]	12.7% [10.8%, 15.0%]	-0.1% [-0.5%, 0.3%]

*An ‘ever user’ of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 4 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), bidis, and kreteks.

[a] For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Table 5-9 includes a subset of the youth in Table 5-8 because there are no Wave 4 interview data for the Wave 5 youth who were shadow youth at Wave 4.

[c] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

Table 5-8 compares Wave 4 demographic characteristics between the Wave 5 youth interview respondents and those eligible for the Wave 5 youth interview. The percentages sum to 100 percent[20] over the categories associated with each characteristic. For Wave 4 sex, race/ethnicity, and census region, the "before Wave 5 weighting adjustment" distributions among the Wave 5 youth interview respondents were almost the same as the distributions among those eligible for the Wave 5 youth interview. Similar to the result shown in Table 4-6 for the Wave 1 Cohort, the most noticeable underrepresentation among the Wave 5 youth respondents from the Wave 4 Cohort was the Wave 4 "under 12" age group. Possible reasons for this result were discussed in Section 5.2.1.

For the "after Wave 5 weighting adjustment" differences, the estimated confidence intervals (in the last column of Table 5-8) included zero or were in very close proximity to zero for Wave 4 sex, race/ethnicity, and census region. For Wave 4 age, a slight overrepresentation of the "14-17" age group persisted after Wave 5 weighting adjustments, and there were consequent slight underrepresentations of the "under 12" and "12-13" age groups; however, the differences were not substantively meaningful.

Table 5-9 shows the comparison of Wave 4 "ever tobacco-use" rates between Wave 4 youth respondents who completed the Wave 5 youth interview and all the Wave 4 youth respondents eligible for the Wave 5 youth interview (among the Wave 4 Cohort). Table 5-9 includes a subset of the youth in Table 5-8 because there are no Wave 4 interview data for the Wave 5 youth who were shadow youth at Wave 4.) For both the "before Wave 5 weighting adjustment" and the "after Wave 5 weighting adjustment" estimates, differences between respondents and those eligible for the interview were no more than 0.7 percent. The largest difference was after the Wave 5 weighting adjustments for the Wave 4 "ever tobacco-use" measure among youth from the Northeast census region at Wave 4. The corresponding confidence interval barely excluded zero, unlike all of the other confidence intervals in Table 5-9.

Overall, these results reflect the use of both demographic variables and tobacco-use measures from Wave 4 for calibrating the Wave 5 single-wave weight for the Wave 4 Cohort. Assuming that the Wave 4 demographic and tobacco-use characteristics in Tables 5-8 and 5-9 are correlated with key tobacco- and health-related outcome measures in Wave 5 of the PATH Study, these results indicate

[20] The sum may be not exactly 100 percent due to rounding.

little if any nonresponse bias in the youth interview estimates for the Wave 4 Cohort due to attrition from Wave 4 to Wave 5.

5.2.3 Comparison of Youth Cigarette-Smoking Estimates to Other National Studies

Tables 5-10 and 5-11 compare estimates for two common measures of youth cigarette smoking based on the Wave 5 youth interview respondents from the Wave 4 Cohort of the PATH Study to similar estimates based on data from NSDUH 2018, NHANES 2017-2018, MTF 2017, NYTS 2019, and NYRBS 2017. To improve comparability, different subgroups were used when computing estimates for different studies. In Table 5-10, comparisons to NHANES and NSDUH are provided by age (12-13, 14-15, and 16-17) and comparisons to MTF are provided by grade (8, 10, and 12). In Table 5-11, estimates from the PATH Study are compared to NYTS and NYRBS estimates by school level (middle school and high school). As mentioned in Section 3.2.4, the data for all five external studies were restricted to ages 12 to 17 to match the PATH Study definition of youth. The primary measure of cigarette smoking among youth is whether the youth has ever tried smoking a cigarette; even one or two puffs. Another measure is current smoking, defined as having smoked at all in the past 30 days. (See Appendix A for details.)

As noted above, PATH Study estimates, accompanied by 95 percent confidence intervals, are shown for various youth subgroups relating to age, grade, and school level. The point estimates for the PATH Study were calculated using the Wave 5 single-wave weight for the Wave 4 Cohort and confidence intervals required the corresponding replicate weights for variance estimation. Point estimates and 95 percent confidence intervals are reported for the other national studies as well.

Table 5-10. Comparison of youth cigarette-smoking estimates by age and grade from PATH Study Wave 5 youth interview (Wave 4 Cohort) and other national studies

Cigarette use by age					Cigarette use by grade			
Wave 5 cigarette-smoking measure and subgroup	PATH Study Wave 5 (2018-2019) sample size	Percentage from PATH Study Wave 5 (2018-2019) [95% confidence interval]	Percentage from NSDUH 2018 [95% confidence interval]	Percentage from NHANES 2017-2018 [95% confidence interval]	Wave 5 cigarette-smoking measure and subgroup	PATH Study Wave 5 (2018-2019) sample size	Percentage from PATH Study Wave 5 (2018-2019) [95% confidence interval]	Percentage from MTF 2017 [95% confidence interval]
Ever tried cigarette smoking, even one or two puffs					**Ever tried cigarette smoking, even one or two puffs**			
Age 12-13, overall	3,175	2.1% [1.6%, 2.8%]	2.7% [2.2%, 3.3%]	2.8% [1.4%, 5.6%]	8th Grade	1,706	3.8% [2.8%, 5.1%]	9.4% [8.6%, 10.3%]
Age 12-13, male	1,690	2.1% [1.5%, 2.9%]	2.0% [1.5%, 2.7%]	2.3% [0.8%, 6.4%]	8th Grade, male	922	3.8% [2.7%, 5.4%]	9.0% [8.1%, 10.1%]
Age 12-13, female	1,470	2.1% [1.4%, 3.2%]	3.4% [2.5%, 4.6%]	3.4% [1.4%, 7.9%]	8th Grade, female	775	3.8% [2.5%, 5.8%]	9.6% [8.6%, 10.8%]
Age 14-15, overall	4,327	6.9% [6.0%, 7.9%]	8.9% [7.8%, 10.2%]	12.5% [7.7%, 19.7%]	10th Grade	2,135	9.1% [7.9%, 10.4%]	15.8% [14.7%, 17.0%]
Age 14-15, male	2,284	7.1% [6.0%, 8.3%]	8.5% [7.1%, 10.3%]	16.7% [9.5%, 27.7%]	10th Grade, male	1,058	9.3% [7.7%, 11.2%]	15.9% [14.5%, 17.4%]
Age 14-15, female	2,026	6.7% [5.6%, 8.0%]	9.2% [7.6%, 11.2%]	8.0% [3.4%, 17.8%]	10th Grade, female	1,067	8.8% [7.0%, 11.0%]	15.4% [14.1%, 16.8%]
Age 16-17, overall	4,444	15.5% [14.3%, 16.6%]	17.2% [15.8%, 18.7%]	18.8% [12.2%, 28.0%]	12th Grade	1,266	15.2% [13.2%, 17.5%]	23.2% [21.2%, 25.3%]
Age 16-17, male	2,237	16.5% [15.0%, 18.1%]	18.4% [16.4%, 20.6%]	23.4% [13.3%, 37.8%]	12th Grade, male	587	15.3% [12.3%, 18.8%]	24.5% [21.9%, 27.2%]
Age 16-17, female	2,191	14.3% [12.6%, 16.1%]	16.0% [14.4%, 17.8%]	14.3% [8.2%, 23.7%]	12th Grade, female	674	14.9% [12.4%, 17.9%]	21.9% [19.8%, 24.2%]
Has smoked cigarettes in past 30 days, age 12–17	11,962	2.3% [1.9%, 2.7%]	2.7% [2.3%, 3.1%]	2.0% [0.9%, 4.1%]	**Has smoked cigarettes in past 30 days, 12th grade**	1,269	3.4% [2.5%, 4.6%]	7.2% [6.1%, 8.5%]

Table 5-11. Comparison of youth cigarette-smoking estimates by school level from PATH Study Wave 5 youth interview (Wave 4 Cohort) and other national studies

Wave 5 cigarette-smoking measure and subgroup	PATH Study Wave 5 (2018-2019) sample size	Percentage from PATH Study Wave 5 (2018-2019) [95% confidence interval]	Percentage from NYTS 2019 [95% confidence interval]	Percentage from NYRBS 2017 [95% confidence interval]
Ever tried cigarette smoking, even one or two puffs				
Middle school	3,701	2.9% [2.3%, 3.6%]	9.0% [7.7%, 10.4%]	N/A
Middle school, male	1,984	3.4% [2.7%, 4.3%]	9.2% [7.9%, 10.6%]	N/A
Middle school, female	1,699	2.4% [1.7%, 3.4%]	8.8% [7.2%, 10.7%]	N/A
High school	7,363	10.6% [9.9%, 11.4%]	21.3% [17.9%, 25.1%]	27.0% [24.1%, 30.2%]
High school, male	3,728	11.1% [10.1%, 12.1%]	24.5% [19.1%, 30.9%]	28.7% [25.8%, 31.8%]
High school, female	3,612	10.1% [9.0%, 11.3%]	17.8% [15.5%, 20.4%]	25.5% [22.1%, 29.2%]
Has smoked cigarettes in past 30 days, high school	7,376	2.9% [2.4%, 3.3%]	5.0% [3.8%, 6.4%]	7.8% [6.4%, 9.6%]

When comparing by age, the PATH Study cigarette-smoking estimates were generally lower than the estimates from the other studies. However, the confidence intervals overlapped for nearly all estimates between Wave 5 of the PATH Study, NHANES 2017-2018, and NSDUH 2018 presented in Table 5-10. None of the confidence intervals overlapped between Wave 5 of the PATH Study and MTF 2017 for the grade-level estimates of cigarette smoking in Table 5-10, with the MTF estimates being consistently higher. Similarly, the PATH Study estimates of cigarette smoking by school level were well below those from the comparison studies in Table 5-11, with none of the NYTS 2019 or NYRBS 2017 estimates having confidence intervals that overlapped with those from the PATH Study.

It is unclear whether the PATH Study estimates of youth cigarette smoking are generally lower due to nonresponse bias in one or more of the estimates compared, or for other reasons. The disparities in Table 5-10 and 5-11 can be explained by a number of potential reasons, similar to those discussed in Section 5.1.3. In addition to the varying degrees of sampling and measurement errors, the surveys differ in mode of administration, context, question order and wording, and year of data collection. The PATH Study, NHANES, and NSDUH use ACASI for the questions about tobacco use by youth, and these are administered individually in a household or mobile examination center setting. The NYTS, MTF, and NYRBS are all school-based surveys self-administered in the classroom during normal school hours. The NYTS is completed by students in grades 6 through 12 using an electronic device, whereas both the MTF and NYRBS are pencil-and-paper surveys.[21] The MTF is administered to 8th, 10th, and 12th graders and the NYRBS is a survey of only high school students. Currivan et al. (2004) found that even when telephone ACASI was used, estimates of youth smoking prevalence were lower for a telephone survey of youth smoking than for a school-based survey of the same population (see also Fowler and Stringfellow, 2001, for a discussion of higher smoking rates in school-based surveys). In addition, higher estimates of cigarette use in school-based studies as compared to home-based studies do not necessarily mean the school-based estimates are more accurate. As noted in Boyd et al. (2020), school-based studies are often implemented in a group setting and the fact that peers could observe their answers may make respondents feel it more socially desirable to over-report risky behaviors.

The contexts and purposes of these surveys also differ: NHANES is a general health survey; and NSDUH and MTF are cross-sectional surveys on substance use (including tobacco use) and health. Unlike the cross-sectional prevalence surveys, the PATH Study uses a longitudinal cohort design to assess within-person changes and between-person differences in tobacco-use behaviors and health over time. Other differences among the questions used in the instruments of these different studies are outlined in Appendix A.

5.2.4 Comparison of Youth ENDS-Use Estimates to Other National Studies

Tables 5-12 and 5-13 compare estimates for youth ENDS use based on the Wave 5 youth interview respondents from the Wave 4 Cohort of the PATH Study to similar estimates based on data from MTF 2017, NYTS 2019, and NYRBS 2017. NYTS 2019 asks about "e-cigarette" use, MTF 2017 asks respondents if they have ever "vaped nicotine," and NYRBS 2017 asks about use of "electronic vapor products." However, the definitions provided to respondents immediately prior to asking these questions are virtually identical to the definition of "electronic nicotine products" provided to PATH Study respondents.

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[21] The MTF 2017 data used in this report were collected via pencil and paper. In 2019, MTF began transitioning from pencil-and-paper administration to electronic device administration.

To improve comparability, estimates were computed for grades 8, 10, and 12 to compare to MTF 2017 (Table 5-12), and for middle school and high school to compare to NYTS 2019 and for high school to compare to NYRBS 2017 (Table 5-13). The primary measure of ENDS use among youth is whether the youth has ever tried using ENDS. A second measure is current ENDS use, defined as having used ENDS at all in the past 30 days. (See Appendix B for details.)

As noted above, PATH Study estimates and 95 percent confidence intervals are provided for various youth subgroups relating to grade and school level. The point estimates for the PATH Study were calculated using the Wave 5 single-wave weight for the Wave 4 Cohort and confidence intervals required the corresponding replicate weights for variance estimation. Point estimates and 95 percent confidence intervals are reported for MTF 2017, NYTS 2019, and NYRBS 2017.

Table 5-12. Comparison of youth ENDS-use estimates by grade from PATH Study Wave 5 youth interview (Wave 4 Cohort) and MTF 2017

Wave 5 ENDS-use measure and subgroup	PATH Study Wave 5 (2018-2019) sample size	Percentage from PATH Study Wave 5 (2018-2019) [95% confidence interval]	Percentage from MTF 2017 [95% confidence interval]
Ever tried ENDS, even one or two puffs			
8th Grade	1,698	9.4% [8.1%, 10.9%]	11.0% [10.0%, 12.2%]
8th Grade, male	917	9.2% [7.4%, 11.5%]	11.5% [9.9%, 13.4%]
8th Grade , female	773	9.6% [7.6%, 11.9%]	10.7% [9.3%, 12.2%]
10th Grade	2,126	25.3% [23.2%, 27.6%]	22.1% [20.3%, 24.0%]
10th Grade , male	1,054	27.1% [24.2%, 30.3%]	20.9% [18.6%, 23.4%]
10th Grade, female	1,063	23.8% [20.9%, 26.9%]	23.1% [20.8%, 25.5%]
12th Grade	1,264	39.4% [36.3%, 42.7%]	23.9% [20.9%, 27.2%]
12th Grade, male	587	39.0% [34.7%, 43.6%]	27.0% [22.2%, 32.4%]
12th Grade, female	672	39.9% [35.7%, 44.3%]	22.1% [18.8%, 25.9%]
Has used ENDS in past 30 days, 12th grade	1,267	18.4% [16.1%, 20.9%]	9.1% [7.3%, 11.3%]

Table 5-13. Comparison of youth ENDS-use estimates by school level from PATH Study Wave 5 youth interview (Wave 4 Cohort) and other national studies

Wave 5 ENDS-use measure and subgroup	PATH Study Wave 5 (2018-2019) sample size	Percentage from PATH Study Wave 5 (2018-2019) [95% confidence interval]	Percentage from NYTS 2019 [95% confidence interval]	Percentage from NYRBS 2017 [95% confidence interval]
Ever tried ENDS, even one or two puffs				
Middle school	3,691	6.6% [5.8%, 7.4%]	21.6% [20.0%, 23.4%]	N/A
Middle school, male	1,980	7.5% [6.4%, 8.7%]	21.3% [19.2%, 23.5%]	N/A
Middle school, female	1,694	5.5% [4.6%, 6.7%]	22.0% [20.0%, 24.0%]	N/A
High school	7,339	27.8% [26.6%, 29.0%]	45.8% [43.0%, 48.7%]	40.6% [37.7%, 43.6%]
High school, male	3,721	28.3% [26.7%, 29.9%]	46.2% [42.1%, 50.4%]	42.8% [40.5%, 45.2%]
High school, female	3,597	27.3% [25.8%, 28.9%]	45.6% [42.6%, 48.5%]	38.5% [34.4%, 42.7%]
Has used ENDS in past 30 days, high school	7,361	12.8% [12.0%, 13.6%]	26.2% [24.1%, 28.5%]	11.9% [10.1%, 14.0%]

The PATH Study estimates of ever use of ENDS were higher than those from MTF 2017 for 12th grade students, with no overlap in confidence intervals. However, with the exception of a higher PATH Study estimate for 10th grade males, estimates of ever use of ENDS among 8th and 10th graders were similar between the two studies with all confidence intervals overlapping. The PATH Study estimate of past 30-day ENDS use among 12th graders was higher than the MTF estimate, with no overlap in confidence intervals. Estimates of ever use of ENDS from the PATH Study by school level were noticeably lower than those from both the NYTS 2019 and NYRBS 2017. However, the NYTS and NYRBS ever use estimates had overlapping confidence intervals for the high-school subgroups. The PATH Study and NYRBS estimates of ENDS use in the past 30 days among high-school students were similar, but both were lower than the NYTS estimate. As discussed in Section 5.2.3, some differences in the estimates may be attributable to differences in context, question order and wording, mode of administration, time frames of the surveys, and measurement error. In particular, MTF, NYTS, and NYRBS are administered in the school whereas the PATH Study generally is administered in the home. It is uncertain which factors are the key drivers of the differences in ENDS-use estimates observed between the PATH Study and other national studies; however, the survey reference year is likely important given changes in the prevalence of ENDS use in recent years.

6. Results of Biospecimen Collection Nonresponse Analyses at Wave 5

6.1 Adult Biospecimen Collection Response Rates Conditioning on Wave 5 Interview Response

This section discusses unweighted response rates for the urine specimen collection and blood specimen collection among Wave 5 adult interview respondents. Each response rate was calculated as a ratio, with the numerator being the number of Wave 5 adults who provided a biospecimen and the denominator being the number of Wave 5 adult respondents from whom a biospecimen was requested.

Tables 6-1 and 6-2 show the unweighted response rates for urine collection and blood collection, respectively. In addition to the overall rows, the response rates are tabulated by Wave 5 socio-demographic characteristics and tobacco-use status, as well as recruitment wave for both urine and blood collection. For urine collection, response rates are also presented by whether Wave 5 was the first time the respondent had completed a PATH Study adult interview. Adults with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic.

As shown in Table 6-1, the unweighted response rate for urine collection was 96.0 percent overall, with little variation except by age group and whether or not it was the participant's first time completing an adult interview. Adults ages 18 to 24 were the least likely to provide urine specimens (93.6 percent) among the age groups and those completing their first adult interview were less likely to provide urine specimens (92.8 percent) than other adult interview respondents (97.8 percent); these are highly correlated characteristics because all first-time adult interview respondents were in the "18-24" age group. This last finding is not surprising because all of the adults who had previously completed a PATH Study adult interview and were asked to provide a urine specimen at Wave 5, were members of the Wave 1 Biomarker Core (meaning they had provided urine at Wave 1). Despite the differences in response propensities across subgroups, the response rates for the subgroups examined were no less than 92.8 percent.

Table 6-1. Unweighted urine collection response rates among Wave 5 adult interview respondents selected to provide a urine specimen

Characteristic[a]	Number of adult respondents selected to provide urine (n)	Number of adult respondents who provided urine (n)	Unweighted response rate for urine collection (%)
Overall	12,600	12,102	96.0
Wave 5 sex			
Male	6,479	6,237	96.3
Female	6,105	5,851	95.8
Wave 5 age group			
18-24	5,126	4,798	93.6
25-44	4,342	4,243	97.7
45-64	2,441	2,396	98.2
65 and above	691	665	96.2
Wave 5 race/ethnicity[b]			
Hispanic	2,764	2,637	95.4
Non-Hispanic White alone	6,731	6,462	96.0
Non-Hispanic Black alone	1,910	1,871	98.0
Non-Hispanic other race or multiple races	1,001	946	94.5
Wave 5 education			
Less than high school or GED	2,914	2,828	97.0
High school	3,870	3,658	94.5
Some college, no degree	4,160	4,020	96.6
Bachelor degree and above	1,594	1,540	96.6
Wave 5 health insurance			
Yes	9,905	9,511	96.0
No	2,524	2,439	96.6
Wave 5 tobacco-use status[c]			
Current established user	5,800	5,667	97.7
Not current established user	6,770	6,405	94.6
Wave 5 first time completing adult interview			
Yes	4,433	4,114	92.8
No	8,167	7,988	97.8
Recruitment wave			
Wave 1	11,703	11,270	96.3
Wave 4	897	832	92.8

[a] With the exception of recruitment wave, all the characteristics apply to the Wave 5 adult interview responses; however, respondents are asked to self-report their sex and race/ethnicity only once, during their first youth or adult interview. For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

[c] A 'current established user' of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 5 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

Table 6-2. Unweighted blood collection response rates among Wave 5 first-time adult interview respondents

Characteristic[a]	Number of first-time adult interview respondents at Wave 5 (n)	Number of blood providers among first-time adult interview respondents at Wave 5 (n)	Unweighted response rate for blood collection (%)
Overall	4,433	2,040	46.0
Wave 5 sex			
Male	2,267	1,012	44.6
Female	2,153	1,023	47.5
Wave 5 race/ethnicity[b]			
Hispanic	1,356	693	51.1
Non-Hispanic White alone	1,993	895	44.9
Non-Hispanic Black alone	591	246	41.6
Non-Hispanic other race or multiple races	408	169	41.4
Wave 5 education			
Less than high school or GED	1,039	494	47.5
High school	2,074	918	44.3
Some college, no degree	1,273	613	48.2
Bachelor degree and above	18	3	16.7
Wave 5 health insurance			
Yes	3,325	1,563	47.0
No	999	437	43.7
Wave 5 tobacco-use status[c]			
Current established user	863	388	45.0
Not current established user	3,554	1,647	46.3
Recruitment wave			
Wave 1	3,254	1,529	47.0
Wave 4	897	400	44.6

[a] With the exception of recruitment wave, all the characteristics apply to the Wave 5 adult interview; however, respondents are asked to self-report their sex and race/ethnicity only once, during their first youth or adult interview. For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

[c] A 'current established user' of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 5 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

As shown in Table 6-2, the unweighted response rate for blood collection among first-time adult interview respondents was 46.0 percent. Females (47.5 percent) were more likely to provide blood specimens than males (44.6 percent). Hispanic adults (51.1 percent) were more likely to provide blood specimens than non-Hispanic adults (with unweighted response rates ranging from 41.4 to 44.9 percent by race). Adults with at least a bachelor's degree had a low response rate to the blood collection but there were very few first-time adult interview respondents in this subgroup. Blood collection response rates were also higher for adults with health insurance (47.0 percent) compared to those without (43.7 percent). The modest overall response rate for blood collection combined with moderate variation in response propensities by subgroups indicates some potential for nonresponse bias with respect to these data when unweighted.

6.2 Youth Urine Specimen Collection Response Rates Conditioning on Wave 5 Interview Response

This section discusses unweighted response rates for the urine specimen collection among Wave 5 youth interview respondents who were asked to provide a biospecimen. Each response rate was calculated as a ratio, with the numerator being the number of Wave 5 youth who provided a urine specimen and the denominator being the number of Wave 5 youth interview respondents from whom a biospecimen was requested.

Table 6-3 shows the unweighted response rates for urine collection. In addition to the overall row, the response rates are tabulated by Wave 5 demographic characteristics and tobacco-use status, as well as recruitment wave. Youth with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic.

Table 6-3. Unweighted urine collection response rates among Wave 5 youth interview respondents

Characteristic[a]	Number of respondents selected to provide urine (n)	Number of respondents who provided urine (n)	Unweighted response rate for urine collection (%)
Overall	11,214	10,584	94.4
Wave 5 sex			
Male	5,960	5,717	95.9
Female	5,209	4,826	92.6
Wave 5 age group			
12-13	3,175	2,789	87.8
14-17	8,039	7,795	97.0
Wave 5 race/ethnicity[b]			
Hispanic	3,287	3,136	95.4
Non-Hispanic White alone	4,921	4,577	93.0
Non-Hispanic Black alone	1,411	1,373	97.3
Non-Hispanic other race or multiple races	1,061	999	94.2
Wave 5 tobacco-use status[c]			
Ever user	2,695	2,622	97.3
Never user	8,159	7,615	93.3
Recruitment wave			
Wave 1	6,132	5,948	97.0
Wave 4	5,082	4,636	91.2

[a] With the exception of recruitment wave, all the characteristics apply to the Wave 5 youth interview responses; however, respondents are asked to self-report their sex and race/ethnicity only once, during their first youth interview. For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

[b] Hispanic refers to Hispanic or Latino; Black refers to Black or African-American.

[c] An 'ever user' of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 5 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), bidis, and kreteks.

The unweighted response rate for youth urine collection was 94.4 percent overall, with the "14-17" age group more likely to provide a specimen (97.0 percent) than the "12-13" age group (87.8 percent). This is likely because most of the youth asked to provide a urine specimen for the first time at Wave 5 (i.e., those who completed their first youth interview at Wave 4.5 or Wave 5) were in the younger age group and most of those who consented to urine collection at Wave 4 were in the older age group. Males (95.9 percent) and youth who had ever used tobacco (97.3 percent) were more likely to provide a urine specimen than females (92.6 percent) and never users of tobacco (93.3 percent). The response rates varied by 4.3 percentage points across race/ethnicity groups, with non-Hispanic Black alone youth being the most likely to provide a specimen (97.3 percent). Youth from the Wave 4 replenishment sample were less likely to provide urine specimens.

7. Summary of Findings

This chapter summarizes the PATH Study's Wave 5 nonresponse bias analysis findings separately for the Wave 1 Cohort interview data, the Wave 4 Cohort interview data, and the Wave 5 biospecimen collections (ignoring cohort membership). Each of the following components of the analysis is reviewed:

- Wave 5 interview response rates.
- The national representativeness of Wave 5 interview respondents and statistical weighting adjustments to reduce potential nonresponse bias.
- Wave 5 biospecimen collection response rates among Wave 5 interview respondents.

7.1 Wave 1 Cohort

7.1.1 Adult Interview

The Wave 1 Cohort unweighted all-waves response rate was slightly lower than the weighted all-waves response rate for the Wave 5 adult interview (conditioning on Wave 1 response) overall and by subgroups. There were also differences in weighted all-waves response rates across subgroups.

- The weighted overall all-waves response rate was 65.1 percent.
- The weighted all-waves response rate was lower among males (62.5 percent) than among females (67.3 percent); this pattern is consistent with most household surveys.
- The Wave 1 "18-24" age group had a lower weighted all-waves response rate (59.2 percent) than the older age groups (which had response rates ranging from 64.2 percent to 68.8 percent).
- The weighted all-waves response rates were 5.0 percentage points lower for Wave 1 adult current established tobacco users than for Wave 1 adults who were not current established tobacco users, and 6.0 percentage points lower for Wave 1 youth who had ever used tobacco than for Wave 1 youth who were never users.

However, differences in all-waves response rates do not necessarily indicate nonresponse bias in the Wave 5 adult estimates for the Wave 1 Cohort.

As a result of comparing select demographic and socio-economic characteristics between the Wave 5 adult interview all-waves respondents and those eligible to be Wave 5 adult interview all-waves respondents, the noticeable underrepresentation due to nonresponse since the formation of the Wave 1 Cohort was for the male population, and the most evident overrepresentations were of adults ages 45 to 64 at Wave 1 and those with at least a bachelor's degree at Wave 1. When similar comparisons were made with respect to tobacco-use characteristics, the differences generally indicated an underrepresentation of Wave 1 tobacco users among the Wave 5 adult interview all-waves respondents.

After Wave 5 longitudinal weighting adjustments, the differences between the Wave 5 adult interview all-waves respondents and those eligible to be Wave 5 adult interview all-waves respondent were negligible with three exceptions: (1) a slight underrepresentation of Wave 1 current established tobacco users among those who were non-Hispanic adults of other or multiple races at Wave 1; (2) a slight underrepresentation of Wave 1 ever tobacco users among non-Hispanic White alone adults who were youth at Wave 1; and (3) a slight overrepresentation of Wave 1 ever tobacco users among non-Hispanic Black alone adults who were youth at Wave 1. Despite the use of both demographic variables and tobacco-use measures from Wave 1 to calibrate the Wave 5 all-waves weight, these results hint at the challenges faced by panel studies as waves of nonresponse accrue.

7.1.2 Youth Interview

The Wave 1 Cohort unweighted all-waves response rate was similar to the weighted all-waves response rate for the Wave 5 youth interview (conditioning on Wave 1 response) overall and by subgroups, although there were differences in weighted all-waves response rates across subgroups.

- The weighted overall all-waves response rate was 68.4 percent.
- Moderate differences (i.e., of no more than 5 percent) were found by Wave 1 sex, race/ethnicity, census region, and (for those who were youth at Wave 1) ever use of tobacco.
- The weighted all-waves response rate for those under age 12 at Wave 1 (66.7 percent) was noticeably lower than for those ages 12 to 13 (73.5 percent). This is likely because the persons in the youngest age group were shadow youth at Wave 1 and were not required to complete a PATH Study interview at that time in order to become a study participant. These shadow youth were first asked to complete a youth interview at a

later wave once they turned twelve years old. (A similar response rate pattern has been observed in previous waves.)

These differences in response rates do not necessarily indicate nonresponse bias in the Wave 5 youth estimates for the Wave 1 Cohort.

As a result of comparing select demographic and tobacco-use characteristics between the Wave 5 youth interview all-waves respondents and those eligible to be Wave 5 youth interview all-waves respondents, the only slight underrepresentations due to nonresponse since the formation of the Wave 1 Cohort were for the Wave 1 "under 12" age group and the Wave 1 ever users of tobacco among females who were youth at Wave 1.

After Wave 5 longitudinal weighting adjustments, the differences between the Wave 5 youth interview all-waves respondents and those eligible to be Wave 5 youth interview all-waves respondents were negligible; however, the underrepresentation of Wave 1 shadow youth was not entirely eliminated. Separate analyses (not shown here) confirmed that the Wave 1 age group distribution was preserved when the Wave 5 all-waves weight was applied to all Wave 5 all-waves respondents (i.e., youth and adults) together. This suggests that the findings for Wave 5 youth need to be interpreted in light of the fact that not all study participants were exactly 5 years older at the time of Wave 5, compared to their Wave 1 age.

7.2 Wave 4 Cohort

7.2.1 Adult Interview

The Wave 4 Cohort unweighted single-wave response rate was similar to the weighted single-wave response rate for the Wave 5 adult interview (conditioning on Wave 4 response) overall and for most subgroups. There were differences in weighted single-wave response rates across subgroups.

- The weighted overall single-wave response rate was 88.0 percent.
- Similar to the weighted all-waves response rates for Wave 1 Cohort adults, the weighted single-wave response rates for Wave 4 Cohort adults were lower for males (86.9 percent) than females (89.1 percent) and for the Wave 4 "18-24" age group (82.5 percent) than the other age groups (which had response rates ranging from 85.0 percent to 90.3 percent).

- The weighted single-wave response rate was higher for Wave 5 adults who responded to the Wave 4.5 special data collection (91.3 percent) than for those who did not respond (41.1 percent) or who were not a part of Wave 4.5 (88.1 percent).
- There was also a large difference in the weighted single-wave response rate for those recruited in Wave 4 (77.0 percent) compared to those recruited in Wave 1 (89.9 percent). The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years and all completed an interview at Wave 4. The Wave 4 replenishment sample members of the Wave 4 Cohort do not have the same history of study participation.

However, differences in single-wave response rates do not necessarily indicate nonresponse bias in the Wave 5 adult estimates for the Wave 4 Cohort.

As a result of comparing select demographic and socio-economic characteristics between the Wave 5 adult interview respondents and those eligible for the Wave 5 adult interview, the most noticeable overrepresentation due to panel attrition among the Wave 4 Cohort was for the Wave 4 "45-64" age group, with corresponding slight underrepresentations for the "under 18" and "18-24" age groups. When similar comparisons were made with respect to tobacco-use characteristics, the differences indicated slight underrepresentation of Wave 4 adult current established tobacco users among the Wave 5 adult respondents.

After Wave 5 longitudinal weighting adjustments, the differences between those eligible to be Wave 5 adult interview respondents and Wave 5 adult interview respondents from the Wave 4 Cohort were mostly negligible and all of the corresponding 95 percent confidence intervals included zero. These results reflect the use of both demographic variables and tobacco-use measures from Wave 4 to calibrate the Wave 5 single-wave weight.

There is no evidence of nonresponse bias in the Wave 4 Cohort with respect to current cigarette-smoking behavior among adults, in the sense that the PATH Study's estimates were all within the range of estimates from other national studies. In general, the estimates of adult current cigarette-smoking rates based on the Wave 4 Cohort at Wave 5 were similar to estimates from NHANES 2017-2018 and NSDUH 2018, and higher than estimates from TUS-CPS 2018-2019 and NHIS 2018.

Similar to the findings for current cigarette smoking, the PATH Study's estimates of adult current ENDS use based on the Wave 4 Cohort at Wave 5 were higher than those from TUS-CPS 2018-

2019 and NHIS 2018. In particular, the current ENDS-use estimate for the "18-24" age group was higher for Wave 5 of the PATH Study (13.7 percent) than TUS-CPS 2018-2019 (5.3 percent) or NHIS 2018 (7.6 percent). However, given that the estimates of past 30-day ENDS use among 12th graders were above 9 percent for both the PATH Study at Wave 5 and MTF 2017, the PATH Study's current ENDS-use estimate for 18- to 24-year-olds appears reasonable. The disparities in ENDS-use estimates may also be explained by a number of factors including varying degrees of sampling and measurement errors as well as differences in question order, context, mode of administration, and year of data collection.

7.2.2 Youth Interview

The Wave 4 Cohort unweighted single-wave response rate was similar to the weighted single-wave response rate for the Wave 5 youth interview (conditioning on Wave 4 response) overall and by subgroups, although there were differences in weighted response rates across subgroups.

- The weighted overall single-wave response rate was 83.5 percent.
- The weighted single-wave response rates varied by 16.6 percentage points for Wave 4 age and 14.3 percentage points for recruitment wave. Response rates were lower for the Wave 4 "under 12" age group (72.6 percent) and among youth recruited in Wave 4 (76.5 percent) – these are highly correlated characteristics because all Wave 4 shadow youth in the Wave 4 Cohort were from the Wave 4 replenishment sample. Shadow youth were not required to complete a PATH Study interview at the time of their enrollment in order to become a study participant. Wave 4 shadow youth were first asked to complete a youth interview at the Wave 4.5 special data collection or at Wave 5, upon turning 12 years old.
- The weighted single-wave response rate was higher for Wave 5 youth who responded to the Wave 4.5 special data collection (93.6 percent) than for those who did not respond (25.4 percent) or who were not a part of Wave 4.5 (70.8 percent).

These differences in response rates do not necessarily indicate nonresponse bias in the Wave 5 youth estimates for the Wave 4 Cohort.

As a result of comparing select demographic and tobacco-use characteristics between the Wave 5 youth interview respondents and those eligible for the Wave 5 youth interview, the noticeable underrepresentation due to panel attrition among the Wave 4 Cohort was for the Wave 4 "under 12" age group, i.e., Wave 4 shadow youth.

After Wave 5 longitudinal weighting adjustments, the underrepresentation of Wave 4 shadow youth was almost eliminated. However, the estimate of Wave 4 ever tobacco use among youth from the Northeast census region at Wave 4 was slightly higher for the Wave 5 youth respondents than for those eligible for the Wave 5 youth interview.

When comparing by age, grade, or school level, the PATH Study's youth cigarette-smoking estimates based on the Wave 4 Cohort at Wave 5 were generally lower than the estimates from the other studies. Although the confidence intervals overlapped between Wave 5 of the PATH Study, NHANES 2017-2018, and NSDUH 2018 for most estimates by age, none of the confidence intervals overlapped between Wave 5 of the PATH Study, MTF 2017, NYTS 2019, and NYRBS 2017 for estimates by grade or school level.

Based on the Wave 4 Cohort, the differences in youth ENDS-use estimates between Wave 5 of the PATH Study and other national studies were not always in the same direction. The PATH Study estimates of ever ENDS use and past 30-day ENDS use at Wave 5 were higher for youth in the 12th grade than estimates from MTF 2017 and none of the confidence intervals overlapped. However, with the exception of a higher PATH Study estimate for 10th grade males, the ever ENDS-use confidence intervals for youth in the 8th and 10th grades all overlapped. In comparison to NYTS 2019 and NYRBS 2017 by school level, the PATH Study estimates of ever ENDS use at Wave 5 were noticeably lower. The past 30-day ENDS use estimates among high school students were similar for Wave 5 of the PATH Study and NYRBS 2017, but both were lower than the NYTS 2019 estimate. It is uncertain whether the differences are due to nonresponse bias in one or more of the studies compared or a variety of other differences between the studies, including the reference year.

7.3 Unweighted Biospecimen Response Rates

7.3.1 Adult Biospecimen Collections

Some Wave 5 adult interview respondents were asked to provide biospecimens. Members of the Wave 1 Biomarker Core were asked to provide a urine specimen and all first-time adult interview respondents were asked to provide urine and blood specimens. Unweighted response rates were calculated conditioning on Wave 5 interview completion.

- The unweighted overall response rate for urine collection was 96.0 percent.
- The unweighted urine collection response rate was noticeably lower for those completing the adult interview for the first time (92.8 percent).
- The unweighted overall response rate for blood collection was 46.0 percent.
- Hispanic adults were more likely to provide blood specimens (51.1 percent) than non-Hispanic adults.

7.3.2 Youth Urine Specimen Collection

A subset of Wave 5 youth interview respondents were asked to provide urine specimens: those who first completed a youth interview at Wave 4.5 or Wave 5, and those who consented to urine collection at Wave 4. Unweighted response rates were calculated conditioning on Wave 5 interview completion. The unweighted overall response rate for urine collection was 94.4 percent. Youth who were 14 to 17 years old at Wave 5 were more likely to provide a urine specimen (97.0 percent) than those who were 12 to 13 years old (87.8 percent); however, these age ranges essentially distinguish the youth who had consented to urine collection previously from the youth who were asked to provide a urine specimen for the first time.

7.4 General Conclusions

Assuming that the demographic, socio-economic, and tobacco-use characteristics examined in this report are correlated with key tobacco- and health-related outcome measures, these results raise no serious concern about potential nonresponse bias in estimates from Wave 5 of the PATH Study. However, they hint at the challenges faced by cohort studies as attrition accumulates and confirm the timeliness of the introduction of the Wave 4 replenishment sample to the study.

The findings for both the Wave 1 Cohort and Wave 4 Cohort suggest that the longitudinal weighting adjustments satisfactorily address the potential bias due to nonresponse. However, at Wave 5, the Wave 4 Cohort produces adult estimates of current ENDS use that are higher than those from other national studies, and youth estimates of ever cigarette smoking that are lower than those from other national studies. Comparisons of youth ENDS-use estimates across studies produced mixed findings depending on the outcome measure (ever use vs. past 30-day use) and subgroup. It is difficult to determine the primary reason(s) for these results due to the many differences between the PATH Study and the surveys compared.

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Appendix A
Cigarette-Smoking Questions in the PATH Study and Other Surveys

Table A-1 lists the questions used to ask about the current cigarette-smoking status of adults in the PATH Study as well as in the surveys used for comparison and describes the populations included in the estimates from those surveys.

Note that although the questions used to define current cigarette smoking are similar among the surveys, small differences could have an effect on the answers given. For example, in the PATH Study, the question used to establish whether a respondent has smoked at least 100 cigarettes in his or her lifetime has closed response categories:

"How many cigarettes have you smoked in your entire life? A pack usually has 20 cigarettes in it."

For adults:

1. 1 or more puffs but never a whole cigarette;
2. 1 to 10 cigarettes (about ½ pack total);
3. 11 to 20 cigarettes (about ½ pack to 1 pack);
4. 21 to 50 cigarettes (more than 1 pack but less than 3 packs);
5. 51 to 99 (more than 2½ packs but less than 5 packs); and
6. 100 or more cigarettes (5 packs or more).

For youth:

1. 1 or more puffs but never a whole cigarette;
2. 1 cigarette;
3. 2 to 10 cigarettes (about ½ pack total);
4. 11 to 20 cigarettes (about ½ pack to 1 pack);
5. 21 to 50 cigarettes (more than 1 pack but less than 3 packs);

6. 51 to 99 (more than 2½ packs but less than 5 packs); and
7. 100 or more cigarettes (5 packs or more).

In TUS-CPS, NHIS, NHANES, and NSDUH, however, the question "Have you smoked at least 100 cigarettes in your entire life?" calls for a yes/no response.

Positioning of the questions also differs among the surveys. In the PATH Study, the cigarette-smoking questions are near the beginning of the adult questionnaire, and the respondent knows that the questionnaire is about tobacco-use behaviors. In TUS-CPS, the cigarette-smoking questions are near the beginning of the adult questionnaire on tobacco, but the survey is administered as part of the CPS. In NHIS, the cigarette-smoking questions follow a long series of questions on health problems (breathing problems, diabetes, hernias, hemorrhoids, etc.). These differences among the surveys may be associated with differences in responses to the cigarette-smoking questions.

Table A-2 lists the questions used to ask about youth cigarette smoking in the PATH Study as well as in the surveys used for comparison and describes the populations included in the estimates from those surveys.

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study, TUS-CPS, NHIS, NHANES, and NSDUH

PATH Study	TUS-CPS*	NHIS	NHANES	NSDUH (original definition)	NSDUH (modified definition)**
Questions to define current cigarette smoking (answers defining current cigarette smoking given in parentheses)					
“How many cigarettes have you smoked in your entire life? A pack usually has 20 cigarettes in it.” (Wave 1 – Wave 5, 100 or more cigarettes (5 packs or more)) “Have you ever smoked a cigarette, even one or two puffs?” (Wave 1, Wave 4, Wave 5 adult, yes) “Have you ever tried cigarette smoking, even one or two puffs?” (Wave 1 – Wave 4.5 youth, yes) “In the past 30 days, have you smoked a cigarette, even one or two puffs?” (Wave 2 – Wave 5 adult, yes) “In the past 12 months, have you smoked a cigarette, even one or two puffs” (Wave 2 – Wave 5, yes) “Do you now smoke cigarettes every day, some days, or not at all?” (every day, some days)	“(Have/has) (you/name) smoked at least 100 cigarettes in (your/his/her) entire life?” (yes) “(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?” (every day, some days)	“Have you smoked at least 100 cigarettes in your entire life?” (yes) “Do you NOW smoke cigarettes every day, some days or not at all?” (every day, some days)	“{Have you/Has SP} smoked at least 100 cigarettes in {your/his/her} entire life?” (yes) “{Do you/Does SP} now smoke cigarettes every day, some days, or not at all?” (every day, some days)	“Have you ever smoked part or all of a cigarette?” (yes) “During the past 30 days, have you smoked part or all of a cigarette?” (yes) “How long has it been since you last smoked part or all of a cigarette?” (within the past 30 days)	“Have you smoked at least 100 cigarettes in your entire life?” (yes) “Have you ever smoked part or all of a cigarette?” (yes) “During the past 30 days, have you smoked part or all of a cigarette?” (yes) “How long has it been since you last smoked part or all of a cigarette?” (within the past 30 days)

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study, TUS-CPS, NHIS, NHANES, and NSDUH (continued)

PATH Study	TUS-CPS*	NHIS	NHANES	NSDUH (original definition)	NSDUH (modified definition)**
Coding of "Yes" and "No" answers					
If R05R_A_ESTD_CIGS = 1 then "Yes" Else if R05R_A_ESTD_CIGS = 2 then "No"	If PEA1 = 1 and PEA3 in (1 2) then "Yes" Else if PEA1 = 2 or PEA3 = 3 then "No"	If SMKEV = 1 and SMKNOW in (1 2) then "Yes" Else if SMKEV = 2 or SMKNOW = 3 then "No"	If SMQ020 = 1 and SMQ040 in (1 2) then "Yes" Else if SMQ020 = 2 or SMQ040 = 3 then "No"	If CIGEVER = 1 and CIGREC = 1 then "Yes" Else if CIGEVER = 2 or CIGREC in (2 3 4 14 19 29) then "No"	If CIG100LF in (1 5) and CIGEVER = 1 and CIGREC = 1 then "Yes" Else if CIG100LF = 2 or CIGEVER = 2 or CIGREC in (2 3 4 14 19 29) then "No"
Questions to define every-day cigarette smoking (answers defining every-day cigarette smoking given in parentheses)					
"Do you now smoke cigarettes every day, some days, or not at all?" (every day)	"(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?" (every day)	"Do you NOW smoke cigarettes every day, some days or not at all?" (every day)	"{Do you/Does SP} now smoke cigarettes every day, some days, or not at all?" (every day)	"During the past 30 days, have you smoked part or all of a cigarette?" (yes) "How long has it been since you last smoked part or all of a cigarette?" (within the past 30 days) "What is your best estimate of the number of days you smoked part or all of a cigarette during the past 30 days?" (all 30 days) "During the past 30 days, that is since [DATEFILL], on how many days did you smoke part or all of a cigarette?" (30)	"During the past 30 days, have you smoked part or all of a cigarette?" (yes) "How long has it been since you last smoked part or all of a cigarette?" (within the past 30 days) "What is your best estimate of the number of days you smoked part or all of a cigarette during the past 30 days?" (all 30 days) "During the past 30 days, that is since [DATEFILL], on how many days did you smoke part or all of a cigarette?" (30)

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study, TUS-CPS, NHIS, NHANES, and NSDUH (continued)

PATH Study	TUS-CPS*	NHIS	NHANES	NSDUH (original definition)	NSDUH (modified definition)**
Coding of "yes" and "No" answers					
If R05R_A_EDY_CIGS = 1 then "Yes" Else if R05R_A_EDY_CIGS = 2 then "No"	If PEA1 = 1 and PEA3 = 1 then "Yes" Else if PEA1 = 2 or PEA3 in (2 3) then "No"	If SMKEV = 1 and SMKNOW = 1 then "Yes" Else if SMKEV = 2 or SMKNOW in (2 3) then "No"	If SMQ020 = 1 and SMQ040 = 1 then "Yes" Else if SMQ020 = 2 or SMQ040 in (2 3) then "No"	If CIGEVER = 1 and CIGREC = 1 and (CG30EST = 6 or CIG30USE = 30) then "Yes" Else if CIGEVER = 2 or CIGREC in (2 3 4 14 19 29) or (CG30EST in (91 93) and CIG30USE in (91 93)) or (CIGEVER = 1 and CIGREC = 1 and (1 <= CG30EST <= 5 or 1 <= CIG30USE <= 29)) then "No"	If CIG100LF in (1 5) and CIGEVER = 1 and CIGREC = 1 and (CG30EST = 6 or CIG30USE = 30) then "Yes" Else if CIG100LF = 2 or CIGEVER = 2 or CIGREC in (2 3 4 14 19 29) or (CG30EST in (91 93) and CIG30USE in (91 93)) or (CIGEVER = 1 and CIGREC = 1 and (1 <= CG30EST <= 5 or 1 <= CIG30USE <= 29)) then "No"
Questions to define some-days cigarette smoking (answers defining some-days cigarette smoking given in parentheses)					
"Do you now smoke cigarettes every day, some days, or not at all?" (some days)	"(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?" (some days)	"Do you NOW smoke cigarettes every day, some days or not at all?" (some days)	"{Do you/Does SP} now smoke cigarettes every day, some days, or not at all?" (some days)	"During the past 30 days, have you smoked part or all of a cigarette?" (yes) "What is your best estimate of the number of days you smoked part or all of a cigarette during the past 30 days?" (1 or 2 days, 3 to 5 days, 6 to 9 days, 10 to 19 days, 20 to 29 days) "During the past 30 days, that is since [DATEFILL], on how many days did you smoke part or all of a cigarette?" (1-29)	"During the past 30 days, have you smoked part or all of a cigarette?" (yes) "What is your best estimate of the number of days you smoked part or all of a cigarette during the past 30 days?" (1 or 2 days, 3 to 5 days, 6 to 9 days, 10 to 19 days, 20 to 29 days) "During the past 30 days, that is since [DATEFILL], on how many days did you smoke part or all of a cigarette?" (1-29)

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study, TUS-CPS, NHIS, NHANES, and NSDUH (continued)

PATH Study	TUS-CPS*	NHIS	NHANES	NSDUH (original definition)	NSDUH (modified definition)**
Coding of “yes” and “No” answers					
If R05R_A_SDY_CIGS = 1 then “Yes” Else if R05R_A_SDY_CIGS = 2 then “No”	If PEA1 = 1 and PEA3 = 2 then “Yes” Else if PEA1 = 2 or PEA3 in (1 3) then “No”	If SMKEV = 1 and SMKNOW = 2 then “Yes” Else if SMKEV = 2 or SMKNOW in (1 3) then “No”	If SMQ020 =1 and SMQ040 = 2 then “Yes” Else if SMQ020 = 2 or SMQ040 in (1 3) then “No”	If CIGEVER = 1 and CIGREC = 1 and (1 <= CG30EST <= 5 or 1 <= CIG30USE <= 29) then “Yes” Else if CIGEVER = 2 or CIGREC in (2,3,4,14,19,29) or (CG30EST in (91 93) and CIG30USE in (91 93)) or (CIGEVER = 1 and CIGREC = 1 and (CG30EST = 6 or CIG30USE = 30)) then “No”	If CIG100LF in (1 5) and CIGEVER = 1 and CIGREC = 1 and (1 <= CG30EST <= 5 or 1 <= CIG30USE <= 29) then “Yes” Else if CIG100LF = 2 or CIGEVER = 2 or CIGREC in (2,3,4,14,19,29) or (CG30EST in (91 93) and CIG30USE in (91 93)) or (CIGEVER = 1 and CIGREC = 1 and (CG30EST = 6 or CIG30USE = 30)) then “No”
Age range included in estimates					
18+	18+	18+	18+	18+	18+

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study, TUS-CPS, NHIS, NHANES, and NSDUH (continued)

PATH Study	TUS-CPS*	NHIS	NHANES	NSDUH (original definition)	NSDUH (modified definition)**
Exclusions from population					
The Wave 4 Cohort target population at Wave 4 included only the U.S. civilian, noninstitutionalized population. The adult target population for the Wave 4 Cohort at Wave 5 was the Wave 4 target population residing in the U.S. and at least 18 years old at Wave 5, except for those who were incarcerated at that time. Thus, it includes Wave 4 respondents who were on active duty or living in a health care institution (e.g., a mental health hospital) but not those in a correctional facility at Wave 5.	Includes only the U.S. civilian, noninstitutionalized population age 18 or older at the time of the interview.	Includes only the civilian, noninstitutionalized population residing in the U.S. at the time of the interview. Several segments of the population are excluded, such as: persons in long-term care institutions; persons on active duty with the Armed Forces; persons in correctional facilities; and U.S. nationals living in foreign countries.	Includes only the U.S. civilian, non-institutionalized population.	Includes only the U.S. civilian, non-institutionalized population. Excludes homeless persons who do not use shelters, military personnel on active duty, and residents of institutional group quarters, such as jails and hospitals.	Includes only the U.S. civilian, non-institutionalized population. Excludes homeless persons who do not use shelters, military personnel on active duty, and residents of institutional group quarters, such as jails and hospitals.
Proxy responses allowed					
No	Yes	Yes, for individuals physically or mentally incapable of responding.	Yes, when there is a serious physical or mental condition, a proxy respondent may be used to conduct the interview.	No	No

*A proxy response is allowed only after four failed callback attempts to the self-respondent or if the self-respondent will not return before closeout. See https://cancercontrol.cancer.gov/sites/default/files/2020-06/cpsmay19.pdf, p7-1.

**The modified definition is given in Ryan et al. (2012).

Table A-2. Questions used to define youth cigarette smoking in the PATH Study, NHANES, NSDUH, NYTS, NYRBS, and MTF

PATH Study	NHANES	NSDUH	NYTS	NYRBS	MTF
Questions to define ever tried cigarette smoking (answers defining ever tried cigarette smoking given in parentheses)					
"Have you ever tried cigarette smoking, even one or two puffs?" (Wave 1 – Wave 5, yes) "In the past 12 months, have you smoked a cigarette, even one or two puffs?" (Wave 2 – Wave 5, yes)	"About how many cigarettes have you smoked in your entire life?" (1 or more puffs but never a whole cigarette, 1 cigarette, 2 to 5 cigarettes, 6 to 15 cigarettes, 16 to 25 cigarettes, 26 to 99 cigarettes, 100 or more cigarettes)	"Have you ever smoked part or all of a cigarette?" (yes)	"Have you ever tried cigarette smoking, even one or two puffs?" (yes)	"Have you ever tried cigarette smoking, even one or two puffs?" (yes)	"Have you ever smoked cigarettes?" (yes)
Coding of "Yes" and "No" answers					
If R05R_Y_EVR_CIGS = 1 then "Yes" Else if R05R_Y_EVR_CIGS= 2 then "No"	If SMQ621 in (2 3 4 5 6 7 8) then "Yes" Else if SMQ621 = 1 then "No"	If CIGEVER = 1 then "Yes Else if CIGEVER = 2 then "No"	If QN6 = 1 then "Yes" Else if QN6 = 2 then "No"	If QN30 = 1 then "Yes" Else if QN30 = 2 then "No"	For 8th and 10th grades: If V7101D = 1 then "Yes" Else if V7101D = 2 then "No" For 12th grade: If V2101D = 1 then "Yes" Else if V2101D = 2 then "No"

Table A-2. Questions used to define youth cigarette smoking in the PATH Study, NHANES, NSDUH, NYTS, NYRBS, and MTF (continued)

PATH Study	NHANES	NSDUH	NYTS	NYRBS	MTF
Questions to define past 30 day cigarette use (answers defining past 30-day cigarette use in parentheses)					
"Have you ever tried cigarette smoking, even one or two puffs?" (Wave 1 – Wave 5, yes) "In the past 12 months, have you smoked a cigarette, even one or two puffs?" (Wave 2 – Wave 5, yes) "When was the last time you smoked a cigarette, even one or two puffs?" (earlier today, not today but sometime in the past 7 days, not in the past 7 days but sometime in the past 30 days)	"About how many cigarettes have you smoked in your entire life?" "During the past 30 days, on how many days did you smoke cigarettes?" (1-30)	"Have you ever smoked part or all of a cigarette?" (yes) "During the past 30 days, have you smoked part or all of a cigarette?" (yes) "What is your best estimate of the number of days you smoked part or all of a cigarette during the past 30 days?" (1 or 2 days, 3 to 5 days, 6 to 9 days, 10 to 19 days, 20 to 29 days, all 30 days) "During the past 30 days, that is since [DATEFILL], on how many days did you smoke part or all of a cigarette?" (1-30)	"During the past 30 days, on how many days did you smoke cigarettes?" (1-30)	"During the past 30 days, on how many days did you smoke cigarettes?" (1-30)	"How frequently have you smoked cigarettes during the past 30 days?" (less than 1 cigarette per day, 1-5 cigarettes per day, about one-half pack per day, about one pack per day, about one and one-half packs per day, two packs or more per day)
Coding of "Yes" and "No" answers					
If R05R_Y_CUR_CIGS = 1 then "Yes" Else if R05R_Y_CUR_CIGS= 2 then "No"	If SMD641 in (1 2 … 30) then "Yes" Else if SMQ621 = 1 or SMD641 = 0 then "No"	If CIGEVER = 1 and (CIGREC = 1 or 1 <= CG30EST <= 6 or 1 <= CIG30USE <= 30) then "Yes" Else if CIGEVER=2 or CIGREC in (2 3 4 14 19 29) or (CG30EST in (91 93) and CIG30USE in (91 93)) then "No"	If QN9 in (1 2 … 30) then "Yes" Else if QN9 in (0, .S) then "No"	If QN32 = 1 then "Yes" Else if QN32 = 2 then "No"	If V2102D = 1 then "Yes" Else if V2102D = 0 then "No"
Age range included in estimates					
12-17	12-17	12-17	12-17	12-17	12-17

Table A-2. Questions used to define youth cigarette smoking in the PATH Study, NHANES, NSDUH, NYTS, NYRBS, and MTF (continued)

PATH Study	NHANES	NSDUH	NYTS	NYRBS	MTF
Exclusions from population					
The Wave 4 Cohort target population at Wave 4 included only the U.S. civilian, noninstitutionalized population. The youth target population for the Wave 4 Cohort at Wave 5 was the Wave 4 target population residing in the U.S. and 12 to 17 years old at Wave 5, except for those who were incarcerated at that time. Thus, it includes Wave 4 respondents who were on active duty or living in a health care institution (e.g., a mental health hospital) but not those in a correctional facility at Wave 5.	Includes only the U.S. civilian, noninstitutionalized population.	Includes only the U.S. civilian, noninstitutionalized population. Excludes homeless persons who do not use shelters, military personnel on active duty, and residents of institutional group quarters, such as jails and hospitals.	Includes only youth who are public and private school students enrolled in regular middle schools and high schools in grades 6 through 12 in the 50 U.S. States and the District of Columbia. Alternative schools, special education schools, Department of Defense operated schools, Bureau of Indian Affairs schools, vocational schools that serve only pull-out populations, and students enrolled in regular schools unable to complete the questionnaire without special assistance are excluded.	Includes only students in grades 9 to 12 who are attending public and private schools. US territories and the Virgin Islands are excluded from the sampling frame.	Includes only youth who are attending public and private schools within the 48 contiguous states.
Other comments					
			Self-administered survey in classroom.	Self-administered survey in classroom.	Separate surveys for 8th and 10th grade vs. 12th grade, self-administered in classroom.

Appendix B
ENDS-Use Questions in the PATH Study and Other Surveys

Table B-1 lists the questions used to ask about current ENDS use of adults in the PATH Study, TUS-CPS, and NHIS. The observations discussed in Appendix A regarding the difference in positioning of the cigarette-smoking questions among the surveys also apply to the questions about use of ENDS.

Table B-2 lists the questions used to ask about ever and past 30-day ENDS use of youth in the PATH Study as well as in the surveys used for comparison. Administration of the questions differs between the PATH Study, MTF, NYRBS, and NYTS. In the PATH Study, the questions generally are asked in the home where the parent or guardian may be present. Questions in MTF, NYRBS, and NYTS are self-administered by students at their school.

The information provided in Tables A-1 (for adults) and A-2 (for youth) about age restrictions, population exclusions, proxy responses, and "other comments" applies to both the cigarette-smoking and ENDS-use estimates and so is not repeated in Tables B-1 and B-2.

Table B-1. Questions and algorithms used to define adult current ENDS use in the PATH Study, TUS-CPS, and NHIS

PATH Study	TUS-CPS	NHIS
Questions to define current ENDS use (answers defining current ENDS use given in parentheses)		
"Have you ever used an e-cigarette, such as NJOY, Blu, or Smoking Everywhere, even one or two times?" (Wave 1, yes) "Have you ever used an electronic nicotine product, even one or two times?" (Wave 2, Wave 4, Wave 5, yes) "In the past 30 days, have you used an electronic nicotine product, even one or two times?" (Wave 3 – Wave 5 adult, yes) "In the past 12 months, have you used an electronic nicotine product, even one or two times?" (Wave 3 – Wave 5, yes) "Have you ever used e-cigarettes fairly regularly?" (Wave 1, yes) "Have you ever used electronic nicotine products fairly regularly?" (Wave 2 – Wave 5, yes) "Do you now use electronic nicotine products every day, some days, or not at all?" (every day, some days)	"(Have/Has) (you/name) ever used e-cigarettes even one time?" (yes) "(Do you/Does [name]) now use an e-cigarette every day, some days, or not at all?" (every day, some days)	"Have you ever used an e-cigarette even one time?" (yes) "Do you now use e-cigarettes every day, some days, or not at all?" (every day, some days)
Coding of "Yes" and "No" answers		
If R05R_A_CUR_ESTD_EPRODS = 1 then "Yes" Else if R05R_A_CUR_ESTD_EPRODS = 2 then "No"	If PEJ1A3_5 = 1 and PEJ2A3_5 in (1 2) then "Yes" Else if PEJ1A3_5 = 2 or PEJ2A3_5 = 3 then "No"	If ECIGEV2 = 1 and ECIGCUR2 in (1 2) then "Yes" Else if ECIGEV2 = 2 or ECIGCUR2 = 3 then "No"
Questions to define every-day ENDS use (answers defining every-day ENDS use given in parentheses)		
"Do you now use electronic nicotine products every day, some days, or not at all?" (every day)	"(Do you/Does [name]) now use an e-cigarette every day, some days, or not at all?" (every day)	"Do you now use e-cigarettes every day, some days, or not at all?" (every day)
Coding of "Yes" and "No" answers		
If R05R_A_EDY_EPRODS = 1 then "Yes" Else if R05R_A_EDY_EPRODS = 2 then "No"	If PEJ1A3_5 = 1 and PEJ2A3_5 = 1 then "Yes" Else if PEJ1A3_5 = 2 or PEJ2A3_5 in (2 3) then "No"	If ECIGEV2 = 1 and ECIGCUR2 =1 then "Yes" Else if ECIGEV2 = 2 or ECIGCUR2 in (2 3) then "No"
Questions to define some-days ENDS use (answers defining some-days ENDS use given in parentheses)		
"Do you now use electronic nicotine products every day, some days, or not at all?" (some days)	"(Do you/Does [name]) now use an e-cigarette every day, some days, or not at all?" (some days)	"Do you now use e-cigarettes every day, some days, or not at all?" (some days)
Coding of "Yes" and "No" answers		
If R05R_A_SDY_EPRODS = 1 then "Yes" Else if R05R_A_SDY_EPRODS = 2 then "No"	If PEJ1A3_5 = 1 and PEJ2A3_5 = 2 then "Yes" Else if PEJ1A3_5 = 2 or PEJ2A3_5 in (1 3) then "No"	If ECIGEV2 = 1 and ECIGCUR2 =2 then "Yes" Else if ECIGEV2 = 2 or ECIGCUR2 in (1 3) then "No"

Table B-2. Questions and algorithms used to define youth ENDS use in the PATH Study, MTF, NYRBS, and NYTS

PATH Study	MTF	NYRBS	NYTS
Questions to define ever ENDS use (answers defining ever ENDS use given in parentheses)			
“Have you ever used an e-cigarette, such as NJOY, Blu, or Smoking Everywhere, even one or two times?” (Wave 1, yes) “Have you ever used an electronic nicotine product, even one or two times?” (Wave 2 – Wave 5, yes) “In the past 12 months, have you used an electronic nicotine product, even one or two times?” (Wave 3 – Wave 5, yes)	“On how many occasions (if any) have you vaped NICOTINE in your lifetime?” (1-2 occasions, 3-5 occasions, 6-9 occasions, 10-19 occasions, 20-39 occasions, 40 or more)	“Have you ever used an electronic vapor product?” (yes)	“Have you ever used an e-cigarette, even once or twice?” (yes)
Coding of “Yes” and “No” answers			
If R05R_Y_EVR_EPRODS =1 then “Yes” Else if R05R_Y_EVR_EPRODS= 2 then “No”	For 8th and 10th grades: If V7649 in (2 3 4 5 6 7) then “Yes” Else if V7649 = 1 then “No” For 12th grade: If V2567 in (2 3 4 5 6 7) then “Yes” Else if V2567 = 1 then “No”	If QN34 = 1 then “Yes” Else if QN34 = 2 then “No”	If QN34 = 1 then “Yes” Else if QN34 = 2 then “No”
Question to define past 30-day ENDS use (answers defining past 30-day ENDS use given in parentheses)			
“When was the last time you used an electronic nicotine product, even one or two times?” (earlier today, not today but sometime in the past 7 days, not in the past 7 days but sometime in the past 30 days)	“On how many occasions (if any) have you vaped NICOTINE during the last 30 days” (1-2 occasions, 3-5 occasions, 6-9 occasions, 10-19 occasions, 20-39 occasions, 40 or more)	“During the past 30 days, on how many days did you use an electronic vapor product?” (1-30)	“During the past 30 days, on how many days did you use e-cigarettes?” (1 or more days)
Coding of “Yes” and “No” answers			
If R05R_Y_CUR_EPRODS = 1 then “Yes” Else if R05R_Y_CUR_EPRODS = 2 then “No”	If V2569 in (2 3 4 5 6 7) then “Yes” Else if V2569 = 1 then “No”	If QN35 = 1 then “Yes” Else if QN35 = 2 then “No”	If QN37 in (1 2 … 30) then “Yes” Else if QN37 in (0 .S) then “No”